Main Category	Category	Terms
Disease	OMIM_DISEASE	Breast cancer, male, susceptibility to
Disease	OMIM_DISEASE	Medulloblastoma
Disease	OMIM_DISEASE	Prostate cancer
Disease	OMIM_DISEASE	Wilms tumor
Disease	OMIM_DISEASE	Fanconi anemia, complementation group D1
Disease	OMIM_DISEASE	Breast-ovarian cancer, familial, 2
Disease	OMIM_DISEASE	Glioblastoma 3
Disease	OMIM_DISEASE	Pancreatic cancer 2
Disease	UP_KW_DISEASE	Tumor suppressor
Disease	UP_KW_DISEASE	Disease variant
Disease	UP_KW_DISEASE	Fanconi anemia
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Cell cycle
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA damage
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA recombination
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	DNA repair
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoskeleton
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Membrane
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	DNA-binding
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic and acidic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Polar residues

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:BRCA2 OB1
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:BRCA2 OB3
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Breast cancer type 2 susceptibility protein helical
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Tower
Functional Annotations	UP_SEQ_FEATURE	MOTIF:Nuclear export signal; masked by interaction with SEM1
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:S->E: Impaired interaction with RAD51.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Loss of phosphorylation by CHEK1 and CHEK2 (in vitro).
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:W->A: Disrupts interaction with SEM1.
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with FANCD2
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with HSF2BP
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with NPM1
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with PALB2
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with POLH
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with RAD51
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with SEM1
Functional Annotations	UP_SEQ_FEATURE	REGION:Required for stimulation of POLH DNA polymerization activity
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 1
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 2
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 3

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 4
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 5
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 6
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 7
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BRCA2 8
Functional Annotations	UP_SEQ_FEATURE	TRANSMEM:Helical
Gene Ontology	GOTERM_BP_DIRECT	telomere maintenance via recombination
Gene Ontology	GOTERM_BP_DIRECT	double-strand break repair via homologous recombination
Gene Ontology	GOTERM_BP_DIRECT	oocyte maturation
Gene Ontology	GOTERM_BP_DIRECT	inner cell mass cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	DNA repair
Gene Ontology	GOTERM_BP_DIRECT	nucleotide-excision repair
Gene Ontology	GOTERM_BP_DIRECT	double-strand break repair
Gene Ontology	GOTERM_BP_DIRECT	DNA recombination
Gene Ontology	GOTERM_BP_DIRECT	chromatin remodeling
Gene Ontology	GOTERM_BP_DIRECT	regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	DNA damage response
Gene Ontology	GOTERM_BP_DIRECT	male meiosis I
Gene Ontology	GOTERM_BP_DIRECT	spermatogenesis
Gene Ontology	GOTERM_BP_DIRECT	brain development
Gene Ontology	GOTERM_BP_DIRECT	cell population proliferation

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	female gonad development
Gene Ontology	GOTERM_BP_DIRECT	intrinsic apoptotic signaling pathway in response to DNA damage
Gene Ontology	GOTERM_BP_DIRECT	response to X-ray
Gene Ontology	GOTERM_BP_DIRECT	response to UV-C
Gene Ontology	GOTERM_BP_DIRECT	response to gamma radiation
Gene Ontology	GOTERM_BP_DIRECT	hemopoiesis
Gene Ontology	GOTERM_BP_DIRECT	DNA damage response, signal transduction by p53 class mediator
Gene Ontology	GOTERM_BP_DIRECT	replication fork processing
Gene Ontology	GOTERM_BP_DIRECT	regulation of cytokinesis
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of mammary gland epithelial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	intrinsic apoptotic signaling pathway in response to DNA damage by p53 class mediator
Gene Ontology	GOTERM_BP_DIRECT	chordate embryonic development
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	chromosome organization
Gene Ontology	GOTERM_BP_DIRECT	centrosome duplication
Gene Ontology	GOTERM_BP_DIRECT	establishment of protein localization to telomere
Gene Ontology	GOTERM_BP_DIRECT	hematopoietic stem cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	cellular response to ionizing radiation
Gene Ontology	GOTERM_BP_DIRECT	stem cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	cellular senescence

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	mitotic recombination-dependent replication fork processing
Gene Ontology	GOTERM_BP_DIRECT	regulation of DNA damage checkpoint
Gene Ontology	GOTERM_CC_DIRECT	nuclear ubiquitin ligase complex
Gene Ontology	GOTERM_CC_DIRECT	chromosome, telomeric region
Gene Ontology	GOTERM_CC_DIRECT	lateral element
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	chromosome
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	centrosome
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	cytoskeleton
Gene Ontology	GOTERM_CC_DIRECT	membrane
Gene Ontology	GOTERM_CC_DIRECT	secretory granule
Gene Ontology	GOTERM_CC_DIRECT	protein-containing complex
Gene Ontology	GOTERM_CC_DIRECT	BRCA2-MAGE-D1 complex
Gene Ontology	GOTERM_CC_DIRECT	DNA repair complex
Gene Ontology	GOTERM_MF_DIRECT	protease binding
Gene Ontology	GOTERM_MF_DIRECT	DNA binding
Gene Ontology	GOTERM_MF_DIRECT	single-stranded DNA binding
Gene Ontology	GOTERM_MF_DIRECT	protein binding

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	histone H3 acetyltransferase activity
Gene Ontology	GOTERM_MF_DIRECT	histone H4 acetyltransferase activity
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding
Gene Ontology	GOTERM_MF_DIRECT	gamma-tubulin binding
General Annotations	CHROMOSOME	13
General Annotations	CYTOGENETIC_LOCATION	13q13.1
General Annotations	ENTREZ_GENE_SUMMARY	Inherited mutations in BRCA1 and this gene, BRCA2, confer increased lifetime risk of developing breast or ovarian cancer. Both BRCA1 and BRCA2 are involved in maintenance of genome stability, specifically the homologous recombination pathway for double-strand DNA repair. The largest exon in both genes is exon 11, which harbors the most important and frequent mutations in breast cancer patients. The BRCA2 gene was found on chromosome 13q12.3 in human. The BRCA2 protein contains several copies of a 70 aa motif called the BRC motif, and these motifs mediate binding to the RAD51 recombinase which functions in DNA repair. BRCA2 is considered a tumor suppressor gene, as tumors with BRCA2 mutations generally exhibit loss of heterozygosity (LOH) of the wild-type allele. [provided by RefSeq, May 2020]
General Annotations	UP_COMMENT_DISEASE	Breast cancer (BC) [MIM:114480]: A common malignancy originating from breast epithelial tissue. Breast neoplasms can be distinguished by their histologic pattern. Invasive ductal carcinoma is by far the most common type. Breast cancer is etiologically and genetically heterogeneous. Important genetic factors have been indicated by familial occurrence and bilateral involvement. Mutations at more than one locus can be involved in different families or even in the same case. {ECO:0000269|PubMed:10399947, ECO:0000269|PubMed:10978364, ECO:0000269|PubMed:11139248, ECO:0000269|PubMed:11149425, ECO:0000269|PubMed:11241844, ECO:0000269|PubMed:11948477, ECO:0000269|PubMed:12145750, ECO:0000269|PubMed:12373604, ECO:0000269|PubMed:12442274, ECO:0000269|PubMed:12442275, ECO:0000269|PubMed:12569143, ECO:0000269|PubMed:12938098, ECO:0000269|PubMed:14722926, ECO:0000269|PubMed:15026808, ECO:0000269|PubMed:15172753, ECO:0000269|PubMed:15365993, ECO:0000269|PubMed:16793542, ECO:0000269|PubMed:17924331, ECO:0000269|PubMed:23108138, ECO:0000269|PubMed:24013206, ECO:0000269|PubMed:29161300, ECO:0000269|PubMed:8640237, ECO:0000269|PubMed:9150152, ECO:0000269|PubMed:9609997, ECO:0000269|PubMed:9654203, ECO:0000269|

Main Category	Category	Terms
		PubMed:9971877}. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Breast-ovarian cancer, familial, 2 (BROVCA2) [MIM:612555]: A condition associated with familial predisposition to cancer of the breast and ovaries. Characteristic features in affected families are an early age of onset of breast cancer (often before age 50), increased chance of bilateral cancers (cancer that develop in both breasts, or both ovaries, independently), frequent occurrence of breast cancer among men, increased incidence of tumors of other specific organs, such as the prostate. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Fanconi anemia complementation group D1 (FANCD1) [MIM:605724]: A disorder affecting all bone marrow elements and resulting in anemia, leukopenia and thrombopenia. It is associated with cardiac, renal and limb malformations, dermal pigmentary changes, and a predisposition to the development of malignancies. At the cellular level it is associated with hypersensitivity to DNA-damaging agents, chromosomal instability (increased chromosome breakage) and defective DNA repair. {ECO:0000269|PubMed:12065746, ECO:0000269|PubMed:14670928, ECO:0000269|PubMed:16825431, ECO:0000269|PubMed:17924331, ECO:0000269|PubMed:21719596, ECO:0000269|PubMed:23108138}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Glioma 3 (GLM3) [MIM:613029]: Gliomas are benign or malignant central nervous system neoplasms derived from glial cells. They comprise astrocytomas and glioblastoma multiforme that are derived from astrocytes, oligodendrogliomas derived from oligodendrocytes and ependymomas derived from ependymocytes. {ECO:0000269|PubMed:15689453}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Medulloblastoma (MDB) [MIM:155255]: Malignant, invasive embryonal tumor of the cerebellum with a preferential manifestation in children. {ECO:0000269|PubMed:29753700}. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Pancreatic cancer 2 (PNCA2) [MIM:613347]: A malignant neoplasm of the pancreas. Tumors can arise from both the exocrine and endocrine portions of the pancreas, but 95% of them develop from the exocrine portion, including the ductal epithelium, acinar cells, connective tissue, and lymphatic tissue. {ECO:0000269|PubMed:9140390}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_FUNCTION	Involved in double-strand break repair and/or homologous recombination. Binds RAD51 and

Main Category	Category	Terms
		potentiates recombinational DNA repair by promoting assembly of RAD51 onto single-stranded DNA (ssDNA). Acts by targeting RAD51 to ssDNA over double-stranded DNA, enabling RAD51 to displace replication protein-A (RPA) from ssDNA and stabilizing RAD51-ssDNA filaments by blocking ATP hydrolysis. Part of a PALB2-scaffolded HR complex containing RAD51C and which is thought to play a role in DNA repair by HR. May participate in S phase checkpoint activation. Binds selectively to ssDNA, and to ssDNA in tailed duplexes and replication fork structures. May play a role in the extension step after strand invasion at replication-dependent DNA double-strand breaks; together with PALB2 is involved in both POLH localization at collapsed replication forks and DNA polymerization activity. In concert with NPM1, regulates centrosome duplication. Interacts with the TREX-2 complex (transcription and export complex 2) subunits PCID2 and SEM1, and is required to prevent R-loop-associated DNA damage and thus transcription-associated genomic instability. Silencing of BRCA2 promotes R-loop accumulation at actively transcribed genes in replicating and non-replicating cells, suggesting that BRCA2 mediates the control of R-loop associated genomic instability, independently of its known role in homologous recombination (PubMed:24896180). {ECO:0000269|PubMed:15115758, ECO:0000269|PubMed:15199141, ECO:0000269|PubMed:15671039, ECO:0000269|PubMed:18317453, ECO:0000269|PubMed:20729832, ECO:0000269|PubMed:20729858, ECO:0000269|PubMed:20729859, ECO:0000269|PubMed:21084279, ECO:0000269|PubMed:21719596, ECO:0000269|PubMed:24485656, ECO:0000269|PubMed:24896180}.
General Annotations	UP_COMMENT_INTERACTION	P51587; P51587: BRCA2; NbExp=3; IntAct=EBI-79792, EBI-79792; P51587; Q14565: DMC1; NbExp=12; IntAct=EBI-79792, EBI-930865; P51587; Q9BXW9: FANCD2; NbExp=16; IntAct=EBI-79792, EBI-359343; P51587; Q9BXW9-2: FANCD2; NbExp=3; IntAct=EBI-79792, EBI-596878; P51587; Q9P0W2: HMG20B; NbExp=8; IntAct=EBI-79792, EBI-713401; P51587; Q9GZQ8: MAP1LC3B; NbExp=2; IntAct=EBI-79792, EBI-373144; P51587; Q86YC2: PALB2; NbExp=30; IntAct=EBI-79792, EBI-1222653; P51587; Q9NTI5: PDS5B; NbExp=26; IntAct=EBI-79792, EBI-1175604; P51587; Q9Y253: POLH; NbExp=6; IntAct=EBI-79792, EBI-2827270; P51587; Q06609: RAD51; NbExp=46; IntAct=EBI-79792, EBI-297202; P51587; Q06609-1: RAD51; NbExp=12; IntAct=EBI-79792, EBI-15557721; P51587; P60896: SEM1; NbExp=10; IntAct=EBI-79792, EBI-79819; P51587; P04637: TP53; NbExp=7; IntAct=EBI-79792, EBI-366083;
General Annotations	UP_COMMENT_PTM	Phosphorylated by ATM upon irradiation-induced DNA damage. Phosphorylation by CHEK1 and CHEK2 regulates interaction with RAD51. Phosphorylation at Ser-3291 by CDK1 and CDK2 is low in S phase when recombination is active, but increases as cells progress towards mitosis; this phosphorylation prevents

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		homologous recombination- dependent repair during S phase and G2 by inhibiting RAD51 binding. {ECO:0000269|PubMed:15199141, ECO:0000269|PubMed:15800615, ECO:0000269|PubMed:18317453}.
General Annotations	UP_COMMENT_PTM	Ubiquitinated in the absence of DNA damage; this does not lead to proteasomal degradation. In contrast, ubiquitination in response to DNA damage leads to proteasomal degradation. {ECO:0000269|PubMed:15314155}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cytoplasm, cytoskeleton, microtubule organizing center, centrosome {ECO:0000256|ARBA:ARBA00004300}. Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000269|PubMed:24013206, ECO:0000269|PubMed:26833090, ECO:0000305|PubMed:21276791}. Cytoplasm, cytoskeleton, microtubule organizing center, centrosome {ECO:0000269|PubMed:21276791}. Note=Colocalizes with ERCC5/XPG to nuclear foci following DNA replication stress. {ECO:0000269|PubMed:26833090}.
General Annotations	UP_COMMENT_SUBUNIT	Monomer and dimer (PubMed:20729858). Interacts with RAD51; regulates RAD51 recruitment and function at sites of DNA repair (PubMed:12442171, PubMed:15800615, PubMed:18317453, PubMed:20729832, PubMed:20729859). Interacts with WDR16, USP11, DMC1, ROCK2 and NPM1 (PubMed:15314155, PubMed:15967112, PubMed:20729832, PubMed:21084279). Interacts with SEM1; the interaction masks a nuclear export signal in BRCA2 (PubMed:10373512, PubMed:16205630, PubMed:21719596, PubMed:24013206). Interacts with both nonubiquitinated and monoubiquitinated FANCD2; this complex also includes XRCC3 and phosphorylated FANCG (PubMed:15115758, PubMed:15199141, PubMed:18212739). Part of a BRCA complex containing BRCA1, BRCA2 and PALB2 (PubMed:19369211). Component of the homologous recombination repair (HR) complex composed of ERCC5/XPG, BRCA2, PALB2, DSS1 and RAD51 (PubMed:26833090). Within the complex, interacts with ERCC5/XPG and PALB2 (PubMed:26833090). Interacts directly with PALB2 which may serve as a scaffold for a HR complex containing PALB2, BRCA2, RAD51C, RAD51 and XRCC3 (PubMed:16793542, PubMed:19369211, PubMed:19609323, PubMed:24141787, PubMed:26833090, PubMed:28319063). Interacts with BRCA1 only in the presence of PALB2 which serves as the bridging protein (PubMed:19369211). Interacts with POLH; the interaction is direct (PubMed:24485656). Interacts with the TREX-2 complex subunits PCID2 and SEM1 (PubMed:21719596, PubMed:24896180). Interacts with HSF2BP and BRME1; the interaction with HSF2BP is direct and allows the formation of a ternary complex (PubMed:31242413). The complex BRME1:HSF2BP:BRCA2 interacts with SPATA22, MEIOB and RAD51 (By similarity). {ECO:0000250|UniProtKB:P97929, ECO:0000269|PubMed:10373512, ECO:0000269|PubMed:12442171, ECO:0000269|

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		PubMed:15115758, ECO:0000269|PubMed:15199141, ECO:0000269|PubMed:15314155, ECO:0000269|PubMed:15800615, ECO:0000269|PubMed:15967112, ECO:0000269|PubMed:16205630, ECO:0000269|PubMed:16793542, ECO:0000269|PubMed:18212739, ECO:0000269|PubMed:18317453, ECO:0000269|PubMed:19369211, ECO:0000269|PubMed:19609323, ECO:0000269|PubMed:20729832, ECO:0000269|PubMed:20729858, ECO:0000269|PubMed:20729859, ECO:0000269|PubMed:21084279, ECO:0000269|PubMed:21719596, ECO:0000269|PubMed:24013206, ECO:0000269|PubMed:24141787, ECO:0000269|PubMed:24485656, ECO:0000269|PubMed:24896180, ECO:0000269|PubMed:26833090, ECO:0000269|PubMed:28319063, ECO:0000269|PubMed:31242413}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Highest levels of expression in breast and thymus, with slightly lower levels in lung, ovary and spleen.
Identifiers	ENTREZ_GENE_ID	675
Identifiers	ENSEMBL_GENE_ID	ENSG00000139618
Identifiers	UNIPROT_ID	Q9H4L2_HUMAN
Identifiers	UNIPROT_ID	Q9H4L3_HUMAN
Identifiers	UNIPROT_ID	Q9H265_HUMAN
Identifiers	UNIPROT_ID	Q9BX99_HUMAN
Identifiers	UNIPROT_ID	Q8IU77_HUMAN
Identifiers	UNIPROT_ID	Q8IU82_HUMAN
Identifiers	UNIPROT_ID	Q8IU64_HUMAN
Identifiers	UNIPROT_ID	K4JWF8_HUMAN
Identifiers	UNIPROT_ID	K4JXU1_HUMAN
Identifiers	UNIPROT_ID	K4JUC8_HUMAN
Identifiers	UNIPROT_ID	K4K7W0_HUMAN
Identifiers	UNIPROT_ID	K4JUC3_HUMAN
Identifiers	UNIPROT_ID	K4JXT6_HUMAN
Identifiers	UNIPROT_ID	K4JTT6_HUMAN
Identifiers	UNIPROT_ID	K4K7V6_HUMAN
Identifiers	UNIPROT_ID	K4JTS7_HUMAN
Identifiers	UNIPROT_ID	K4JTT2_HUMAN
Identifiers	UNIPROT_ID	K4JWF3_HUMAN
Identifiers	UNIPROT_ID	H0YE37_HUMAN
Identifiers	UNIPROT_ID	H0YD86_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	BRCA2_HUMAN
Identifiers	UNIPROT_ID	A1YBP1_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BGN4_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BGN0_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BGQ6_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BGN2_HUMAN
Identifiers	UNIPROT_ID	A0A8V8TQQ4_HUMAN
Identifiers	UNIPROT_ID	A0A8V8TPZ2_HUMAN
Identifiers	UNIPROT_ID	A0A8E6I3X3_HUMAN
Identifiers	UNIPROT_ID	A0A8A2FPR6_HUMAN
Identifiers	UNIPROT_ID	A0A8A2FLM5_HUMAN
Identifiers	UNIPROT_ID	A0A7P0TAP7_HUMAN
Identifiers	UNIPROT_ID	A0A7P0T9D7_HUMAN
Identifiers	UNIPROT_ID	A0A7G9ZE26_HUMAN
Identifiers	UNIPROT_ID	A0A7G9ZE27_HUMAN
Identifiers	UNIPROT_ID	A0A7G8KP82_HUMAN
Identifiers	UNIPROT_ID	A0A7G8KP83_HUMAN
Identifiers	UNIPROT_ID	A0A7G6J4N9_HUMAN
Identifiers	UNIPROT_ID	A0A6G8IU23_HUMAN
Identifiers	UNIPROT_ID	A0A6G8IU57_HUMAN
Identifiers	UNIPROT_ID	A0A649UI40_HUMAN
Identifiers	UNIPROT_ID	A0A649UI09_HUMAN
Identifiers	UNIPROT_ID	A0A649UIZ9_HUMAN
Identifiers	UNIPROT_ID	A0A649UI46_HUMAN
Identifiers	UNIPROT_ID	A0A590UJU6_HUMAN
Identifiers	UNIPROT_ID	A0A590UJI7_HUMAN
Identifiers	UNIPROT_ID	A0A590UJ24_HUMAN
Identifiers	UNIPROT_ID	A0A4P8DL22_HUMAN
Identifiers	UNIPROT_ID	A0A4P8DKU7_HUMAN
Identifiers	UNIPROT_ID	A0A4P8DL54_HUMAN
Identifiers	UNIPROT_ID	A0A4P8DKN2_HUMAN
Identifiers	UNIPROT_ID	A0A386IN22_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	A0A386INK6_HUMAN
Identifiers	UNIPROT_ID	A0A386IQ54_HUMAN
Identifiers	UNIPROT_ID	A0A386IN16_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV8_HUMAN
Identifiers	UNIPROT_ID	A0A386IP79_HUMAN
Identifiers	UNIPROT_ID	A0A386IQ62_HUMAN
Identifiers	UNIPROT_ID	A0A386IQ50_HUMAN
Identifiers	UNIPROT_ID	A0A386IPY1_HUMAN
Identifiers	UNIPROT_ID	A0A386ING9_HUMAN
Identifiers	UNIPROT_ID	A0A386INL2_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV2_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV9_HUMAN
Identifiers	UNIPROT_ID	A0A386ING3_HUMAN
Identifiers	UNIPROT_ID	A0A386IPX8_HUMAN
Identifiers	UNIPROT_ID	A0A386IN05_HUMAN
Identifiers	UNIPROT_ID	A0A386IPJ4_HUMAN
Identifiers	UNIPROT_ID	A0A386IPX7_HUMAN
Identifiers	UNIPROT_ID	A0A386INM2_HUMAN
Identifiers	UNIPROT_ID	A0A386IP96_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV1_HUMAN
Identifiers	UNIPROT_ID	A0A386IPI4_HUMAN
Identifiers	UNIPROT_ID	A0A386INE3_HUMAN
Identifiers	UNIPROT_ID	A0A386INF6_HUMAN
Identifiers	UNIPROT_ID	A0A386IP91_HUMAN
Identifiers	UNIPROT_ID	A0A386IP85_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV5_HUMAN
Identifiers	UNIPROT_ID	A0A386IQ46_HUMAN
Identifiers	UNIPROT_ID	A0A386IPY0_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV3_HUMAN
Identifiers	UNIPROT_ID	A0A386IPK0_HUMAN
Identifiers	UNIPROT_ID	A0A386IN23_HUMAN
Identifiers	UNIPROT_ID	A0A386IND4_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	A0A386IN02_HUMAN
Identifiers	UNIPROT_ID	A0A386IN15_HUMAN
Identifiers	UNIPROT_ID	A0A386IPH1_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV4_HUMAN
Identifiers	UNIPROT_ID	A0A386IPV6_HUMAN
Identifiers	UNIPROT_ID	A0A386IPH8_HUMAN
Identifiers	UNIPROT_ID	A0A2S1WCJ3_HUMAN
Identifiers	UNIPROT_ID	A0A248X3Y4_HUMAN
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Identifiers	UNIPROT_ID	A0A248X3Z1_HUMAN
Identifiers	UNIPROT_ID	A0A248X3Z0_HUMAN
Identifiers	UNIPROT_ID	A0A248X3Z4_HUMAN
Identifiers	UNIPROT_ID	A0A248X3Y9_HUMAN
Identifiers	UNIPROT_ID	A0A248X3Z6_HUMAN
Identifiers	UNIPROT_ID	A0A248X3Y7_HUMAN
Identifiers	UNIPROT_ID	A0A248X3Z2_HUMAN
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Identifiers	UNIPROT_ID	A0A0F6TMZ6_HUMAN
Identifiers	UNIPROT_ID	A0A0F6RAD4_HUMAN
Identifiers	UNIPROT_ID	A0A0F6RAU4_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	A0A0F6RAD6_HUMAN
Identifiers	UNIPROT_ID	A0A0F6RAU8_HUMAN
Identifiers	UNIPROT_ID	A0A0F6RAU6_HUMAN
Identifiers	UNIPROT_ID	A0A0F6RAD2_HUMAN
Identifiers	UNIPROT_ID	A0A0A7E6R6_HUMAN
Identifiers	UNIPROT_ID	A0A068BFX3_HUMAN
Identifiers	UNIPROT_ID	A0A068BI50_HUMAN
Identifiers	UNIPROT_ID	A0A068BFY4_HUMAN
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Identifiers	UNIPROT_ID	A0A068BDS6_HUMAN
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Identifiers	UNIPROT_ID	A0A068BDU2_HUMAN
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Identifiers	UNIPROT_ID	A0A068BFU7_HUMAN
Identifiers	UNIPROT_ID	A0A068BEW3_HUMAN
Identifiers	UNIPROT_ID	A0A068BFW6_HUMAN
Identifiers	UNIPROT_ID	A0A068BDV3_HUMAN
Identifiers	UNIPROT_ID	A0A068BFW8_HUMAN
Identifiers	UNIPROT_ID	A0A068BI27_HUMAN
Identifiers	UNIPROT_ID	A0A068BDU4_HUMAN
Identifiers	UNIPROT_ID	A0A068BDT0_HUMAN
Identifiers	UNIPROT_ID	A0A068BEX2_HUMAN
Identifiers	UNIPROT_ID	A0A068BC92_HUMAN
Identifiers	UNIPROT_ID	A0A068BEV8_HUMAN
Identifiers	UNIPROT_ID	A0A068BI30_HUMAN
Identifiers	UNIPROT_ID	A0A068BDS2_HUMAN
Identifiers	UNIPROT_ID	A0A068BCA9_HUMAN
Literature	GENERIF_SUMMARY	Suppression of tumorigenicity of rat liver tumor cells by human chromosome 13

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Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations in Russian familial breast cancer
Literature	GENERIF_SUMMARY	Germline mutation in BRCA2 associated with hypersensitivity to radiation
Literature	GENERIF_SUMMARY	A low frequency of recurrent BRCA2 mutations has been found in breast and ovarian cancers in Spain.
Literature	GENERIF_SUMMARY	Three Chinese cases carrying the recurrent BRCA2 mutation 3337C>T show sharing of some alleles, suggesting some degree of shared ancestry but not sufficient to demonstrate founder effect.
Literature	GENERIF_SUMMARY	minor role of exon 1 among Sudanesse breast cancer patients
Literature	GENERIF_SUMMARY	mutagen sensitivity of blood lymphocytes from women carrying brca2 mutatioon
Literature	GENERIF_SUMMARY	Contribution of BRCA2 germline mutations to hereditary breast/ovarian cancer in Germany.
Literature	GENERIF_SUMMARY	BRCA2 N372H polymorphism associated with modest recessively inherited risk of breast cancer in Australian women
Literature	GENERIF_SUMMARY	somatic mutations in BRCA2 and high frequency of allelic loss occurs in sporadic male breast cancer
Literature	GENERIF_SUMMARY	Expression of BRCA1 and BRCA2 in different tumor cell lines with various growth status
Literature	GENERIF_SUMMARY	Cells from carriers of mutations in one allele of the BRCA1 or BRCA2 genes have no gross defects in their ability to rejoin radiation-induced DNA breaks.
Literature	GENERIF_SUMMARY	cell lines derived from Fanconi anemia B and D1 patients have biallelic mutations in BRCA2 and express truncated BRCA2 proteins
Literature	GENERIF_SUMMARY	BRCA2 exons 10 & 11 were studied by the protein truncation test, & BRCA2 exons 9, 17, 18 and 23 with the SSCP assay on genomic DNA from early-onset breast cancer patients by PCR. 2 frameshifts, 3 missense mutations, and a polymorphism were found.
Literature	GENERIF_SUMMARY	Results demonstrate the importance of BRCA2 in the development of ovarian cancer in this Turkish population.
Literature	GENERIF_SUMMARY	The 999del5 mutation in the BRCA2 gene explains a substantial proportion of familial risk of breast cancer in Iceland.
Literature	GENERIF_SUMMARY	first BRCA2 missense mutation shown to be a predicted deleterious protein-truncating mutation and suggests a potentially useful method for determining the clinical significance of a subset of the many unclassified variants in BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	Contribution of BRCA1 and BRCA2 mutations to breast

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		and ovarian cancer in Pakistan.
Literature	GENERIF_SUMMARY	eight of the most common reported missense mutations in BRCA1 and BRCA2 occurring in patients tested for hereditary risk of breast and ovarian cancers
Literature	GENERIF_SUMMARY	crystal structure of a COOH terminal BRCA2 domain bound to DSS1; demonstrate that this BRCA2 domain binds single-stranded DNA and show that BRCA2 stimulates RAD51-mediated recombination in vitro
Literature	GENERIF_SUMMARY	relation of gene to various cancers, especially breast and ovarian cancers
Literature	GENERIF_SUMMARY	inactivated in ovarian cancer
Literature	GENERIF_SUMMARY	Survival in prospectively ascertained familial breast cancer
Literature	GENERIF_SUMMARY	recent findings regarding BRCA2 in breast cancer - review
Literature	GENERIF_SUMMARY	BBRCA2 gene mutation in unselected women with ovarian and early breast cancer in Mongolia predicts the genetic predisoposition to these diseases.
Literature	GENERIF_SUMMARY	crystal structure of a complex between an evolutionarily conserved sequence in BRCA2 (the BRC repeat) and the RecA-homology domain of RAD51
Literature	GENERIF_SUMMARY	Two BRCA2 missense variants were identified in breast cancer patients in India.
Literature	GENERIF_SUMMARY	BRCA2 mutation, 7883delTTAA, was identified in breast cancer patients in China.
Literature	GENERIF_SUMMARY	BRCA2 2663-2664insA and rare variants were identified in breast cancer patients in Mexico.
Literature	GENERIF_SUMMARY	Germline mutations in BRCA2 account for breast neoplasms predisposition in the majority of families.
Literature	GENERIF_SUMMARY	highly recurrent BRCA2 splice site mutation (IVS16-2A>G) in three breast cancer-only families
Literature	GENERIF_SUMMARY	BRCA2 has a more specific role in DNA repair, regulating the activity of RAD51(review)
Literature	GENERIF_SUMMARY	cancer-predisposing mutation compromises interaction between BRCA2 and replication protein A
Literature	GENERIF_SUMMARY	The BRCA2 gene plays a significant role in the familial breast cancer phenotype in the Cypriot population.
Literature	GENERIF_SUMMARY	mutation analysis of this gene in a case of familial endometriosis
Literature	GENERIF_SUMMARY	Our data support an important role for BRCA2 germline mutations in a subpopulation of families with familial pancreatic cancer.
Literature	GENERIF_SUMMARY	S-phase RAD51 foci form normally in CAPAN-1 cells

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		expressing truncated BRCA2. The observed BRCA2-dependent & independent formation of RAD51 foci shows that intact BRCA2 is not required for RAD51 focus formation per se.
Literature	GENERIF_SUMMARY	BRCA2 germline mutation is associated with fallopian tube cancer.
Literature	GENERIF_SUMMARY	polymorphisms in BRCA2 is associated with esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	Changes in the topoisomerase I activity in V-C8 cells are due to the defective function of the Brca2 gene.
Literature	GENERIF_SUMMARY	Average cumulative risks in BRCA2-mutation carriers by age 70 years were 45% (95% confidence interval 31%-56%) for breast cancer and 11% (2.4%-19%) for ovarian cancer
Literature	GENERIF_SUMMARY	germline BRCA2 mutations are not associated with an increased risk for lymphoid malignancies
Literature	GENERIF_SUMMARY	Founder mutations are present within the Scottish/ Northern Irish population and have implications for the organisation of molecular screening services.
Literature	GENERIF_SUMMARY	wild-type BRCA2, but not a tumor-specific truncated mutant BRCA2, synergizes with the nuclear receptor coactivator p160 GRIP1 to enhance transcriptional activation by androgen receptor
Literature	GENERIF_SUMMARY	major role of Brca2 in mediating cell survival after irradiation is in the S and G(2) phases of the cell cycle
Literature	GENERIF_SUMMARY	Androgen receptor expression is absent in BRCA1-mutated breast tumors when compared to BRCA2-mutated cases, indicating that BRCA2-mutated tumors may be different.
Literature	GENERIF_SUMMARY	BRCA2 has a role in modulating M phase progression
Literature	GENERIF_SUMMARY	None of the BRCA 1 or 2 mutations were detected in the ovarian cancer patient group.
Literature	GENERIF_SUMMARY	Recurrent BRCA2 mutations have no role in predisposition to prostate cancer in Finland.
Literature	GENERIF_SUMMARY	The product of the BRCA2 gene has a function relating to the differentiation of epithelial tissue in the breast in response to the hormones of pregnancy.
Literature	GENERIF_SUMMARY	Eight novel BRCA2 mutations identified in breast and ovarian cancers may be deleterious cancer predisposing mutations.
Literature	GENERIF_SUMMARY	the spectrum of BRCA2 mutations in African Americans
Literature	GENERIF_SUMMARY	In general, OC use, childbearing and breastfeeding do not differ between BRCA1/2 carriers and non-carriers with ovarian cancer. However, the effects of tubal ligation may differ between BRCA1 carriers and non-carriers.

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Literature	GENERIF_SUMMARY	BRCA2 is epistatic to FA genes for ICL repair, but not for damage-induced modification of FANCD2 and may act downstream form FANCD2.
Literature	GENERIF_SUMMARY	study from South India, on BRCA1, BRCA2 & CHEK2 mutations in patients with a family history of breast and/or ovarian cancer and early onset breast/ovarian cancer
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with serous carcinoma of ovary
Literature	GENERIF_SUMMARY	risks of breast and ovarian cancer were determined for Ashkenazi Jewish women with inherited mutations in the tumor suppressor genes BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	Data report the isolation of a holoenzyme complex termed BRCC containing BRCA1, BRCA2, and RAD51, which displays increased association with p53 following DNA damage and ubiquitinates p53 in vitro.
Literature	GENERIF_SUMMARY	inactivation of the FANC-BRCA pathway is relatively common in solid tumors and may be related to tobacco and alcohol exposure
Literature	GENERIF_SUMMARY	fiver germline mutations in cases of familial esophageal squamous cell carcinoma suggests role in genetic susceptibility
Literature	GENERIF_SUMMARY	The remarkable clinical overlap between sporadic EMSY amplification and familial BRCA2 deletion implicates a BRCA2 pathway in sporadic breast and ovarian cancer.
Literature	GENERIF_SUMMARY	In transgenic mice, rescues embryonic lethality and mice develop normally.
Literature	GENERIF_SUMMARY	Molecular analysis revealed biallelic BRCA2/FANCD1 mutations in a kindred with solid tumors in childhood.
Literature	GENERIF_SUMMARY	Novel germline deleterious pathogenic, protein truncating frameshift and non-sense mutations were detected in exon 11 of BRCA2 in breast-ovarian cancer families.
Literature	GENERIF_SUMMARY	heterozygosity for germ-line mutations in BRCA2 results in development of progesterone receptor A predominance.
Literature	GENERIF_SUMMARY	Man with a history of clinically diagnosed right breast cancer who subsequently tested positive for the breast cancer susceptibility gene BRCA2 and received a diagnosis of mammographically detected left breast cancer at screening.
Literature	GENERIF_SUMMARY	Allelic loss at the BRCA2 locus may be of use as a negative predictor for metastases-free and overall survival in breast cancer patients.
Literature	GENERIF_SUMMARY	BRCA1-related breast cancers are more frequently estrogen receptor (ER) negative than are either BRCA2-related or nonhereditary breast cancers.

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Literature	GENERIF_SUMMARY	identified 6 children in 5 kindreds exhibiting the co-occurrence of BRCA2 mutations, FA, and early onset acute leukemia
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations may have a role in progression of ovarian cancer
Literature	GENERIF_SUMMARY	BRCA mutations were present in 12.7% of the high risk patients, compared with 2.8% of the unselected patients.
Literature	GENERIF_SUMMARY	Our results suggest that there is a field effect of early genetic events preceding morphologic changes in the mammary glands of BRCA mutation carriers.
Literature	GENERIF_SUMMARY	FANCD2 and BRCA2 can be coimmunoprecipitated from cell extracts of both human and Chinese hamster wild-type cells, thus confirming that the interaction occurs in vivo
Literature	GENERIF_SUMMARY	Cancer variation associated with the position of the mutation in the BRCA2 gene in 7 different neoplasms and the differences in cancer risk remain to be explored.
Literature	GENERIF_SUMMARY	BRCA2 mutations could not be detected among unrelated non-Ashkenazi-Jewish high risk families in Israel.
Literature	GENERIF_SUMMARY	Identification and evaluation of 55 genetic variations in the BRCA1 and the BRCA2 genes of patients from 50 Japanese breast cancer families
Literature	GENERIF_SUMMARY	RAD51D polymorphism is not associated with BRCA1 or 2 genes in breast cancer.
Literature	GENERIF_SUMMARY	results demonstrate that monoubiquitination of FANCD2, which is regulated by the FA pathway, promotes BRCA2 loading into chromatin complexes. These complexes appear to be required for normal homology-directed DNA repair.
Literature	GENERIF_SUMMARY	BRCA2 germline mutations may have a role in primary cancer of the fallopian tube
Literature	GENERIF_SUMMARY	Results suggest that BRCA2 expression levels are regulated by ubiquitination in response to DNA damage and that USP11 participates in DNA damage repair functions within the BRCA2 pathway independently of BRCA2 deubiquitination.
Literature	GENERIF_SUMMARY	Patients with bilateral breast cancer having BRCA2 mutations are significantly younger than non-carriers.
Literature	GENERIF_SUMMARY	A novel pathogenic germline mutations,BRCA2 c.2259delT, was found in a screening of sporadic Korean breast cancer patients, along with 4 polymorphic and 8 intronic variants of unknown clinical significance.
Literature	GENERIF_SUMMARY	results show that BRCA2 deficiency impairs the completion of cell division by cytokinesis

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Literature	GENERIF_SUMMARY	prevalence of BRCA2 mutations in a large hospital based series of unselected breast cancer cases
Literature	GENERIF_SUMMARY	Quantitative-PCR and Northern analysis confirmed down-regulation of UCRP and UBE2L6 with BRCA2 knockdown, respectively.
Literature	GENERIF_SUMMARY	FANCD2 mediates double strand DNA break repair independently of BRCA2- and Rad51-associated homologous recombination
Literature	GENERIF_SUMMARY	Identification of a potential novel domain in the BRCA2 protein.
Literature	GENERIF_SUMMARY	results reveal unique characteristics of BRCA1/2 mutation, genotype-phenotype & prognosis in moderate- & low-risk individuals of Greek ancestry; breast cancer due to mutations in BRCA1 & BRCA2 appears to be a heterogeneous syndrome in Greek population
Literature	GENERIF_SUMMARY	SLUG is a negative regulator for BRCA2 gene expression
Literature	GENERIF_SUMMARY	results indicate that S3291 phosphorylation might provide a molecular switch to regulate RAD51 recombination activity, providing new insight into why BRCA2 C-terminal deletions lead to radiation sensitivity and cancer predisposition
Literature	GENERIF_SUMMARY	BRCA1/2 mutation screening should be considered for all women diagnosed before age 41.
Literature	GENERIF_SUMMARY	No association was found for the N372H polymorphism in BRCA2 and breast cancer among Caucasian women.
Literature	GENERIF_SUMMARY	There is no significant effect of AIB1 genetic variation on breast cancer risk in BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	role of BRCA1 and BRCA2 mutation in pre-disposition to ovarian cancer.
Literature	GENERIF_SUMMARY	Expression of BRCA2 and MAGE-D1 synergistically suppresses cell proliferation independently of the p53 pathway.
Literature	GENERIF_SUMMARY	Germline mutations within breast cancer susceptibility genes, such as BRCA1 and BRCA2 are associated with a major risk of breast cancer during lifetime.
Literature	GENERIF_SUMMARY	BRCA2 BRC motifs bind distinct regions, and form stable complexes with RAD51-DNA nucleoprotein filaments.
Literature	GENERIF_SUMMARY	Distribution of BRCA1 and BRCA2 mutations in a cohort of young women with breast cancer, compared as a function with race.
Literature	GENERIF_SUMMARY	women with a BRCA1 protein mutation and 4 or more children had a 38% decrease in breast cancer risk compared to nulliparous women, while among BRCA2 protein carriers, increasing parity was associated with

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		an increased risk of breast cancer
Literature	GENERIF_SUMMARY	ells lacking the fully functional protein have consistently been found to show increased sensitivity to a variety of DNA-damaging agents.
Literature	GENERIF_SUMMARY	analysis of Alu element insertions within the BRCA1/2 coding sequences
Literature	GENERIF_SUMMARY	Mutational analysis of BRCA1/2 genes in 151 high-risk patients characterized the spectrum of gene alterations and demonstrated the dominant role of the BRCA1 c.5266dupC allele in hereditary breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Conclusive evidence that the S384F variant of BRCA2 is not a disease causing mutation in breast cancer.
Literature	GENERIF_SUMMARY	Weight loss in early adult life (age 18 to 30) protects against early-onset BRCA2 associated breast cancers.
Literature	GENERIF_SUMMARY	large genomic deletions inactivate the BRCA2 gene in breast cancer families
Literature	GENERIF_SUMMARY	results of this study suggested that the polymorphism N372H in BRCA2 gene may be associated with idiopathic male infertility with azoospermia or severe oligozoospermia
Literature	GENERIF_SUMMARY	BRCA2 mutations are associated with breast and ovarian cancer
Literature	GENERIF_SUMMARY	the Met1915Thr polymorphism in the BRCA2 gene may be considered as an independent marker of breast cancer
Literature	GENERIF_SUMMARY	The frequency of hereditary ovarian carcinoma is attributed to BRCA2 gene mutation.
Literature	GENERIF_SUMMARY	The data suggest applying an increased level of clinical alertness to those with defects in BRCA-related pathways.
Literature	GENERIF_SUMMARY	the data are not compatible with selective pressures during tumorigenesis promoting the functional loss of BRCA2 and MRE11 in microsatellite unstable tumors, but fit closely with an absence of selective pressures acting on BRCA2 and MRE11 gene status
Literature	GENERIF_SUMMARY	there is a decrease in p53 apoptotic rate associated with the No Ins-72Pro haplotype in BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Salpingo-oophorectomy, despite being quite a radical preventive method, might offer protection for the carriers against life-threatening silently-developing cancer.
Literature	GENERIF_SUMMARY	Recurrent BRCA2 mutations in French Canadian breast and/ovarian cancer families could be attributed to common founders.

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Literature	GENERIF_SUMMARY	Carrying a BRCA1 or BRCA2 mutation is not a risk factor for spontaneous abortions but may be associated with frequency of induced abortion.
Literature	GENERIF_SUMMARY	Preeminent associations were identified in SNPs mapping to genes pivotal in the DNA damage-response and cell-cycle pathways, including ATM F858L and P1054R, CHEK2 I157T, BRCA2 N372H, and BUB1B Q349R.
Literature	GENERIF_SUMMARY	The effects of BRCA2 intron variants on mRNA splicing and expression.
Literature	GENERIF_SUMMARY	FANCG-XRCC3 and FANCG-BRCA2 interactions did not require the presence of other FA proteins from the core complex
Literature	GENERIF_SUMMARY	Carrier of mutant BRCA2 gene undergoes prophylactic mastectomy.
Literature	GENERIF_SUMMARY	Women who had developed breast cancer under the age of 40 and who were identified as BRCA1 or BRCA2 mutation carriers experienced devastation, loneliness and isolation.
Literature	GENERIF_SUMMARY	These results provide insight into the recently discovered diversity of BRCA2 domain structures in different organisms.
Literature	GENERIF_SUMMARY	Analysis resulted in the identification of 25 and 52 variants in the BRCA1 and BRCA2 genes, respectively in breast or ovarian cancer.
Literature	GENERIF_SUMMARY	two recurrent BRCA2 mutations predisposing to male breast cancer may facilitate the analyses aimed at the identification of mutation carriers in our geographic area
Literature	GENERIF_SUMMARY	BRCA2 mutation is common (9%) among unselected young breast cancer patients undergoing BCT.
Literature	GENERIF_SUMMARY	No BRCA1/2 genomic rearrangements found in high-risk French-Canadian breast/ovarian cancer families.
Literature	GENERIF_SUMMARY	Missense mutation in exon 13 in BRCA2, c.7235G>A, results in skipping of exon 13.
Literature	GENERIF_SUMMARY	PALB2 colocalizes with BRCA2 in nuclear foci, promotes its localization and stability in key nuclear structures (e.g., chromatin and nuclear matrix), and enables its recombinational repair and checkpoint functions
Literature	GENERIF_SUMMARY	The small group of patients with biallelic mutations in BRCA2 is distinctive in the severity of the phenotype, and early onset and high rates of leukaemia and specific solid tumours, and may comprise an extreme variant of Fanconi anaemia.
Literature	GENERIF_SUMMARY	Study identified a specific spectrum of germline BRCA1/BRCA2 mutations in Portuguese families with inherited predisposition to breast/ovarian cancer and

Main Category	Category	Terms
		found evidence for genetic anticipation regarding age of diagnosis in succeeding generations.
Literature	GENERIF_SUMMARY	The objective in this study was to determine the frequency of large genomic rearrangements in BRCA1 and BRCA2 in a large group of Danish families with increased risk of breast and ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 participates in repair of replication-mediated double-strand breaks generated when replication forks encounter interstrand cross-link
Literature	GENERIF_SUMMARY	Identification of 13 novel variants including two deleterious truncating mutations and two potentially pathogenic missense mutations on the BRCA1 and BRCA2 genes
Literature	GENERIF_SUMMARY	Results suggest that breast tumours with mutations in BRCA2 or TP53 could be promising candidates for Aurora-A targeted treatment.
Literature	GENERIF_SUMMARY	BRCA2 mutation may have a role in developing hematologic malignancy
Literature	GENERIF_SUMMARY	Testing strategy with an initial test using a panel of reported recurrent mutations, followed by full sequencing predicts prevalence of breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Prevalence of BRCA2 mutations in breast cancer cases among racial and age groups and show key predictors of carrier status for both White and Black women and women.
Literature	GENERIF_SUMMARY	Human BRCA2 has multiple BRC repeats, motifs of approximately 30 residues, that associate in a ternary complex with RAD51 on single-stranded DNA and double-stranded DNA.
Literature	GENERIF_SUMMARY	Ninteen percent of the women who developed both invasive breast and ovarian tumors carried one of the analyzed BRCA1 gene mutations but none of the women were positive for the analyzed BRCA2 mutation.
Literature	GENERIF_SUMMARY	analysis of BRCA1 and BRCA2 mutations from Korean breast cancer patients using denaturing HPLC
Literature	GENERIF_SUMMARY	Novel and distinct BRCA2 deletions were detected in three families and their boundaries were determined.
Literature	GENERIF_SUMMARY	Effective strategies have been developed to reduce the risk for the development of breast and ovarian cancer in women with BRCA1/2 mutations, making genetic testing for these mutations an important part of the management. [REview]
Literature	GENERIF_SUMMARY	Breast cancers arising in BRCA1 and BRCA2 mutation carriers appear to have specific pathological and gene expression profiles, which show a high level of concordance.BRCA2 overexpress some DNA repair. [REVIEW]

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Literature	GENERIF_SUMMARY	Findings show that BRCA2 mutations account for a substantial proportion of hereditary breast/ovarian cancer and early-onset breast and ovarian cancer cases in Pakistan.
Literature	GENERIF_SUMMARY	Tumor cells having disruptions in BRCA1/2 network genes and TP53 together are more sensitive to cisplatin than cells with either disruption alone.
Literature	GENERIF_SUMMARY	BRCA2 mutations appear to account for a lower proportion of breast cancer patients at increased risk of harboring such mutations in Northern India
Literature	GENERIF_SUMMARY	The 5164del4 BRCA2 mutation is not likely to be a founder mutation in non-Ashkenazi Jewish families with high risk of breast and ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutation status have roles in inter-cell-line phenotypic variability after irradiation of lymphoblastoid cell lines
Literature	GENERIF_SUMMARY	analysis of BRCA1 and BRCA2 mutations in breast cancer patients from Brazil
Literature	GENERIF_SUMMARY	findings show that BRCA mutations account for a substantial proportion of hereditary breast/ovarian cancer in Colombia
Literature	GENERIF_SUMMARY	These results identify a novel role for FoxM1 in the transcriptional response during DNA damage/ checkpoint signaling and show a novel mechanism by which Chk2 protein regulates expression of DNA repair enzymes.
Literature	GENERIF_SUMMARY	Large genomic BRCA2 rearrangements were not found among our 36 Finnish male breast cancer patients
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations may be more frequent in general populations than previously thought and may be associated with various types of cancers.
Literature	GENERIF_SUMMARY	These data suggest that BRCA2 mutation carriers with ovarian cancer may have better survival than BRCA1 carriers and non-carriers.
Literature	GENERIF_SUMMARY	A novel BRCA2 frame-shift mutation, 3951del3insAT, which produces a protein truncated at codon 1258, was observed in six patients with BC from the same village in Sardinia.
Literature	GENERIF_SUMMARY	investigation of the contribution of BRCA-1 and BRCA-2 germline mutations to the clinical features and outcome in 66 Italian women with early-onset breast cancer
Literature	GENERIF_SUMMARY	BRCA2 mutations are associated with breast ans ovarian cancer
Literature	GENERIF_SUMMARY	Results implicate BRCA2 in the regulation of the centrosome cycle and provide new insight into the aneuploid nature of many breast cancers.

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Literature	GENERIF_SUMMARY	low mRNA and protein expression in the BRCA1/BRCA2 and XRCC5 genes occur in lung adenocarcinoma and squamous cell carcinoma, respectively, and promoter hypermethylation is the predominant mechanism in deregulation of these genes
Literature	GENERIF_SUMMARY	no evidence was found in this study for an association between BRCA2 mutations and susceptibility to hereditary prostate cancer in men selected from high-risk families
Literature	GENERIF_SUMMARY	substantial levels of aberrant methylation, in the fluid from the breasts of healthy BRCA mutation carriers
Literature	GENERIF_SUMMARY	BRCA2 genomic rearrangements is associated with breast and ovarian cancer
Literature	GENERIF_SUMMARY	Our findings suggest that in Central Sudan BRCA1/2 represent an important etiological factor of breast cancer in males and young women less exposed to pregnancy and lactation.
Literature	GENERIF_SUMMARY	Borderline ovarian tumors are neither part of the BRCA1- nor the BRCA2- related tumor spectrum.
Literature	GENERIF_SUMMARY	unlikely that annual screening will reduce mortality from ovarian cancer in BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	common polymorphisms in the ATM, BRCA1, BRCA2, CHEK2 and TP53 cancer susceptibility genes are not shown to increase breast cancer risk
Literature	GENERIF_SUMMARY	the genomic instability observed in normal cells from BRCA1 and BRCA2 mutation carriers is associated with a down-regulation of nuclear BRCA1 protein accumulation in dot like structures
Literature	GENERIF_SUMMARY	a total of 287 Greek breast/ovarian cancer families, 46 and 13 carry a deleterious mutation in BRCA1 and BRCA2, respectively.
Literature	GENERIF_SUMMARY	presence of BRCA1 and BRCA2 rearrangements among Asian patients with early onset or familial history of breast or ovarian cancer
Literature	GENERIF_SUMMARY	Regular surveillance in women at increased familial risk of breast cancer is associated with a good outcome if they carry BRCA2 mutations
Literature	GENERIF_SUMMARY	among carriers of BRCA1 or BRCA2 mutations, the cumulative lifetime risk of developing breast cancer is 50-60% and the equivalent risk of ovarian cancer is 20-40% in Australian women
Literature	GENERIF_SUMMARY	Novel germline mutations in BRCA2 gene is associated with hereditary breast and breast-ovarian cancer
Literature	GENERIF_SUMMARY	Productive recombination results from the functional balance between the different RAD51-binding modes of the BRC repeat and exon 27 regions of BRCA2.
Literature	GENERIF_SUMMARY	Interactions of the BRCA2 C-terminal region with RAD51

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		may facilitate efficient nucleation of RAD51 multimers on DNA and thereby stimulate recombination-mediated repair.
Literature	GENERIF_SUMMARY	BRCA2 is a universal regulator of RAD51/DMC1 recombinase actions
Literature	GENERIF_SUMMARY	The output 3D molecular structure from the combination between BRCA2-RAD51 is derived
Literature	GENERIF_SUMMARY	BRCA2 c.9079 G>A is not a predisposing variant for early onset breast cancer
Literature	GENERIF_SUMMARY	the Icelandic BRCA2 999del5 founder mutation was strongly associated with rapidly progressing lethal prostate cancer
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers without cancer had increased breaks as well as breaks and gaps per cell
Literature	GENERIF_SUMMARY	review of some of the most well-known and significant examples of founder mutations in BRCA genes found in European and non-European populations [review]
Literature	GENERIF_SUMMARY	XRCC1 down-regulation in HeLa cells leads to a decrease in the DNA ligase 3 protein level, significantly increased sensitivity to alkylating agents, elevated level of sister chromatid exchange and decrease in the survival of BRCA2-deficient cells.
Literature	GENERIF_SUMMARY	High-risk patients with BRCA1-negative tumors should be screened first for BRCA2 gene.
Literature	GENERIF_SUMMARY	Mutations found in a significant proportion of women with ductal carcinoma in situ who presented for hereditary risk assessment.
Literature	GENERIF_SUMMARY	The researchers found 17 pathogenic changes in the BRCA2 genes in 64 families with a pedigree of male breast neoplasms.
Literature	GENERIF_SUMMARY	families appear to exhibit features most consistent with BRCA1 and BRCA2 carrier status
Literature	GENERIF_SUMMARY	PTEN germline mutations are rare
Literature	GENERIF_SUMMARY	BRCA2-8765delAG has an independent origin in geographically and ethnically distinct populations, acting as a founder mutation in North but not in South Sardinia.
Literature	GENERIF_SUMMARY	From a population database of BRCA1 and 2 mutation carriers in Southwestern Ontario, Canada, we identified three women with advanced-stage endometrial cancer.
Literature	GENERIF_SUMMARY	The identified BRCA2 c.7806-9T > G [Genbank
Literature	GENERIF_SUMMARY	BRCA1/2 rearrangements is not advantageous in male breast neoplasm (MBC) cases not belonging to high-risk breast cancer families and that common CHEK2 mutations play an irrelevant role in MBC predisposition in Italy.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Results point to a critical role for BACH1 helicase activity not only in the timely progression through the S phase by association with BRCA1/BRCA2, but also in maintaining genomic stability.
Literature	GENERIF_SUMMARY	Higher BTAK expression was found in ovarian cancer cells compared to ovaries without cancer but with known BRCA1/2 mutation or strong family history.
Literature	GENERIF_SUMMARY	Inactivation of a single gene within the BRCA2 pathway can increase risks for multiple cancers and inactivation of a different gene in the same pathway may have similar effects.
Literature	GENERIF_SUMMARY	ectopically BRCA2-expressing cells have different intracellular levels of Aurora A, Aurora B, p21, E2F-1, and pRb, suggesting a BRCA2-mediated suppression of polyploidy via stabilization of the checkpoint proteins levels
Literature	GENERIF_SUMMARY	Data indicate that BRCA2 are the major susceptibility genes for ovarian cancer but that other susceptibility genes may exist.
Literature	GENERIF_SUMMARY	Results suggest that protein-truncating BRCA2 mutations confer an elevated relative risk of early-onset prostate cancer
Literature	GENERIF_SUMMARY	A suspicious deleterious BRCA2 variant is identified in 15 of 197 cases of esophageal squamous cell carcinoma in the Iranian Turkmen population.
Literature	GENERIF_SUMMARY	The greatest proportion of serous cancer risk in BRCA mutation-positive women should be assigned to the fimbria rather than the ovary.
Literature	GENERIF_SUMMARY	the BRCA2 HH homozygous genotype might be positively associated with an increased risk of male breast cancer in men younger than 60 years.
Literature	GENERIF_SUMMARY	analysis of BRCA1 and BRCA2 mutations in Eastern Finnish breast/ovarian cancer families
Literature	GENERIF_SUMMARY	The prevalence of BRCA1 and BRCA2 germline mutations in high-risk breast cancer patients of Chinese Han nationality.
Literature	GENERIF_SUMMARY	Study found that Fanconi anemia pathway activation is triggered mainly by the HPV type 16 (HPV-16) E7 oncoprotein and is associated with an enhanced formation of large FANCD2 foci and recruitment of FANCD2 as well as FANCD1/BRCA2 to chromatin.
Literature	GENERIF_SUMMARY	analysis of a subset of unclassified sequence variants in BRCA2 that may adversely affect splicing and thereby contribute to BRCA2 disruption
Literature	GENERIF_SUMMARY	Self-image and self-disclosure concerning prophylactic mastectomy (PM) for women with a BRCA1/2 mutation.
Literature	GENERIF_SUMMARY	prevalence of BRCA1 & BRCA2 mutations in breast cancer patients with affected relatives in Tunisia; two

Main Category	Category	Terms
		frameshift mutations (c.1309del4 and c.5682insA) were observed in BRCA2
Literature	GENERIF_SUMMARY	The clinical significance of 1,433 sequence variants of unknown significance (VUSs) in the BRCA genes, was assessed.
Literature	GENERIF_SUMMARY	Tumors are unlikely to arise directly from BRCA2 heterozygous cells without other genetic events such as loss of the wild-type BRCA2 allele and/or loss of p53 function or other cell cycle inhibitors.
Literature	GENERIF_SUMMARY	5' UTR polymorphism in BRCA2 is associated with breast cancer, which is further influenced by the germline genetic backgrounds of codon 72 polymorphism of p53
Literature	GENERIF_SUMMARY	The site of first distant metastasis is different between BRCA1- and BRCA2-associated and sporadic breast cancer patients.
Literature	GENERIF_SUMMARY	High frequency of BRCA1/2 and p53 somatic inactivation in sporadic ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA1 or BRCA2 mutations have roles in breast cancer in smokers
Literature	GENERIF_SUMMARY	We found six differentially expressed proteins; among them, the checkpoint mediator protein MDC1 whose expression was disrupted in FANCC-/- cells.
Literature	GENERIF_SUMMARY	RAD51 is the first gene to be reliably identified as a modifier of risk among BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	MDM2 SNP309 accelerates breast and ovarian carcinogenesis in BRCA1 and BRCA2 carriers of Jewish-Ashkenazi descent.
Literature	GENERIF_SUMMARY	Germline mutations in the BRCA1 or BRCA2 tumour-suppressor genes are strong predictors of breast and/or ovarian cancer development.
Literature	GENERIF_SUMMARY	study describes a family with multiple cases of MEN1-associated cancers as well as pancreatic adenocarcinoma, ovarian cancer, and male breast cancer, in which we identified germline mutations in both MEN1 and BRCA2
Literature	GENERIF_SUMMARY	evaluation of the risks of developing breast carcinoma for male BRCA1 and BRCA2 mutation carriers; both the relative and cumulative risks were higher for BRCA2 mutation carriers than for BRCA1 mutation carriers
Literature	GENERIF_SUMMARY	there is no evidence of sex ratio skewing in offspring of female BRCA mutation carriers
Literature	GENERIF_SUMMARY	BRCA1/2 mutations predispose to early onset breast and ovarian cancers, modified by factors such as IGF-I.
Literature	GENERIF_SUMMARY	depletion of normal BRCA2 proteins in the cytoplasm leads to centrosome amplification and binucleated cells. Our results suggest that disruption of NES

Main Category	Category	Terms
		function by genetic changes results in deregulation of BRCA2 export, which leads to centrosome disorder
Literature	GENERIF_SUMMARY	BRCA2 mutations are associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA1/2 mutations are significantly more common in Italian women who developed breast cancer
Literature	GENERIF_SUMMARY	analysis of breast cancer genes that may modify risk in BRCA1/2 mutation carriers
Literature	GENERIF_SUMMARY	Loss of heterozygosity of the BRCA2 gene region was found to be common in the cutaneous squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Sixty-four Polish families with a history of breast and/or ovarian cancer were screened for mutations in the BRCA1/2 genes
Literature	GENERIF_SUMMARY	models used to analyze age-incidence curve of breast cancer in women carrying gerrmline BRCA1 or BRCA2 mutations; results suggest in carriers there are 2 events which may occur at rates similar to mutation rates for normal cells leading to breast cancer
Literature	GENERIF_SUMMARY	Founder BRCA2 mutation is associated with male breast cancer
Literature	GENERIF_SUMMARY	study confirms that, among Ashkenazi ovarian cancer patients, BRCA1/2 mutations are associated with improved long-term survival
Literature	GENERIF_SUMMARY	BRCA2 protein is lost in carcinoma cells compared to normal and hyperplastic prostate epithelium
Literature	GENERIF_SUMMARY	The CHEK2 1100delC mutation is not present in Korean patients with breast cancer cases tested for BRCA1 and BRCA2 mutation
Literature	GENERIF_SUMMARY	heterogeneous ethnicity increases the variety of BRCA1 and BRCA2 mutations that can be found in Spanish populations
Literature	GENERIF_SUMMARY	In summary, BRCA2 clearly accounts for a proportion of LFS/LFL (Li-Fraumeni syndrome, LFS-like) families negative for TP53. mutations.
Literature	GENERIF_SUMMARY	Since BRCA1/2 mutation carrier status is associated with more aggressive disease, it is a prognostic factor for PRCA outcome.
Literature	GENERIF_SUMMARY	Mutation of BRCA2 gene is an indication of susceptibility to breast and ovarian neoplasms.
Literature	GENERIF_SUMMARY	Allelic imbalance affecting BRCA1 and to a lesser extent BRCA2 may contribute to both familial and non-familial forms of breast cancer.
Literature	GENERIF_SUMMARY	FANCG promotes formation of a newly identified protein complex containing BRCA2, FANCD2 and XRCC3.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The 4088insA mutation appears to be associated with a favourable clinical course of breast and ovarian cancer
Literature	GENERIF_SUMMARY	incidence of mutations in the BRCA1 and BRCA2 genes in the studied sampling of 74 patients with ovarian cancer was 19%; majority of mutations (86%) were detected in BRCA1 gene, where 5382insC mutation predominated (58%)
Literature	GENERIF_SUMMARY	review of BRCA1/2 associated hereditary breast cancer [review]
Literature	GENERIF_SUMMARY	we review the DNA-damage response network consisting of FA and BRCA proteins and what is known about their involvement in breast cancer susceptibility
Literature	GENERIF_SUMMARY	This articles reviews the evidence for the association of BRCA2 polymorphisms and the development of breast neoplasms.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer
Literature	GENERIF_SUMMARY	Association between BRCA2 mutations and the presence of breast cancer in a Cuban population.
Literature	GENERIF_SUMMARY	The BRCA2 N372H and the IVS21-66T>C polymorphisms increase breast cancer risk.
Literature	GENERIF_SUMMARY	A novel BRCA2-interacting protein, BJ-HCC-20A, which is reported to be a potential cancer-testis antigen, was idenified.
Literature	GENERIF_SUMMARY	56 mutations were identified in BRCA2 in families containing at least one reported ovarian cancer diagnosed less than 50 years or at any age with family history of breast or ovarian cancer for mutations in BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	Chk1 and Chk2 regulate the functional associations between hBRCA2 and Rad51 in response to DNA damage
Literature	GENERIF_SUMMARY	Cyclin D1 expression analysis in familial breast cancers may discriminate BRCAX from BRCA2-linked cases.
Literature	GENERIF_SUMMARY	Tubal p53 signature merits serious consideration as an important early event in serous carcinogenesis in BRCA+ women.
Literature	GENERIF_SUMMARY	The differences in the effects of the FGFR2 and MAP3K1 SNPs between BRCA1 and BRCA2 carriers point to differences in the biology of BRCA1 and BRCA2 breast cancer tumors and confirm the distinct nature of breast cancer in BRCA1 mutation carriers.
Literature	GENERIF_SUMMARY	contribution of Alu insertion in BRCA2 exon 3(c.156_157insAlu) to inherited predisposition to breast/ovarian cancer in families mostly from northern/central Portugal; findings show it accounts for more that one-fourth of deleterious BRCA1/BRCA2 mutations
Literature	GENERIF_SUMMARY	We studied 10 BRCA1 and 12 BRCA2 variants identified

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		in Australian families with breast cancer.
Literature	GENERIF_SUMMARY	Down-regulation of BRCA2 leads to radio-sensitization mainly through the inhibition of homologous recombination repair type double-strand break repair.
Literature	GENERIF_SUMMARY	The most current evidence suggests that it may be more beneficial in those with BRCA2 mutations because tumors associated with these mutations are likely to be estrogen-receptor positive.
Literature	GENERIF_SUMMARY	BRCA2 may play an important role in the familial breast cancer in eastern Shandong Chinese population, but not in the early-onset breast cancer.
Literature	GENERIF_SUMMARY	Survival from diagnosis of ovarian cancer was calculated using Kaplan-Meier and compared for proven BRCA1/2 carriers with non-carriers; 5 & 10 year survival in BRCA1/2 mutation carriers was 58.6% and 36%, which was sig. worse than for non-BRCA carriers.
Literature	GENERIF_SUMMARY	Dinucleotide CA repeat polymorphism at RAD51 and BRCA2 gene regions might be associated with genetic susceptibility to breast cancer.
Literature	GENERIF_SUMMARY	A total of 14 BRCA1 and 17 BRCA2 sequence alterations, of which eight are novel, are reported.
Literature	GENERIF_SUMMARY	BRCA2 mutation may not contribute to increases in the risk for both sporadic and familial pancreatic cancer in Korea.
Literature	GENERIF_SUMMARY	caspase 3 cleavage of Rad51 resulted in a functional decrease in Rad51 strand exchange activity and that inhibition of caspase 3 activity increased Rad51 protein levels and Rad51 foci
Literature	GENERIF_SUMMARY	BRCA2 gene mutation was identified in 4 male breast cancer (MBC) patients; 3 carried the Slovenian founder mutation IVS16-2A>G; all 4 mutations were confined to patients with a family history of breast cancer
Literature	GENERIF_SUMMARY	In Ashkenazi Jews, mutations in BRCA1/2 may constitute a major cause for pancreatic cancer.
Literature	GENERIF_SUMMARY	These observations are consistent with the idea that BRCA2, but not BRCA1, is a tumor suppressor of prostate cancer.
Literature	GENERIF_SUMMARY	An excellent method for identifying inactivating missense mutations in the BRCA2 DBD predicting likelihood of cancer in carriers.
Literature	GENERIF_SUMMARY	The mutation distributions are comparable with those from Scandinavian and European studies and indicate that the Danish BRCA1 and BRCA2 mutations are a mixture of Scandinavian mutations and European mutations including two of the Ashkenazi mutations.
Literature	GENERIF_SUMMARY	deleterious genetic variants in the BRCA2 gene in the Czech population

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Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutation carriers had similar expression profiles, with some subclustering of missense mutation carriers.
Literature	GENERIF_SUMMARY	No large genomic rearrangements were identified in BRCA2 in Finnish breast and/or ovarian cancer families
Literature	GENERIF_SUMMARY	A case-control study is reported on infertility, treatment of infertility, and the risk of breast neoplasms among women with BRCA2 mutations.
Literature	GENERIF_SUMMARY	TGFB1 L10P genotype does not modify the risk of breast cancer in BRCA1 or BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Single nucleotide polymorphism in BRCA2 is associated with breast cancer.
Literature	GENERIF_SUMMARY	There is a highly significant reduction in life expectancy in BRCA1 compared to BRCA2 carriers in ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutation prevalence and clinical characteristics of a population-based series of ovarian cancer cases from Denmark.
Literature	GENERIF_SUMMARY	BRCA2-associated cancers were characterized by the higher relative expression of FGF1 and FGFR2
Literature	GENERIF_SUMMARY	our findings suggest no difference in the occurrence of miscarriage between BRCA2 carriers and non-carriers when parental consanguinity was taken into consideration
Literature	GENERIF_SUMMARY	The median survival from diagnosis of prostate cancer was 4.0 years for men with a BRCA2 mutation vs 8.0 years for men with a BRCA1 mutation, and the difference was highly significant.
Literature	GENERIF_SUMMARY	A syngeneic variance library for functional annotation of human variation in BRCA2.
Literature	GENERIF_SUMMARY	the novel BRCA2 splice variant is a de novo mutation introduced in the male spermatozoa that can be classified as a disease causing mutation
Literature	GENERIF_SUMMARY	BRCA2 deficiency is partially rescued by Rad51 overexpression
Literature	GENERIF_SUMMARY	analysis of BRCA1 and BRCA2 mutations in Malaysia
Literature	GENERIF_SUMMARY	alteration in RAD51 region may contribute to the disturbances of DNA repair involving RAD51 and/or BRCA2 penetration and thus enhance the risk of breast cancer development.
Literature	GENERIF_SUMMARY	we determined the methylation statuso f the promoter in putative modifier genes
Literature	GENERIF_SUMMARY	a limited role for the three Ashkenazi BRCA1/2 founder mutations in cutaneous malignant melanoma risk among the Ashkenazi Jewish population

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Literature	GENERIF_SUMMARY	characterized BRCA1 and BRCA2 gene polymorphic variants in familial breast cancer
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 genomic rearrangement testing be considered in all non-Ashkenazi Jewish women with an estimated mutation prevalence >or=10%
Literature	GENERIF_SUMMARY	analysis of breast cancer risk variation in BRCA1 and BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Our aim was to ascertain the pathological effect of the BRCA1 IVS6-1G>A (c. 302-1G>A) and the BRCA2 IVS15+1G>A (c. 7617+1G>A) variants detected in Spanish breast/ovarian cancer families
Literature	GENERIF_SUMMARY	The study demonstrates for the first time that microsatellite-stable FHIT-negative sebaceous gland carcinomas accumulate mutations that target central components of the HRR network.
Literature	GENERIF_SUMMARY	BRCA1/2 in high-risk African-American women with breast cancer
Literature	GENERIF_SUMMARY	Different from ERbeta, p53 interacts with HDAC1 and CtBP1 and forms an inhibiting transcriptional complex that could compete for binding to Sp1 sites with ERalpha transcriptional complex and inhibit BRCA2 transcription more significantly
Literature	GENERIF_SUMMARY	Methylation status of CpG islands at sites -59 to +96 in exon 1 varies in mammary tissue among women with sporadic breast cancer
Literature	GENERIF_SUMMARY	A high mutation detection rate and the frequent occurrence of a limited array of recurring mutations allow a simple and fast initial test for BRCA1/2 mutation screening in families with Slovenian ancestry
Literature	GENERIF_SUMMARY	a link between CHEK2 and BRCA2 pathways
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with male breast cancer.
Literature	GENERIF_SUMMARY	possible relationship between defects in the FA/BRCA pathway of genomic stability and potential pathogenesis of T and B cell lymphoma.
Literature	GENERIF_SUMMARY	In a population well beyond the average age of breast/ovarian cancer onset, 21 different sequence variants in the BRCA1 gene (one novel) and 36 variants in the BRCA2 gene (7 novel) were detected.
Literature	GENERIF_SUMMARY	no association between duration of epithelial ovarian cancer symptoms and BRCA mutation status
Literature	GENERIF_SUMMARY	C-terminally truncated BRCA2 hinders RAD51 nuclear translocation, possibly contributing to genetic instabilities in homozygous as well as heterozygous individuals.
Literature	GENERIF_SUMMARY	germ-line BRCA1 or BRCA2 mutations may have a role in response to primary platinum-based chemotherapy

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Literature	GENERIF_SUMMARY	Parp-1 down-regulates BRCA2 expression through an interaction with a repression region of the BRCA2 promoter
Literature	GENERIF_SUMMARY	Carriers of mutations in the genes BRCA1/2 may present a specific high risk group for PABC especially at younger ages.
Literature	GENERIF_SUMMARY	no defect in FANCD2 ubiquitination, BRCA2 and FANCJ expression; absence of FANCN protein in three cell lines
Literature	GENERIF_SUMMARY	analysis of novel mutations identified in Slovak HBOC families, c.80 + 3del4 (IVS2 + 3delAGTC) in BRCA1 gene and mutation c.6589delA (6817delA) in BRCA2 gene [case report]
Literature	GENERIF_SUMMARY	Studied BRCA1 and BRCA2 exp'n in young breast cancer patients;BRCA1 & BRCA2 expression correlated in healthy, but not in tumor tissues. Neither BRCA1 nor BRCA2 exp'n was assoc'd with tumor histology, diff'n, nodal metastasis or p53 and HER-2 exp'n.
Literature	GENERIF_SUMMARY	Cross-sectional analysis of germ-line BRCA2 mutations in Japanese patients suspected to have breast/ovarian cancer is reported.
Literature	GENERIF_SUMMARY	BRCA2 T1915M polymorphism alone might be associated with a reduced risk of breast cancer, but among CHEK2 mutation carriers, it may lead to an unexpectedly high risk.
Literature	GENERIF_SUMMARY	analysis of the effectiveness of screening in diagnosing early stage ovarian cancer in BRCA1 and BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Consistent with previous reports, BRCA2 mutations are associated with an increased risk of PAC.
Literature	GENERIF_SUMMARY	BRCA2 gene mutation is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 IVS4-12del5 is a deleterious mutation.
Literature	GENERIF_SUMMARY	The BRCA2 gene, found on the long arm of chromosome 13, encodes for an even larger protein of 3418 amino acids.BRCA 1 and 2 mutations account for 5% to 10% of breast cancer cases.
Literature	GENERIF_SUMMARY	The oral contraceptive pill (OCP) reduces the risk of both endometrial and ovarian cancer in the general population and BRCA carriers
Literature	GENERIF_SUMMARY	Three hundred and forty-five probands were examined for specific mutations of BRCA1/2 genes. The estimated penetrance for the age groups among BRCA1/2 carriers was 31.9% (or =50 years
Literature	GENERIF_SUMMARY	BARD1 has BRCA1-dependent and BRCA1-independent functions in mitosis. BARD1, but not BRCA1, localizes to the midbody at telophase and cytokinesis, where it colocalizes with Aurora B.

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Literature	GENERIF_SUMMARY	Two mutations of BRCA2 gene at exon and splicing site in a woman who underwent oncogenetic counseling. (Case report)
Literature	GENERIF_SUMMARY	It is possible that GATA3 mutations occur earlier in the evolution of breast tumors, compared to BRCA1, BRCA2 or sporadic tumors, and are therefore easier to detect by direct sequencing in the presence of some stromal contamination.
Literature	GENERIF_SUMMARY	Genetic variants in the BRCA2 gene is associated with breast cancer.
Literature	GENERIF_SUMMARY	There was no difference in mean parity between carriers (1.9) and noncarriers (1.9)of BRCA mutations.
Literature	GENERIF_SUMMARY	the CASP8 D302H polymorphism diminishes the high risk of breast cancer conferred by BRCA1 and BRCA2 mutations
Literature	GENERIF_SUMMARY	MDM2SNP309G/G main effect on BRCA1/2 positive mutation carriers is linked to its effect on patients survival.
Literature	GENERIF_SUMMARY	two codon-usage-changing variants were detected in BRCA2 in familial breast cancer but showed no association with breast cancer risk
Literature	GENERIF_SUMMARY	In Sardinian breast cancer population, effects of BRCA2 germline mutations on survival varied within the first 2 years from diagnosis. After longer follow-up, breast cancer-specific rates of death were similar for BRCA2 mutation carriers and non-carriers.
Literature	GENERIF_SUMMARY	The genotype HH of the N372H locus in BRCA2 is associated with breast cancers.
Literature	GENERIF_SUMMARY	the BRCA2 silent mutation Lys172Lys is a disease-causing mutation.
Literature	GENERIF_SUMMARY	BRCA1 is an upstream regulator of BRCA2 in the DNA-damage response, and PALB2 is the linker between BRCA1 and BRCA2 in breast and ovarian cancer.
Literature	GENERIF_SUMMARY	We report a novel germline 5369delATTT mutation in BRCA2 gene, detected in a 45-year-old woman with bilateral breast cancer. This deletion was also detected in her father with prostatic cancer and her sister with breast cancer.
Literature	GENERIF_SUMMARY	Study shows that the solitary BRC4 amino motif of BRCA2 promotes assembly of RAD51 onto single-stranded DNA, but not dsDNA, to stimulate DNA strand exchange.
Literature	GENERIF_SUMMARY	promoter methylation is a not a frequent ""second-hit"" in tumors from BRCA1 or BRCA2 carriers."
Literature	GENERIF_SUMMARY	We identified seven (7/22, 31.8%) new deleterious mutations and also confirmed and reported a BRCA2 founder mutation in our Hong Kong Chinese cohort

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Literature	GENERIF_SUMMARY	impaired homologous recombination repair is one of the fundamental causes for genomic instability and tumorigenesis observed in patients carrying BRCA1, BRCA2, or PALB2 mutations
Literature	GENERIF_SUMMARY	parity appears to be associated with protection from breast cancer in women with mutations in BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	It was shown that the presence of the mutations in the BRCA1/2 genes among patients with bilateral breast cancer is associated with an earlier occurrence of the first and the second breast cancer than in patients without hereditary mutations.
Literature	GENERIF_SUMMARY	The prevalence of germline mutations in the BRCA1 and BRCA2 genes found was lower than reported on high-risk Brazilian populations.
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	review of available data on ovarian cancers in the context of other investigations of BRCA-related transcriptional alterations , and highlight areas for future research.
Literature	GENERIF_SUMMARY	Review states that referral criteria for BRCA1 genetic testing may differ between European countries based on BRCA2 mutation prevalence.
Literature	GENERIF_SUMMARY	germ line BRCA2 gene mutation may play a role in familial aggregation of ESCC in high-risk region of India.
Literature	GENERIF_SUMMARY	Primary human fibroblasts derived from heterozygous BRCA2 mutation carriers show significantly prolonged cytokinesis.
Literature	GENERIF_SUMMARY	Prevalence of BRCA2 mutation and risk of secondary malignancies across diverse racial groups in young women with breast cancer are reported.
Literature	GENERIF_SUMMARY	The proportion of Korean ovarian cancer patients with a strong family history was significant, and the prevalence of BRCA1 and BRCA2 mutations in such patients was high.
Literature	GENERIF_SUMMARY	two siblings diagnosed with central nervous system embryonal tumors at an early age in association with biallelic BRCA2 inactivation, including the first reported case of a spinal cord primitive neuroectodermal tumor (PNET) in a BRCA2/FANCD1 kindred
Literature	GENERIF_SUMMARY	Results suggest that Rad51 binding by the C-terminal Brca2 motif is dispensable for the execution of homologous DNA recombination but instead links the disassembly of Rad51 complexes to mitotic entry.
Literature	GENERIF_SUMMARY	BRCA mutation carriers, as well as women with a

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		significant family history of breast and ovarian cancer are more vulnerable to exogenous hormones in oral contraceptives
Literature	GENERIF_SUMMARY	Role of MCPH1 in the DNA damage response is in part associated with the ability to localize BRCA2 to sites of DNA double-stand breaks.
Literature	GENERIF_SUMMARY	CBP-mediated post-translational N-glycosylation of BRCA2.
Literature	GENERIF_SUMMARY	Show here that PALB2 physically and functionally connects BRCA1 and BRCA2 into a DNA damage response network that also includes the RAD51 recombinase.
Literature	GENERIF_SUMMARY	Data show that variants in genes that interact biologically with BRCA1 and/or BRCA2 may be associated with modified ovarian cancer risk in women who carry BRCA1/2 mutations.
Literature	GENERIF_SUMMARY	structure explains the effects of both cancer-associated truncating mutants in PALB2 and missense mutations in the amino-terminal region of BRCA2.
Literature	GENERIF_SUMMARY	surveillance and prevention strategies may have different outcomes in BRCA1 and BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	A synonymous variant (Ser51) in TOX3 (previously TNRC9) was associated with an increased risk of breast cancer (OR=1.82, p
Literature	GENERIF_SUMMARY	analysis of BRCA1 and BRCA2 germline mutations in breast cancer susceptibility within Sardinian population
Literature	GENERIF_SUMMARY	immunohistochemistry-based test (which takes only 4 hours) appears to identify BRCA2 hereditary cancer with high accuracy.
Literature	GENERIF_SUMMARY	BRCA2 targets RAD51 to ssDNA while inhibiting dsDNA binding and that these contrasting activities together bolster one another to stimulate homologous DNA recombination.
Literature	GENERIF_SUMMARY	we identified a novel germ line mutation in the BRCA2 gene (c.51dupA) in a patient with early onset bilateral breast cancer and no family history of the disease
Literature	GENERIF_SUMMARY	restoration of BRCA2 due to secondary BRCA2 mutation is involved in acquired drug resistance of BRCA2-mutated ovarian carcinoma.
Literature	GENERIF_SUMMARY	Alterations in BRCA2 gene sequence, copy number, or expression are extremely common in clear cell carcinoma of the ovary and may contribute to a paradoxical better clinical outcome.
Literature	GENERIF_SUMMARY	This study suggests the possibility of the first true founder mutation of BRCA1/BRCA2 identified in the Korean population.

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Literature	GENERIF_SUMMARY	SNP rs13281615 at 8q24 was not associated with breast cancer for either BRCA1 or BRCA2 mutation.
Literature	GENERIF_SUMMARY	Heterozygosity and homozygosity of any of the examined nine BRCA1 and BRCA2 missense polymorphisms cannot explain the increased risk of breast and/or ovarian cancer observed in families with hereditary breast and/or ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA1/2 genomic rearrangement is likely to make only a small contribution to breast cancer in the Korean population.
Literature	GENERIF_SUMMARY	PCNA may function as a molecular platform to facilitate the mono-ubiquitination of FANCD2 and activation of the FA-BRCA pathway
Literature	GENERIF_SUMMARY	There was no evidence that TP53 Arg72Pro or MDM2 309T>G, either singly or in combination, influence breast cancer risk in BRCA1 or BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	BRCA2-plectin interaction plays an important role in the regulation of centrosome localization and also that displacement of the centrosome may result in genomic instability and cancer development.
Literature	GENERIF_SUMMARY	a novel pathogenic mutation in BRCA2; (c.2999delCT,frameshift mutation consists of a deletion of two bases in exon 11)
Literature	GENERIF_SUMMARY	among the 13 known Fanconi Anemia genes, only FANCD1/BRCA2 plays a major role in high-risk breast cancer predisposition
Literature	GENERIF_SUMMARY	Nomenclature for BRCA1/2 DNA test.
Literature	GENERIF_SUMMARY	Recruitment of fanconi anemia and breast cancer proteins to DNA damage sites is differentially governed by replication.
Literature	GENERIF_SUMMARY	there was no significant difference in cyclin D1 expression between BRCA2 and BRCAX cancers
Literature	GENERIF_SUMMARY	Breast cancer in a BRCA2 mutation carrier with a history of prostate cancer is reported.
Literature	GENERIF_SUMMARY	Cancers with a BRCA2 germ line mutation are less frequent and resemble sporadic cases with a comparably higher proliferative activity and a lack of Her2 overexpression.
Literature	GENERIF_SUMMARY	analysis of BRCA1/2 mutation-carrying and non-mutation-carrying high-grade serous carcinomas of ovary
Literature	GENERIF_SUMMARY	Study identified linkage disequilibrium between KL-VS and BRCA2 6174delT mutation on 13q12.
Literature	GENERIF_SUMMARY	BRCA2 germ-line mutation analysis among Indian women from south India identified novel mutations and a high-frequency occurrence of a base-pairing gene

Main Category	Category	Terms
		deletion.
Literature	GENERIF_SUMMARY	AR allelotype length did not correlate with survival in this statistically representative cohort of patients with BRCA1/2 mutations. Associations between short AR and outcome in BRCA2-associated ovarian cancers remain to be determined.
Literature	GENERIF_SUMMARY	BRCA2 mutation testing is associated with ovarian cancer prevention.
Literature	GENERIF_SUMMARY	Study investigated the role of genetic variation in IGF signaling and breast cancer risk in women carrying deleterious mutations in BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	BRCA2 germline mutation in a juvenile breast carcinoma diagnosed at 21 years of age, 4 years after an early full-term pregnancy, in absence of cancer family history
Literature	GENERIF_SUMMARY	Examine contralateral breast cancer risk in BRCA1 and BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	a predisposing mutation in BRCA2,is present in approximately 6% of French-Canadian women with early-onset breast cancer
Literature	GENERIF_SUMMARY	151 consecutive primary ovarian tumors (including 21 with BRCA1/2 mutations and 130 without the mutations) were screened for loss of heterozygosity at loci on chromosomes 17 and 13q.
Literature	GENERIF_SUMMARY	The ITGB3 Leu33Pro polymorphism does not modify breast or ovarian cancer risk in BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	5382insC and 185delAG mutations in BRCA1 and 6174delT in BRCA2 have much less frequency in Iranian breast cancer patients.
Literature	GENERIF_SUMMARY	In this prospective study of women who were unaffected at the time of genetic testing and who were negative for the known familial mutation in BRCA1/2, no excess risk of invasive breast cancer was observed.
Literature	GENERIF_SUMMARY	Multistep level sections to detect occult fallopian tube carcinoma in risk-reducing salpingo-oophorectomies from women with BRCA mutations
Literature	GENERIF_SUMMARY	Data show that although SNPs in BRCA2 and ERCC2 showed stronger associations with high Gleason score, results were not statistically significantly different between clinically aggressive and non-aggressive tumors.
Literature	GENERIF_SUMMARY	the role of a SNP in intron 1 of the ERCC4 gene (rs744154), previously reported to be associated with a reduced risk of breast cancer in the general population, as a breast cancer risk modifier in BRCA1 and BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	Report BRCA2 Spanish proven founder mutations which

Main Category	Category	Terms
		predispose to breast cancer in young women.
Literature	GENERIF_SUMMARY	variant genotypes of the Thr1915Met and Met784Val polymorphisms of the BRCA2 gene may be indicative factors in therapy of ductal breast cancer
Literature	GENERIF_SUMMARY	Multifactorial likelihood analysis indicates that the BRCA1 c.135-1G>T and BRCA2 c.7977-1G>C variants are disease-associated mutations which should be managed clinically in the same fashion as classical truncating mutations.
Literature	GENERIF_SUMMARY	the population of Eastern Spain showed a great heterogeneity in the BRCA1 and BRCA2 mutations
Literature	GENERIF_SUMMARY	The aim of this study is to outline the uptake of risk-reducing surgery in the Danish population of BRCA mutation positive women and to search for factors affecting this decision.
Literature	GENERIF_SUMMARY	analysis of cancer predisposing missense and protein truncating BARD1 mutations in non-BRCA1 or BRCA2 breast cancer families
Literature	GENERIF_SUMMARY	analysis of BRCA1 and BRCA2 deleterious mutations and variants of unknown clinical significance in unilateral and bilateral breast cancer
Literature	GENERIF_SUMMARY	None of the associations between reproductive factors and contralateral breast cancer risk differed between BRCA2 mutation carriers and non-carriers.
Literature	GENERIF_SUMMARY	The prophylactic procedure is highly effective in preventing invasive breast cancer in BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Chemotherapy and tamoxifen reduced the risk of contralateral breast cancer in BRCA1/BRCA2 carriers and non-carriers to a similar relative degree.
Literature	GENERIF_SUMMARY	Meta-analysis suggests that the BRCA2 372H allele may be a low-penetrant risk factor for developing breast cancer.
Literature	GENERIF_SUMMARY	No large genomic rearrangements were detected in the BRCA2 gene in Czech high-risk breast/ovarian cancer patients.
Literature	GENERIF_SUMMARY	These results suggest that, compared to BRCA1 and BRCA2 mutation carriers, non-BRCA1/2 (BRCAX) individuals are genetically heterogeneous.
Literature	GENERIF_SUMMARY	Occult ovarian cancers may be found at the time of risk-reducing salpingo-oophorectomy in BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	the prevalence of BRCA2 mutations in the German familial pancreatic cancer population is less frequent than previously reported
Literature	GENERIF_SUMMARY	BRCA2 gene promoter has bi-directional activity, expressing BRCA2 and a novel C4-type zinc finger

Main Category	Category	Terms
		containing transcription factor ZAR2.
Literature	GENERIF_SUMMARY	The incidence of breast and ovarian cancer in BRCA2 mutation carriers and of ovarian cancer in BRCA1 mutation carriers is still high after 60 years.
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers had an increased risk of prostate cancer and a higher histologic grade, and BRCA1 or BRCA2 mutations were associated with a more aggressive clinical course.
Literature	GENERIF_SUMMARY	A high proportion of DNA variants of BRCA2 is associated with aberrant splicing in breast/ovarian cancer patients.
Literature	GENERIF_SUMMARY	Hereditary breast and ovarian cancer due to mutations in BRCA2 (Review)
Literature	GENERIF_SUMMARY	Major gene rearrangements involving the BRCA1 and BRCA2 genes appear to contribute little to the burden of inherited predisposition to breast and ovarian cancer in the Ashkenazim.
Literature	GENERIF_SUMMARY	Report a novel complex mutation that consists of a deletion of 12 bp and an insertion of 2 bp (c.8402_8413del12ins2bp) in the exon 18 of the BRCA2 gene in a Spanish hereditary male/female breast cancer family.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with increases in breast cancer worry in African American women.
Literature	GENERIF_SUMMARY	Rates of new primaries are predicted to some extent by mutation status
Literature	GENERIF_SUMMARY	The study results suggest that genes that act upstream of p53, or participate in the DNA damage response, may modify the risk of cancer in women with mutant BRCA1/2 alleles.
Literature	GENERIF_SUMMARY	Data suggest that the risk for breast cancer among BRCA1 or BRCA2 carriers may be modified by reproductive characteristics and alcohol use.
Literature	GENERIF_SUMMARY	analysis of locally prevalent BRCA1/2 mutations in all breast cancer patients might extend substantially the percentage of identified mutation carriers.
Literature	GENERIF_SUMMARY	BRCA2 does not regulate cytokinesis in human cells.
Literature	GENERIF_SUMMARY	The risks of subsequent contralateral breast cancer are substantial for women who carry a BRCA1/BRCA2 mutation.
Literature	GENERIF_SUMMARY	Case Reports
Literature	GENERIF_SUMMARY	BRCA2 mutation does not affect either short-term or long-term survival rate in breast cancer. Review.
Literature	GENERIF_SUMMARY	Receipt of BRCA1/2 mutation test results impacts how carriers see the positive aspects of risk reducing oophorectomy/mastectomy and their surgical

Main Category	Category	Terms
		intentions.
Literature	GENERIF_SUMMARY	8 novel molecular changes were identified in the BRCA2 gene in 105 investigated breast cancer patients
Literature	GENERIF_SUMMARY	Letter
Literature	GENERIF_SUMMARY	Data extend the ovarian BRCAness phenotype, imply BRCA1/2-deficient ovarian cancer is biologically distinct, and suggest that patients with visceral metastases should be considered for BRCA1/2 sequencing.
Literature	GENERIF_SUMMARY	study has informed the development of an appropriate support network based on a hub and spoke model to help BRCA1/BRCA2 carriers and their families adapt to living and coping with their genetic risk
Literature	GENERIF_SUMMARY	New loci appeared to interact multiplicatively for breast cancer risk in BRCA1 and BRCA2 carriers.
Literature	GENERIF_SUMMARY	The observed physical association of p53 and BRCA2 may have important functional implications in the p53 transactivation-independent suppression of homologous recombination.
Literature	GENERIF_SUMMARY	plays important role in susceptibility to breast cancer in middle aged Chinese women
Literature	GENERIF_SUMMARY	Authors studied 103 women with fallopian tube cancer (48 with a BRCA1 mutation, 12 with a BRCA2 mutation and 43 with no identified BRCA mutation) and 980 matched controls.
Literature	GENERIF_SUMMARY	we report the first identification of a Danish breast and ovarian cancer family heterozygote for mutations in the BRCA1 and BRCA2 genes
Literature	GENERIF_SUMMARY	Data suggest that breast cancer risk among mutation-negative women from BRCA1/2 mutation-positive families is similar to that observed in the general population.
Literature	GENERIF_SUMMARY	This study assessed the pathological mutation detection rates for BRCA1, BRCA2 and the CHEK2c.1100 delC mutation in 2022 women with breast cancer, including 100 with breast/ovary double primary and 255 with bilateral breast cancer.
Literature	GENERIF_SUMMARY	In total, 5-10% of all breast cancer cases are related to gen mutations. In most cases a mutation in the BRCA1-gen and BRCA2-gen is responsible for insufficient repair of DNA damages that cause breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Analysis of BRCA2 c.8308 G > A by mRNA analysis, and BRCA2 c.8962A > G, BRCA2 c.8972G > A, BRCA2 c.9172A > G, and BRCA2 c.9213G > T by a minigene assay, revealed no evidence for aberrant splicing.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian Neoplasms.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data identified splicing aberration for BRCA1c.4868C>G(p.Ala1623Gly) and BRCA2c.7988A>T(p.Glu2663Val) and c.8168A>G(p.Asp2723Gly).
Literature	GENERIF_SUMMARY	Interaction between BRCA1/BRCA2 and ATM/ATR associate with breast cancer susceptibility.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian cancer predisposition families from a North-Eastern Romanian population.
Literature	GENERIF_SUMMARY	Approximately 30% of women with fallopian tube cancer have a mutation in BRCA1 or BRCA2.
Literature	GENERIF_SUMMARY	BRCA2 genes mutations are responsible for a significant proportion of breast cancer.
Literature	GENERIF_SUMMARY	The currently presented results validate the suggestion that all patients with FANCD1 should be tested for BRCA2 mutations irrespective of whether the family history meets the BRCA cancer criteria.
Literature	GENERIF_SUMMARY	Built a classification method that is able to distinguish BRCA2-mutated from sporadic breast tumors based on their chromosomal aberrations with an accuracy of 86.5%.
Literature	GENERIF_SUMMARY	Study in a multi-ethnic (Malay, Chinese, Indian) cohort suggests that large genomic rearrangements are present at a low frequency but should nonetheless be included in the routine testing for BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	among BRCA1/2 mutation carriers, prophylactic surgery is cost effective compared to chemoprevention and screening
Literature	GENERIF_SUMMARY	the minor allele del of CASP8 rs3834129 is associated under a dominant model with increased breast cancer risk in carriers of BRCA1 mutations but not in carriers of BRCA2 mutations
Literature	GENERIF_SUMMARY	demonstrate significant quantitative differences regarding the production of the BRCA2 full length RNA and the transcript lacking exon 3 in c.156_157insAlu BRCA2 mutation carriers and in controls
Literature	GENERIF_SUMMARY	BRCA2 germline mutation is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with very young age of onset of breast cancer.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	Prioritizing candidate genetic modifiers of BRCA1 and BRCA2 using a combinatorial analysis of global expression and polymorphism association studies.
Literature	GENERIF_SUMMARY	The coding sequence and exon-intron junctions of BRCA1 and BRCA2 were amplified in 24 multiplex PCRs using a single condition.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Development of a scoring method to screen for BRCA1/2 mutations.
Literature	GENERIF_SUMMARY	Functional analysis of BRCA2 variants using a mouse embryonic stem cell-based assay.
Literature	GENERIF_SUMMARY	Of the 173 patients, 43 (25%) were found to have a tubal lesion, including 23% of the BRCA1 mutation carriers and 27% of the BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	BRCA2 is a key mediator of homologous recombination; a molecular basis for understanding how this DNA repair process is disrupted by BRCA2 mutations, which lead to chromosomal instability and cancer
Literature	GENERIF_SUMMARY	BRCA2 binds selectively to ssDNA & to ssDNA in tailed duplexes & replication fork structures. directed the binding of RAD51 recombinase to ssDNA, reduced the binding of RAD51 to duplex DNA and stimulated RAD51-mediated DNA strand exchange.
Literature	GENERIF_SUMMARY	6 RAD51 molecules and promotes RAD51 binding to ssDNA coated by replication protein A in a manner that is stimulated by DSS1.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with menopausal status modifying the association between body weight and breast cancer and disease progression.
Literature	GENERIF_SUMMARY	Testing women with triple negative breast cancers who were younger than 50 years for BRCA mutations is a cost-effective strategy and should be adopted into current guidelines for genetic testing.
Literature	GENERIF_SUMMARY	Multivariate analysis confirmed that the poorer survival of PrCa in BRCA2 mutation carriers is associated with the germline BRCA2 mutation per se
Literature	GENERIF_SUMMARY	Characteristic pathological features and older age of onset among BRCA1-mutated estrogen receptor positive breast cancers.
Literature	GENERIF_SUMMARY	A common genetic variant in BRCA2 (1342A > C) occurred less frequently in deleterious mutation carriers of breast cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with basal cell carcinoma.
Literature	GENERIF_SUMMARY	Targeted prostate cancer screening in men with mutations in BRCA1 and BRCA2 detects aggressive prostate cancer.
Literature	GENERIF_SUMMARY	PALB2 can function synergistically with a BRCA2 chimera (termed piccolo, or piBRCA2) to further promote strand invasion.
Literature	GENERIF_SUMMARY	findings support PALB2 and BRCA2 as risk genes specifically for bipolar disorder, and suggest that altered DNA repair related to neurogenesis may be involved in BD pathophysiology

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	A new human breast cancer xenograft obtained from a BRCA2-mutated patient.
Literature	GENERIF_SUMMARY	BRCA2 frame shift mutations are associated with defective protein translocation.
Literature	GENERIF_SUMMARY	BRCA2 mutations are associated with breast cancer and/or ovarian cancer.
Literature	GENERIF_SUMMARY	Suggest that the frequency of BRCA1/2 gene rearrangements is very low in Turkish breast/ovarian cancer patients.
Literature	GENERIF_SUMMARY	18 individuals (1.51%) who concomitantly carried the 185delAG BRCA1 and the 6174delT BRCA2 mutations and four individuals (0.34%) who carried the 5382insC BRCA1 and the 6174delT mutations
Literature	GENERIF_SUMMARY	males who develop breast cancer may have mutations in BRCA2, and possibly in PALB2
Literature	GENERIF_SUMMARY	Only tumor cell lines with a combination of high Aurora-A and -B expression, BRCA2 mutation and p53 defects showed more sensitivity towards Aurora inhibition than other cell lines.
Literature	GENERIF_SUMMARY	there are no predominant recurring germline mutations in BRCA1 or BRCA2 genes among ethnically diverse Jewish and non Jewish high risk families in Israel.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 genetic test in high risk patients and families.
Literature	GENERIF_SUMMARY	Results indicate that SNPs that modify BRCA2 penetrance identified by an agnostic approach thus far are limited to variants that also modify risk of sporadic BRCA2 wild-type breast cancer.
Literature	GENERIF_SUMMARY	BRCA2 founder mutations are associated with breast cancer and/or ovarian cancer.
Literature	GENERIF_SUMMARY	results suggested that a complex consisting of BRCA2, NPM, and ROCK2 maintains the numerical integrity of centrosomes and accurate cell division and that dysfunction of this regulation might be involved in the tumorigenesis of breast cancer
Literature	GENERIF_SUMMARY	study evaluated the associations of SNPs, rs4973768 in SLC4A7/NEK10, rs6504950 in STXBP4/COX11, and rs10941679 at 5p12, and reanalyzed the previous associations using additional carriers in a sample of 12,525 BRCA1 and 7,409 BRCA2 carriers
Literature	GENERIF_SUMMARY	Higher levels of AI in BRCA1/2 were associated with a 4.22-fold increased risk of familial ovarian cancer.
Literature	GENERIF_SUMMARY	This high rate of BRCA germline mutations in uterine serous carcinoma(USC) patients coupled with a high rate of personal and familial cancer histories may suggest that USC is associated with the hereditary

Main Category	Category	Terms
		breast-ovarian syndrome.
Literature	GENERIF_SUMMARY	A review of MEDLINE from 1989 through 2010 for the terms BRCA1, BRCA2, breast cancer, ovarian cancer, risk assessment, and genetic testing is presented.
Literature	GENERIF_SUMMARY	Rad52 can respond to DNA double-strand breaks and replication stalling independently of BRCA2
Literature	GENERIF_SUMMARY	BRCA1 mutations are associated with breast carcinoma In Situ.
Literature	GENERIF_SUMMARY	analysis of germline mutations of BRCA1 and BRCA2 genes in Turkish breast, ovarian, and prostate cancer patients
Literature	GENERIF_SUMMARY	found no significant associations between any single-single-nucleotide polymorphisms or haplotypes of BRCA1 or BRCA2 and risk of contralateral breast cancer among all women
Literature	GENERIF_SUMMARY	We identified seven different truncating mutations in seven families, five in BRCA2 three in site-specific breast cancer families and two in breast-ovarian cancer families
Literature	GENERIF_SUMMARY	A frequent founder BRCA2 mutation in non-Afrikaner breast cancer patients of the Western Cape of South Africa.
Literature	GENERIF_SUMMARY	BRCA2 germline mutations are associated with uterine serous carcinoma.
Literature	GENERIF_SUMMARY	following a diagnosis of breast cancer, the annual risk of ipsilateral breast cancer risk in BRCA mutation carriers is 1.2% per year.
Literature	GENERIF_SUMMARY	Elevated expression of BRCA2 correlates with TGF-beta1 and HER2 in breast cancer and these three factors may act in synergy to promote cancer.
Literature	GENERIF_SUMMARY	Among the mutations detected in the Slovenian population, 5 mutations in BRCA1 and 4 mutations in BRCA2 have not been described in other populations until now.
Literature	GENERIF_SUMMARY	This multicenter study demonstrates a significant survival advantage in advanced stage ovarian cancer patients with non-AJ BRCA mutations, confirming the previous studies in the Jewish population.
Literature	GENERIF_SUMMARY	analysis of early BRCA1 and 2-associated ovarian and fallopian tube tumorigenesis
Literature	GENERIF_SUMMARY	40 novel sequence variations in BRCA1 and BRCA2 in high risk breast and/or ovarian cancer familiesin Denmark.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations are common in women with invasive ovarian cancer. All women diagnosed with invasive non-mucinous ovarian cancer should be considered to be candidates for genetic testing.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Studies indicate that most of these breast cancers are caused by mutations in the high penetrance genes BRCA1 and BRCA2 accounting for about 5% of all breast cancers.
Literature	GENERIF_SUMMARY	The BRCA2 gene was screened for mutations in well-characterized, Finnish, high-risk hereditary breast and/or ovarian cancer individuals.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	the rate of BRCA1/2 mutations in double primary breast and ovarian cancer (DPBOC) patients is probably less than 60%.
Literature	GENERIF_SUMMARY	This study has assessed the possible pathogenicity of 26 intronic BRCA1 or BRCA2 variants in breast cancer.
Literature	GENERIF_SUMMARY	a novel function for HMG20b in cytokinesis is regulated by its interaction with the BRC repeats of BRCA2
Literature	GENERIF_SUMMARY	EMSY amplification may mimic BRCA2 deficiency, at least by overriding RPA and PALB2, crippling the BRCA2/RAD51 complex at DNA-damage and replication/transcription sites
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with high risk of pelvic serous cancer.
Literature	GENERIF_SUMMARY	7-fold increase in Endometrial Cancer risk with tamoxifen exposure is asscoiated with female family members from BRCA2 mutation.
Literature	GENERIF_SUMMARY	An association was observed between p.Arg280His-rs25489 in XRCC1 and breast cancer risk for BRCA2 mutation carriers, with rare homozygotes at increased risk relative to common homozygotes.
Literature	GENERIF_SUMMARY	AURKA expression is an unfavorable prognostic factor in patients with endometrioid ovarian cancer and BRCA2 is favorable.
Literature	GENERIF_SUMMARY	BRCA2 N372H polymorphism is associated with follicular lymphoma.
Literature	GENERIF_SUMMARY	Loss of function of various proteins involved in double strand break repair other than BRCA1/2 has been suggested to be synthetically lethal with PARP inhibition.
Literature	GENERIF_SUMMARY	Oral contraceptive use and paternal origin of mutation affect breast cancer penetrance in Jewish BRCA1 and BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	Skp2 overexpression reduced BRCA2 protein and promoted cell growth and migration of prostate cancer cells
Literature	GENERIF_SUMMARY	T-C transitions at positions 29 bp and 44 bp in relation to the total sequence of BRCA2 exon 8 were identified.
Literature	GENERIF_SUMMARY	hierarchical statistical modeling of data observed in a

Main Category	Category	Terms
		case-control study of contralateral breast cancer in which all the participants were genotyped for variants in BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	Data show that hyperthermia (41-42.5 degrees C) induces degradation of BRCA2 and sensitize innately HR-proficient tumor cells to PARP-1 inhibitors.
Literature	GENERIF_SUMMARY	The comparison between the two different populations did not show any significant difference regarding BRCA2 methylation but exhibited a trend in the decrease of BRCA2 promoter methylation in peripheral blood DNA of sporadic ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 prevents rather than repairs nucleolytic lesions at stalled replication forks to maintain genomic integrity and hence likely suppresses tumorigenesis through this replication-specific function.
Literature	GENERIF_SUMMARY	FANCD1/BRCA2 played notably important roles in the repair of TMZ-induced DNA damage.
Literature	GENERIF_SUMMARY	A lack of association of the rs2910164
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in BRCA1 and BRCA2 mutation carriers from 42 studies from the Consortium of Investigators of Modifiers of BRCA1/2, were genotyped.
Literature	GENERIF_SUMMARY	More than 50% of the Ashkenazi Jewish chromosomes with the BRCA1 185delAG mutation share an identical 2.1 Mb haplotype and nearly 16% of Ashkenazi Jewish chromosomes carrying the BRCA2 6174delT mutation share a 1.4 Mb haplotype.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations are observed with nearly equal distribution in Ashkenazi Jewish breast-pancreas cancer families, suggesting that both genes are associated with pancreatic cancer risk
Literature	GENERIF_SUMMARY	the interaction between BRC repeat 4 (BRC4) and RAD51 was strengthened by the substitution of a single BRC4 amino acid from valine to isoleucine (V1532I).
Literature	GENERIF_SUMMARY	Spectrum and characteristics of BRCA1/2 mutations in Chinese familial breast cancer exhibit some unique features, and Chinese women with familial breast cancer whose tumors are diagnosed at or before the age of 40 are good candidates for BRCA1/2 testing.
Literature	GENERIF_SUMMARY	BRCA2 and PALB2 are main regulators of G2 checkpoint maintenance following DNA-damage.
Literature	GENERIF_SUMMARY	evaluation of frequency and distribution of common BRCA1 and BRCA2 mutations which associated with breast cancer risk (Meta-Analysis)
Literature	GENERIF_SUMMARY	evaluation of prevalence of de novo mutations in BRCA1 and BRCA2 in a clinic-based cohort of women seeking risk assessment; results suggest de novo mutations in BRCA1 and BRCA2 are less common than in other hereditary cancer syndromes

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	prevalence of BRCA1 mutations in triple negative breast cancer
Literature	GENERIF_SUMMARY	the decreased DSB repair capacity of heterozygous BRCA2 mutations suggests gene dosage effects in G2 fibroblasts
Literature	GENERIF_SUMMARY	propose that the BRC repeats in BRCA2 cooperate in a partially redundant but reinforcing manner to ensure a high probability of RAD51 filament formation
Literature	GENERIF_SUMMARY	Risk-reducing salpingo-oophorectomy (RRSO) in BRCA mutation carriers.
Literature	GENERIF_SUMMARY	Associations were found between risk of breast cancer and linkage disequilibrium blocks in IGF2 for BRCA1 and BRCA2 mutation carriers, HTRA1 for BRCA1 carriers, and MMP3 for BRCA2 carriers.
Literature	GENERIF_SUMMARY	Promoter hypermethylation of PAX6, BRCA2, PAX5, WT1, CDH13 and MSH6 seems to be a frequent early event in breast cancer
Literature	GENERIF_SUMMARY	IVS7 + 2T > G produces an alternatively spliced transcript lacking exons 4-7. BRCA2(Delta105) is proficient in homologous recombination-mediated DNA repair. p.L2510P, p.R2336H, & p.W2626C are deleterious. p.I2490T & p.K2729N probably are neutral.
Literature	GENERIF_SUMMARY	BRCA2 helps restore and maintain integrity of the genome.
Literature	GENERIF_SUMMARY	BRCA2 mutations from multiple breast cancer families are associated with prostate cancer.
Literature	GENERIF_SUMMARY	12 novel BRCA1 and BRCA2 mutations, leading to aberrantly expressed transcripts, were identified in a cohort of hereditary breast and ovarian cancer syndrome patients.
Literature	GENERIF_SUMMARY	Data provide insight into the genomic effects of loss of BRCA2 function.
Literature	GENERIF_SUMMARY	Our results confirm the existence of a central low BC risk region in BRCA1 (LR1) and one in BRCA2 (LR2).
Literature	GENERIF_SUMMARY	Abnormal splicing patterns expected to lead to non-functional proteins were observed for 7 BRCA1 and BRCA2 variants in a cohort of patient screening for hereditary breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Findings reveal how sequence variation in the BRC repeats directly affects affinity with RAD51 and provide significant new insights into the control of RAD51 by human BRCA2.
Literature	GENERIF_SUMMARY	TAS106 induces the down-regulation of BRCA2 and the subsequent abrogation of the homologous recombination pathway, leading to a radiosensitizing effect.
Literature	GENERIF_SUMMARY	Mutations in the BRCA2 gene is associated with

Main Category	Category	Terms
		osteopenia and osteoporosis after risk-reducing salpingo-oophorectomy in breast cancer patients.
Literature	GENERIF_SUMMARY	Mutations in the BRCA2 gene is associated low with sporadic breast cancer of Spanish population.
Literature	GENERIF_SUMMARY	Germline mutations in the BRCA2 gene is associated with Iranian breast cancer patients.
Literature	GENERIF_SUMMARY	Fifteen patients, five male, with a BRCA1 or BRCA2 mutation and PAC and one patient with a BRCA1 mutation and acinar cell carcinoma of the pancreas were identified.
Literature	GENERIF_SUMMARY	first large rearrangement of the BRCA2 gene that includes all of exon 3 and leads to an in frame deletion of exon 3 at the transcriptional level
Literature	GENERIF_SUMMARY	Women with BRCA-associated ovarian tumors demonstrate a greater sensitivity to cytotoxic therapy with Doxil.
Literature	GENERIF_SUMMARY	results suggest that routine testing of early onset PrCa cases for germline BRCA2 mutations will further help to refine the prevalence and risk associated with BRCA2 mutations and may be useful for guiding management options
Literature	GENERIF_SUMMARY	the deletion of the wild-type BRCA2 allele, together with co-occurring changes at 6 q, 11 q, and 17 q, are important events in progression toward advanced breast cancer.
Literature	GENERIF_SUMMARY	polymorphisms in BRCA2 do not contribute to cancer risk in a population of Lynch syndrome patients with colorectal cancer
Literature	GENERIF_SUMMARY	Our results suggest an important role of BRCA1 and BRCA2 promoter methylation in breast cancer development in the Tunisian population.
Literature	GENERIF_SUMMARY	This study suggests that prior appendectomy is more frequently noted in BRCA2 carriers with breast and ovarian cancer than in unaffected mutation carriers.
Literature	GENERIF_SUMMARY	We found that haplotypes in APEX1, BRCA2, ERCC2, and RAD51 were significantly associated with total childhood acute lymphoblastic leukemia
Literature	GENERIF_SUMMARY	Germline mutations in the BRCA2 gene is associated with pancreatic cancer.
Literature	GENERIF_SUMMARY	among women with high-grade serous ovarian cancer, BRCA2 mutation, but not BRCA1 deficiency, was associated with improved survival, improved chemotherapy response, and genome instability compared with BRCA wild-type
Literature	GENERIF_SUMMARY	women who are carriers of BRCA1/2 mutations living in natural fertility conditions have excess fertility as well as excess post-reproductive mortality in relation to controls

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Women who had ductal carcinoma in situ and a family history of ovarian cancer or who had BRCAPRO scores >/=10% had a high rate of BRCA positivity regardless of age at diagnosis.
Literature	GENERIF_SUMMARY	BRCA2 has an important role in telomere stabilization and protection.
Literature	GENERIF_SUMMARY	BRCA2 mutations are associated with unselected Nigerian breast cancer patients.
Literature	GENERIF_SUMMARY	Fifteen mutations were identified (two in BRCA2 and thirteen in BRCA1) representing 15.6% of the total (95% CI
Literature	GENERIF_SUMMARY	The associations of the 12 SNPs with risk for BRCA1 and BRCA2 carriers differ by ER-positive or ER-negative breast cancer status.
Literature	GENERIF_SUMMARY	report of four interactions between mutations in the breast cancer susceptibility genes ATM and CHEK2 with BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	The rate of phenocopy breast cancer in non-carriers from Israeli BRCA1 BRCA2 mutation carrier families is 2.26% with no increased breast cancer risk over the average-risk population.
Literature	GENERIF_SUMMARY	The germline mutation frequency of BRCA2 in esophageal squamous cell carcinoma patients from a low-risk Chinese population was significantly higher than that in healthy controls.
Literature	GENERIF_SUMMARY	Tetraploidy was significantly 3 times more common in BRCA2 breast cancers than sporadic but no differences were found in the overall ploidy distribution between BRCA2-mutation carriers and non-carriers.
Literature	GENERIF_SUMMARY	There was no observed difference in the proportion of male breast cancer tumors with abnormal BRCA2 protein expression in Egypt compared to Morocco.
Literature	GENERIF_SUMMARY	Promoter hypermethylation patterns of of BRCA2 germ line mutation carriers are associated with fallopian tube neoplasms.
Literature	GENERIF_SUMMARY	Pathologic characteristics of BRCA1 and BRCA2 tumors may be useful for improving risk-prediction algorithms and informing clinical strategies for screening and prophylaxis.
Literature	GENERIF_SUMMARY	C-myc up-regulated miR-1245 expression which led to down-regulation of BRCA2.
Literature	GENERIF_SUMMARY	Loss of heterozygosity at BRCA2 locus had no associations with estrogen receptor or progesterone receptor status and prognosis.
Literature	GENERIF_SUMMARY	BRCA2 is required for telomere homeostasis and may be particularly important for the replication of G-rich telomeric lagging strands.

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Literature	GENERIF_SUMMARY	Indicate potential strong associations between morphology and BRCA1/2 genotype in high-grade serous carcinomas.
Literature	GENERIF_SUMMARY	Our results reveal functional redundancy within the BRCA2 protein and emphasize the plasticity of this large protein built for optimal homologous recombination function in mammalian cells
Literature	GENERIF_SUMMARY	functional KL gene variants had no discernable effect on either breast or ovarian cancer risk in a large group of ethnically diverse BRCA1 and BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	the in frame BRCA2 deletion c.9203_9328del126 is a rare mutation strongly associated with familial breast cancer.
Literature	GENERIF_SUMMARY	Overweight BRCA mutation carrier women seem to be at risk of developing more cortical invaginations than women of normal weight.
Literature	GENERIF_SUMMARY	Studies across multiple ethnicities demonstrate that BRCA1 and BRCA2 mutations not only increase the risk of prostate cancer, but also predispose patients to early onset of potentially lethal disease. (Review)
Literature	GENERIF_SUMMARY	The expression of DNA repair proteins, including PARP, FANCD2, BRCA1, PTEN, H2AX, ATM, as well as p53, was studied to see if it correlated with response to treatment, recurrence rate, and survival in ovarian cancer.
Literature	GENERIF_SUMMARY	The results from this study suggest that height, weight, or weight gain do not influence the risk of ovarian cancer among carriers of a BRCA1 or BRCA2 mutation.
Literature	GENERIF_SUMMARY	Among patients with invasive epithelial ovarian cancer, having a germline mutation in BRCA1 or BRCA2 was associated with improved 5-year overall survival. BRCA2 carriers had the best prognosis.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations correlate with TP53 abnormalities and presence of immune cell infiltrates in ovarian high-grade serous carcinoma.
Literature	GENERIF_SUMMARY	Here, the authors use proteomic profiling to identify APRIN (PDS5B), a cohesion-associated protein, as a BRCA2-associated protein.
Literature	GENERIF_SUMMARY	Treatment with 18.5 muM Genistein or 78.5 muM Daidzein might reverse DNA hypermethylation and restore the expression of the oncosuppressor genes BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	Data show that RBMX accumulated at DNA lesions through multiple domains in a poly(ADP-ribose) polymerase 1-dependent manner and promoted HR by facilitating proper BRCA2 expression.
Literature	GENERIF_SUMMARY	New pathogenic mutation in BRCA2 gene identified in women in Croatia.

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Literature	GENERIF_SUMMARY	BRCA2 mutations for non-familial Korean breast cancer patients were detected at a high rate, particularly, in patients with early onset of less than 35 years of age, bilateral breast cancer, and breast and ovarian cancer.
Literature	GENERIF_SUMMARY	The rate of some seemingly recurring germline mutations in the BRCA2 gene in breast/ovarian high risk Israeli women of diverse ethnic origin were determined.
Literature	GENERIF_SUMMARY	The penetrance of the mutation and the aggressiveness of the related tumor could depend on a dosage effect of the mutated allele transcript.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations occurred more frequently in platinum-sensitive EOC than platinum-resistant disease.
Literature	GENERIF_SUMMARY	PWC has the potential to detect occult peritoneal carcinoma in patients with BRCA1 or BRCA2 mutations.
Literature	GENERIF_SUMMARY	Data suggest that BRCA1 and BRCA2 mutations are responsible for a significant proportion of familial breast cancer in Moroccan patients.
Literature	GENERIF_SUMMARY	Age and grade differed significantly between BRCA1 and BRCA2 carriers in our study population.
Literature	GENERIF_SUMMARY	Breast and ovarian cancer risk are high in women with BRCA2 genetic mutation.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in BRCA2 gene modify the association between occupational exposure to solvents and is associated with non-Hodgkin's lymphoma.
Literature	GENERIF_SUMMARY	Four novel structural alterations at the BRCA2 locus in hereditary breast and ovarian cancer syndrome families.
Literature	GENERIF_SUMMARY	Several common alleles have been found to modify breast or ovarian cancer risk for BRCA1 and BRCA2 mutation carriers. These common alleles explain a small proportion of the genetic variability in breast or ovarian cancer risk for mutation carriers.
Literature	GENERIF_SUMMARY	findings not only show that Mre11 activity is required for the survival of BRCA2 mutant cells but also elucidate roles for both the BRCA2 and PARP1 proteins in protecting stalled replication forks
Literature	GENERIF_SUMMARY	No new rearrangements were found in the BRCA2 gene in 472 patients with breast cancer predisposition.
Literature	GENERIF_SUMMARY	three cis-acting BRCA2 polymorphisms that alter the binding of transcription factors at regulatory sites and could possibly affect the risk in mutation carriers through altering expression levels of the wild-type allele
Literature	GENERIF_SUMMARY	Caucasian female with double heterozygosity for disease-causing BRCA1 and BRCA2 mutations carriers seem to develop breast cancer at a younger age and have a more severe disease than carriers of a single

Main Category	Category	Terms
		BRCA mutation.
Literature	GENERIF_SUMMARY	The aim is to determine whether DNA-direct is an acceptable procedure for BRCA1/2 testing, in order to provide customized care to patients with breast cancer
Literature	GENERIF_SUMMARY	high levels of BRCA2 perturb Rad51-mediated homologous recombination
Literature	GENERIF_SUMMARY	For breast cancer, BRCA mutation status is not associated with the rate of optimal tumor debulking at primary surgery after accounting for differences in patient age.
Literature	GENERIF_SUMMARY	PARP inhibitors may represent a new and promising targeted therapy for patients with BRCA1/2 -associated cancer.
Literature	GENERIF_SUMMARY	in patients with triple-negative breast cancer, BRCA1 mutation prevalence was lower, and BRCA2 mutation prevalence was higher, than previously described
Literature	GENERIF_SUMMARY	investigation of the effect on splicing of BRCA2 variants c.8488-1G > A (exon 20) and c.9026_9030del (exon 23), and 41 BRCA2 variants reported in the Breast Cancer Information Core mutation database; a relevant proportion of BRCA2 variants are correlated with splicing disruptions
Literature	GENERIF_SUMMARY	Data indicate taht BRC repeat is a common RAD51 recombinase interaction module that can either promote homologous recombination (HR), as in the case of BRCA2, or to suppress HR, as in RECQL5 helicase.
Literature	GENERIF_SUMMARY	It is concluded that a proportion of patients with therapy related myeloid neoplasms carry cancer susceptibility mutations,including BRCA2 mutations, which is likely to contribute to therapy related leukaemogenesis.
Literature	GENERIF_SUMMARY	In patients diagnosed with breast and ovarian cancer the study found 4 novel deleterious mutations (c.2805_2808delAGAT and c.3124_3133delAGCAATATTA in BRCA1; c.2639_2640delTG and c.5114_5117delTAAA in BRCA2).
Literature	GENERIF_SUMMARY	BRCA2 mutation frequency in triple-negative breast cancer
Literature	GENERIF_SUMMARY	Article presents a comprehensive review of the relevant literature on the role of BRCAness in ovarian cancer with respect to BRCA function.
Literature	GENERIF_SUMMARY	Functional characterization of 12 cancer associated BRCA2 mutations.
Literature	GENERIF_SUMMARY	Data suggest that if survival rates are based on tested patients, all of whom are alive at the time of testing, then the survival experience will be overestimated, both for women who test positive for BRCA1 or BRCA2

Main Category	Category	Terms
		mutations and for non-carriers.
Literature	GENERIF_SUMMARY	The first report identifying BRCA2 mutations in Puerto Rico.
Literature	GENERIF_SUMMARY	8 new missense unclassified variants were identified in the BRCA1 and BRCA2 genes in Algerian breast/ovarian cancer families.
Literature	GENERIF_SUMMARY	Invasive tubal carcinoma and TIC were limited to BRCA-mutation carriers, whereas hyperplasia and minor epithelial atypia were commonly found in both BRCA-mutation carriers and controls.
Literature	GENERIF_SUMMARY	IRS1 Gly972Arg single-nucleotide polymorphism, which affects insulin-like growth factor and insulin signaling, modifies ovarian cancer risk in BRCA1 and BRCA2 mutation carriers and breast cancer risk in BRCA1 class II mutation carriers.
Literature	GENERIF_SUMMARY	Among the 1017 women referred for BRCA testing, 63 women younger than 52 years of age who had been diagnosed with breast cancer were found, subsequent to this diagnosis, to have BRCA1 or BRCA2 gene mutations.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	study genotyped recurrent BRCA1/BRCA2 mutations among Nigerian, African American, and Barbadian breast cancer patients
Literature	GENERIF_SUMMARY	studies suggest substantial mortality benefits in using MRI to screen BRCA1/2 mutation carriers aged 25 to 60 years
Literature	GENERIF_SUMMARY	study examined BRCA1/BRCA2 gene mutations/SNPs and BRCA1 haplotypes in early-onset breast cancer patients of Indian ethnicity; findings indicate a high incidence of BRCA1/BRCA2 gene mutations in the Indian patients
Literature	GENERIF_SUMMARY	Identification of a new mutation located deep into intron 12, c.6937+594T > G, which systematically activates splicing of a cryptic exon introducing a premature termination codon in the coding sequence.
Literature	GENERIF_SUMMARY	Case Report
Literature	GENERIF_SUMMARY	A locus on chromosome 6q22.33 (rs2180341) was evaluated to see if it modifies breast cancer risk in 3,361 BRCA1 and 2,020 BRCA2 carriers.
Literature	GENERIF_SUMMARY	A specific role for BRCA2 in the regulation of midbody structure and function is identified that is separate from DNA damage repair and is an explanation of whole-chromosomal instability in BRCA2-deficient tumors.
Literature	GENERIF_SUMMARY	Low doses of mammography X-rays have the potential to induce chromosomal instability in fibroblasts from BRCA mutation carriers.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	show a repair-independent requirement for FA genes, including FANCD2, and BRCA1 in protecting stalled replication forks from degradation
Literature	GENERIF_SUMMARY	characteristics of BRCA1/2 germline mutations in Korean patients with breast and ovarian cancer; 420 (150 distinct) pathogenic mutations were identified, 211(73 distinct)in BRCA1 and 209 (77 distinct)in BRCA2; majority of distinct mutations resulted in premature termination codon of the BRCA1/2 translation
Literature	GENERIF_SUMMARY	Although there was an association between two SNPs in BRCA2 and cardiovascular disease in a multi-ethnic population, these results were not replicated in two South Asian case-control studies of incident myocardial infarction.
Literature	GENERIF_SUMMARY	High frequency of BRCA2 founder mutations are associated with gastrointestinal and tongue neoplasias in breast cancer.
Literature	GENERIF_SUMMARY	Association of a BRCA2 mutation with inflammatory breast cancer with two members of the same family.
Literature	GENERIF_SUMMARY	Methylation types of the BRCA2 promoter in different genotypes of the MTHFRa1298c polymorphism is not associated with ovarian cancer
Literature	GENERIF_SUMMARY	In TNBC cells that PI3K inhibition leads to DNA damage, downregulation of BRCA1/2, gain in poly-ADP-ribosylation, and subsequent sensitization to PARP inhibition.
Literature	GENERIF_SUMMARY	Novel BRCA1 and BRCA2 pathogenic mutations in Slovene hereditary breast and ovarian cancer families.
Literature	GENERIF_SUMMARY	BRCA2 exon 7 splicing is regulated by multiple exonic elements and is sensitive to disease-associated sequence variations.
Literature	GENERIF_SUMMARY	BRCA1/2 mutation carriers exhibited no significant increase in proliferation in the fallopian tube epithelial cells either in the ampulla or fimbriated ends of the tube.
Literature	GENERIF_SUMMARY	BRCA mutations account for a substantial proportion of hereditary breast/ovarian cancer in Southern Chinese population.
Literature	GENERIF_SUMMARY	Evidence based on spontaneous chromosome abnormalities and epidemiologic data indicates strong effects from exogenous mutagens and does not support the inevitability of cancer in all BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	BRCA1/2 mutations appear embryo-lethal unless rescued by low (CGG n
Literature	GENERIF_SUMMARY	This follow-up study suggests that, contrary to our initial report, this variant is not associated with breast cancer risk among individuals with germline BRCA2

Main Category	Category	Terms
		mutations.
Literature	GENERIF_SUMMARY	HGPIN is considered a precursor to cancer, as no LOH was observed, this assay does not provide a genetic marker that may be considered a positive predictor of tumorigenesis in BRCA2 carriers.
Literature	GENERIF_SUMMARY	In BRCA2 mutation carriers there was significant association between overall breast cancer risk SNP score and age at breast cancer onset.
Literature	GENERIF_SUMMARY	review of progress in understanding of BRCA1 and BRCA2 mutations in ovarian cancers and how these findings affect therapeutics; differing prognostic effects of BRCA2 and BRCA1 mutations is likely due to differing roles of BRCA1 and BRCA2 in homologous recombination repair and stronger association between the BRCA2 mutation and hypermutator phenotype
Literature	GENERIF_SUMMARY	The parental mutation origin did not affect the risk in women with a BRCA2 mutation.
Literature	GENERIF_SUMMARY	Two BRC3 mutations (K1440R and K1440E), reported in human breast cancer patients, occur at amino acids corresponding to those of the K1435R mutation in dogs. These mutations affected the interaction of canine BRC3 with RAD51
Literature	GENERIF_SUMMARY	Sixty-two patients (index cases) with invasive breast cancer belonging to sixty families and their asymptomatic female first-degree relatives (300 cases) were studied for germline mutations of BRCA1 and BRCA2 genes.
Literature	GENERIF_SUMMARY	found BRCA1/2 mutations in 15/103 probands. Among these, a BRCA2 gene mutation has not been described previously. In addition, we identified five novel BRCA2 variants
Literature	GENERIF_SUMMARY	Data show a substantial proportion of germ-line mutations in triple-negative breast cancer (TNBC), with a preponderance of BRCA1 mutations over mutations in BRCA2 or PALB2, but no evidence to implicate BRD7 mutations in the etiology of TNBC.
Literature	GENERIF_SUMMARY	BRCA2 mutation status did not appear to correlate with a distinct clinicopathological phenotype or disease behaviour, and a strong trend was seen suggesting a possible BRCA2 male breast cancer phenotype.
Literature	GENERIF_SUMMARY	Our findings do not support the usefulness of X chromosome inactivation analysis for the identification of BRCA mutation carriers and cancer risk assessment.
Literature	GENERIF_SUMMARY	Human Rad51 paralogs facilitate BRCA2-Rad51-dependent homologous recombination at different stages in the pathway and function independently of Rad52.
Literature	GENERIF_SUMMARY	PARP inhibitor resistance correlated with the emergence of a secondary BRCA2 mutation in patients following an initial favorable response.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Combined SEER database and retrospective study data analysis of lifetime risk of developing ductal carcinoma in situ and BRCA1 and BRCA2 carrier status.
Literature	GENERIF_SUMMARY	Classified 88 missense VUS in BRCA1 and BRCA2 on the basis of an in silico analysis and compared the classification of a subset of 60 VUS of which additional information including family, genetic and tumour data was available.
Literature	GENERIF_SUMMARY	Data indicate that the median levels of Ang-1, Ang-2 and VEGF mRNA were significantly higher in patients harboring BRCA1/2 germline mutations.
Literature	GENERIF_SUMMARY	in a study of US Hispanics, deleterious BRCA mutations were detected in 189 of 746 clinic patients(124 BRCA1, 65 BRCA2); a significant proportion were large rearrangement mutations
Literature	GENERIF_SUMMARY	Results indicate that this workflow performed analysis of BRCA1 and BRCA2 genes point mutations with a specificity of 100% and a sensitivity >/=97.5%.
Literature	GENERIF_SUMMARY	for women with invasive ovarian cancer, the short-term survival advantage of carrying a BRCA1 or BRCA2 mutation does not lead to a long-term survival benefit
Literature	GENERIF_SUMMARY	Older age, absence of a strong family history, and poor survival are all associated with decreased clinical identification of inherited BRCA1/2 mutations in women with ovarian cancer.
Literature	GENERIF_SUMMARY	Data indicate that younger patients with BRCA1 and BRCA2 (BRCA1/2)-mutated ovarian cancers (BMOC) (
Literature	GENERIF_SUMMARY	Results suggest that increased production of E(2) may have a role in the pathogenesis of BRCA2-mutation-related breast cancer.
Literature	GENERIF_SUMMARY	Deletion of chromosomes 13q and 14q is a common feature of tumors with BRCA2 mutations.
Literature	GENERIF_SUMMARY	Two new susceptibility loci. rs11571836 located in the BRCA2 3'-untranslated region.
Literature	GENERIF_SUMMARY	the imbalance in expression of Aurora-A and BRCA2 regulates RAS-induced genomic instability and tumorigenesis
Literature	GENERIF_SUMMARY	in a sample of Black women with early onset breast cancer, study evaluated BRCA mutations; BRCA mutation prevalence and the high prevalence of variants of uncertain significance in participants are consistent with prior studies
Literature	GENERIF_SUMMARY	Data indicate an increase of VEGF, HIF-1alpha expression and microvessel density (MVD) in BRCA1-2 carriers and BRCAX compared to the sporadic control group.
Literature	GENERIF_SUMMARY	review summarizes the recent findings on the epidemiology of BRCA1/2 mutations among ovarian

Main Category	Category	Terms
		carcinoma patients in various geographical areas with special reference to racial/ethnic groups and the common founder mutations
Literature	GENERIF_SUMMARY	data indicate that BRCA1 carriers diagnosed with MBC have worse outcomes compared to BRCA2 carriers and non-carriers. However, the differences in outcome did not reach statistical significance likely due to small sample sizes
Literature	GENERIF_SUMMARY	findings support the hypothesis that the majority of non-BRCA1/BRCA2 breast cancer families might be explained by the action of moderate and/or low penetrance susceptibility alleles
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 germline mutation-related DCIS show a high frequency of overexpression of HIF-1alpha.
Literature	GENERIF_SUMMARY	The idea that BRCA1/2 mutations have antagonistic pleiotropic effects (enhancing fertility while reducing survival) was supported in the natural fertility study.
Literature	GENERIF_SUMMARY	Mean age at natural menopause is 48.8 years for BRCA1 carriers, 49.2 years for BRCA2 carriers, and 50.3 years for control subjects.
Literature	GENERIF_SUMMARY	This study characterises the mutation, expression and methylation status of BRCA1/2 in 41 ovarian cancer cell lines.
Literature	GENERIF_SUMMARY	BRCA2 germline mutation carrier has Ampulla of vater adenocarcinoma.
Literature	GENERIF_SUMMARY	present the proof of principle of a NGS-based mutation screening procedure allowing the detection of inherited Alu insertions within any predefined sequence by investigating 2 cases
Literature	GENERIF_SUMMARY	Eleven variants in BRCA1 and 8 in BRCA2, including 8 not previously characterized at transcript level, were ascertained to affect mRNA splicing
Literature	GENERIF_SUMMARY	The study identified one of seven founder mutations in BRCA1 and BRCA2 in 27% of breast cancer patients in the Bahamas.
Literature	GENERIF_SUMMARY	Germline BRCA1 and BRCA2 mutations were detected in 13% and 7% of the early-onset breast cancer Brazilian patients.
Literature	GENERIF_SUMMARY	Recurrent BRCA2 gene mutation is associated with breast and ovarian cancers.
Literature	GENERIF_SUMMARY	Genotypes of AIB1 polyglutamine polymorphism analyzed do not appear to be associated to a modified risk of breast cancer development in BRCA1 and BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	Results show that age of onset in carriers of BRCA mutations is earlier than expected. Especially for BRCA1

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	BRCA2 gene mutation is associated with breast and ovarian cancers.
Literature	GENERIF_SUMMARY	Our data confirm the previously reported highly skewed distribution of FMR1 genotypes and sub-genotypes toward a high preponderance of low FMR1 alleles in BRCA1/2 carriers.
Literature	GENERIF_SUMMARY	6 (BRCA1) and 8 (BRCA2) microsatellite markers in a multiplex PCR.
Literature	GENERIF_SUMMARY	This comprehensive update of novel and previously reported breast cancer susceptibility loci contributes to the establishment of a panel of SNPs that modify breast cancer risk in BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Data indicate that interruption of ATR signaling strikingly further sensitized BRCA1- and BRCA2-deficient ovarian cancer cells to PARP inhibition.
Literature	GENERIF_SUMMARY	The SIR for BRCA1 carriers was 1.91 (95% CI
Literature	GENERIF_SUMMARY	Tumor size estimation using breast MRI in high risk patients is comparable to its performance in sporadic cancers.
Literature	GENERIF_SUMMARY	BRCA1/2 germ-line mutations confer a more aggressive prostate cancer phenotype with a higher probability of nodal involvement, distant metastasis and poor survival.
Literature	GENERIF_SUMMARY	Heterogeneous nuclear ribonucleoprotein C is a key regulator of BRCA gene expression and homologous recombination-based DNA repair.
Literature	GENERIF_SUMMARY	All patients with BRCA1 and BRCA2 risk factors need large genomic rearrangement screening for breast cancer.
Literature	GENERIF_SUMMARY	Aim of this study was to determine types and frequencies of BRCA mutations in individuals from high-risk families from Serbia.
Literature	GENERIF_SUMMARY	We show a high prevalence of BRCA1/2 mutations with LOH in an Ashkenazi Jewish cohort of surgically resected PDAC and neoplastic lesions, suggesting that these germline mutations are causal in selected individuals.
Literature	GENERIF_SUMMARY	p53 suppresses BRCA2-stimulated ATPase and strand exchange functions of human RAD51 to control homologous recombination.
Literature	GENERIF_SUMMARY	We detected 28 distinct pathogenic mutations (16 in BRCA1 and 12 in BRCA2), of which 3 mutations in BRCA1 and 5 in BRCA2 had not been previously described.
Literature	GENERIF_SUMMARY	A new common BRCA2 pathogenic mutation c.7235_7236insG was detected in Algerian and Moroccan families. [review]

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Rare variants of uncertain significance affecting phosphorylation may be a novel and important mechanism for which BRCA1 and BRCA2 functions are disrupted in breast cancer.
Literature	GENERIF_SUMMARY	Breast tumors from BRCA1 and BRCA2 mutation carriers display characteristic RNA expression patterns.
Literature	GENERIF_SUMMARY	Findings suggest treatment decisions and the clinical management for patients harbouring deleterious BRCA1/BRCA2 mutations.
Literature	GENERIF_SUMMARY	Association between BRCA2 mutation and SULT1A1 gene deletion in male breast cancer emerged.
Literature	GENERIF_SUMMARY	BRCA2 homeostasis in prostate cancer cells has been found to be critical in determining cell fates during prostate cancer progression
Literature	GENERIF_SUMMARY	simplified BRCA2 classifier based on the co-occurrence of Loss of heterozygosity at 13q13 and 14q32 could provide an indication to test for BRCA2 mutation in patients with ER-positive invasive ductal carcinomas.
Literature	GENERIF_SUMMARY	Four BRCA2 mutations including three novel BRCA2 frameshift mutations and one previously known BRCA2 frameshift mutation and one BRCA1 splicing mutation were identified in a group of consecutive breast cancer patients.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with response to therapy in sporadic cancer.
Literature	GENERIF_SUMMARY	Studies indicate that germline mutations in BRCA1/2 and related genes, such as PALB2, are associated with pancreatic cancer.
Literature	GENERIF_SUMMARY	Breast cancer patients who carry the Icelandic founder BRCA2 mutation have inferior long-term survival than noncarriers, but the adverse prognosis is restricted to mutation carriers with diploid
Literature	GENERIF_SUMMARY	Deleterious BRCA2 mutation in women with ovarian cancer was an independent risk factor for carboplatin hypersensitivity reactions.
Literature	GENERIF_SUMMARY	Germline mutations in two major genes, BRCA1 and BRCA2, may have an important influence on the predisposition and development of familial breast cancer.
Literature	GENERIF_SUMMARY	there is emerging evidence of relative resistance of tumors containing BRCA1 or BRCA2 mutations (or BRCAness) to taxanes.
Literature	GENERIF_SUMMARY	provides evidence that tamoxifen use is associated with a reduction in contralateral breast cancer risk for BRCA1 and BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	This article reviews current recommendations and limitations of using MR imaging of the breast to screen asymptomatic women at high risk for breast cancer.

Main Category	Category	Terms
		[review]
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	The goal of this study was to determine the prevalence of PALB2 mutations in a population of BRCA1/BRCA2 negative breast cancer patients selected from either a personal or family history of pancreatic cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutations confer a survival advantage compared with the BRCA1 mutation in BRCA-mutated epithelial ovarian cancer patients.
Literature	GENERIF_SUMMARY	Our results indicate that carriers of BRCA1/2 mutations among non-Caucasian breast cancer patients showed no enhancement in radiation sensitivity.
Literature	GENERIF_SUMMARY	PIK3CA mutations are frequently observed in BRCAX but not BRCA2-associated male breast cancer.
Literature	GENERIF_SUMMARY	The risk of finding a BRCA1/2 mutation in women 40 years or younger is comparable to women of all ages with epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with triple-negative breast cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	findings suggest a mechanism for the regulation of the nucleocytoplasmic distribution of BRCA2 and RAD51 and its impairment by a heterozygous disease-associated mutation
Literature	GENERIF_SUMMARY	The distribution of FMR1 CGG repeat numbers in BRCA mutation carriers and non-carriers is similar in contrast to previous reports.
Literature	GENERIF_SUMMARY	The first comprehensive analysis of large mutations in BRCA1 and BRCA2 genes in Poland, is reported.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with response to chemotherapy in ovarian cancer.
Literature	GENERIF_SUMMARY	found no evidence that BRCA1 and BRCA2 mutation carriers are at higher risk of natural menopause at a given age than their noncarrier relatives
Literature	GENERIF_SUMMARY	As 60%-80% of all advanced ovarian cancers are high-grade serous type, exhibiting HRD in at least 50% (BRCA1/2 mutations) future antitumor strategies may depend on identifying these defects through molecular testing
Literature	GENERIF_SUMMARY	Our data demonstrate a wide variety of BRCA mutations in a highly ethnically diverse EOC population, and confirm that EOC BRCA mutation carriers have better prognosis with longer median survival than patients with nonhereditary disease.
Literature	GENERIF_SUMMARY	Pregnancy concurrent with or after a diagnosis of breast cancer does not appear to adversely affect

Main Category	Category	Terms
		survival among BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	Carriers of BRCA1/BRCA2 mutations are exposed to higher titres of oestradiol and progesterone-known risk-factors for breast cancer.
Literature	GENERIF_SUMMARY	three different cancer susceptibility and FA proteins function in a DNA repair pathway based upon the PALB2 WD40 domain binding to RAD51C and BRCA2.
Literature	GENERIF_SUMMARY	Studies suggest that associations between ever use of oral contraceptive (OC) and ovarian and breast cancer among women who are BRCA1 or BRCA2 mutation carriers are similar to those reported for the general population.
Literature	GENERIF_SUMMARY	these data support the hypothesis that the presence of BRCA mutations does not alter the clinical outcome for hereditary breast cancer patients.
Literature	GENERIF_SUMMARY	genetic associations studies in populations of women in South Korea
Literature	GENERIF_SUMMARY	Female brca2 mutation carriers born in 1958 or later were at a significantly higher risk of developing breast cancer at a younger age (p
Literature	GENERIF_SUMMARY	BRCA2 accumulates DNA damage, which triggers checkpoint signalling and ARF activation
Literature	GENERIF_SUMMARY	new insights into the mechanisms underlying genetic instability mediated by hypoxia and BRCA involvement in sporadic breast cancers.
Literature	GENERIF_SUMMARY	BRCA1/2 mutation carriers seemed to be more at risk of neutropenia after the first cycle of the treatment in early breast cancer patients
Literature	GENERIF_SUMMARY	LUDLU-1 contains somatic mutations in TP53 and BRCA2, allelic imbalance in the expression of two cancer-associated BRCA1 germline polymorphisms and reduced transcription of a potentially endogenous PARP2 inhibitor.
Literature	GENERIF_SUMMARY	For some populations, the number of Fanconi anaemia-patients with biallelic BRCA2 disruption is smaller than that expected from the carrier frequency. [Review]
Literature	GENERIF_SUMMARY	the total mutation burden coupled with BRCA1 or BRCA2 mutations in ovarian cancer is a genomic marker of prognosis and predictor of treatment response.
Literature	GENERIF_SUMMARY	the models reveal novel aspects of cancer evolution in carriers of germline BRCA2 mutations, provide new insights into the tumour suppressive role of BRCA2
Literature	GENERIF_SUMMARY	No association between FMR1 low genotypes and BRCA1/2 mutations in a cohort of 125 Ashkenazi women.
Literature	GENERIF_SUMMARY	both BRCA1 and PALB2 are required for BRCA2-focus

Main Category	Category	Terms
		formation when the DBD is absent.
Literature	GENERIF_SUMMARY	Noncarriers remain at risk in the prospective follow-up of women who tested negative for BRCA1/2.
Literature	GENERIF_SUMMARY	BRCA1 mutation carriers conferred lower overall breast and ovarian cancer risks than reported so far, while the estimates of BRCA2 mutations were among the lowest in a Dutch cohort.
Literature	GENERIF_SUMMARY	The imbalance of DSS1 over-expression associated with BRCA2 expression could affect breast cancer development.
Literature	GENERIF_SUMMARY	Neither BRCA1 nor the BRCA2 studied variant show any significant association with the disease.
Literature	GENERIF_SUMMARY	The SIR for BRCA2 carriers was 0.82.
Literature	GENERIF_SUMMARY	CRCs may be part of the tumour spectrum associated with FANCD1/BRCA2 biallelic mutations and that the presence of such mutations should be considered in families with CRCs, even in the absence of cardinal features of FA.
Literature	GENERIF_SUMMARY	Two cancer syndrome gene variants likely to affect native translation initiation were identified by clinical genetic testing
Literature	GENERIF_SUMMARY	Studies suggest that in the majority of cases with BRCA1 and BRCA2 negative familial breast cancer the etiology is due to interactions of intermediate or low risk alleles with environmental and lifestyle factors.
Literature	GENERIF_SUMMARY	the global spectrum of BRCA2 mutations in breast cancer (Review)
Literature	GENERIF_SUMMARY	variations of the BRCA2may contribute to breast cancer susceptibility along with variations in other low penetrating genes in sporadic types of breast cancer in this cohort of the population
Literature	GENERIF_SUMMARY	Studies may implicate the BRCA2 pathway in tumorigeneis.
Literature	GENERIF_SUMMARY	Twelve pathogenic mutations were detected (25%); three in BRCA1 (c.211dupA in four families, c.5266dupC in three families and c.1504_1508delTTAAA in one family) and two novel mutations in BRCA2 (c.1313dupT in two families and c.7654dupT in two families).
Literature	GENERIF_SUMMARY	This pilot study evaluates the relation between carrying a BRCA1/2 mutation and fertility, using the level of anti-mullerian hormone (AMH), which has been previously shown to be an accurate marker of ovarian reserve and fertility potential.
Literature	GENERIF_SUMMARY	BRCA2 is cleaved into two fragments by membrane type-1 matrix metalloproteinase (MT1-MMP), and knockdown of MT1-MMP prevents the removal of BRCA2 from centrosomes during metaphase.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Report show a Fanconi anemia family with a novel mutation in the splice donor site of intron 1 of BRCA2 leading to reduced expression of a transcript that lacks part of the 5'-UTR.
Literature	GENERIF_SUMMARY	We demonstrate that circulating PE and EM are strongly and positively correlated with tissue-specific PE and EM measured in NAF and DLS among postmenopausal BRCA1/2 mutation carriers
Literature	GENERIF_SUMMARY	BRCA2 mutation is asssociated with low response to chemotherapy in breast cancer.
Literature	GENERIF_SUMMARY	our study did not find association between the three most common BRCA1 & BRCA2 mutations and endometriosis
Literature	GENERIF_SUMMARY	PCAT-1 expression produces a functional deficiency in homologous recombination through its repression of the BRCA2 tumor suppressor.
Literature	GENERIF_SUMMARY	poor prognosis of young BRCA1/2 mutation carriers with breast cancer is mainly explained by the prevalent occurrence of negative prognostic factors rather than mutation status
Literature	GENERIF_SUMMARY	This finding indicates involvement of hereditary factors in non-BRCA1/2 breast cancer families.
Literature	GENERIF_SUMMARY	PALB2 and BRCA2, in addition to their functions in D loop formation, play crucial roles in the initiation of recombination-associated DNA synthesis by Poleta-mediated DNA repair.
Literature	GENERIF_SUMMARY	There was no evidence for association between developing breast/ prostate cancer or being a BRCA1 or BRCA2 mutation carrier and telomere length.
Literature	GENERIF_SUMMARY	This study used multifactorial likelihood analysis and/or bioinformatically-directed mRNA assays to assess pathogenicity of 19 BRCA1 or BRCA2 variants.
Literature	GENERIF_SUMMARY	BRCA mutations in prostate cancer patients.[review]
Literature	GENERIF_SUMMARY	women who are positive for BRCA mutations and who are treated for stage I or II breast cancer with bilateral mastectomy are less likely to die from breast cancer than women who are treated with unilateral mastectomy.
Literature	GENERIF_SUMMARY	BRCA1/2 mutation carriers are potentially at higher risk of cardiovascular diseases due to early surgical menopause, and cardiotoxic effects of adjuvant treatment for breast cancer.
Literature	GENERIF_SUMMARY	In conclusion, our result shows that large genomic rearrangement in the BRCA1 and BRCA2 genes account for a very low proportion of BRCA mutations and does not seem like a major determinant of breast cancer susceptibility in Korean, breast cancer patients.
Literature	GENERIF_SUMMARY	Preventive oophorectomy was associated with an 80%

Main Category	Category	Terms
		reduction in the risk of ovarian, fallopian tube, or peritoneal cancer in BRCA1 or BRCA2 carriers and a 77% reduction in all-cause mortality.
Literature	GENERIF_SUMMARY	Metastatic high-grade serous carcinomas with BRCA1 deficiency exhibit characteristic morphological features that appear more sensitive and specific for BRCA1 and BRCA2 mutations than two other morphologically based prediction systems
Literature	GENERIF_SUMMARY	The article summarizes the multiple functions for BRCA2 in the preservation of chromosome stability which explain its tumour suppressive function. [review]
Literature	GENERIF_SUMMARY	This work highlights the contribution of DNA, RNA, and in silico data to assess the pathogenicity of BRCA1/2 variants of uncertain significance, which, in turn, allows more accurate genetic counseling and clinical management of the families carrying them.
Literature	GENERIF_SUMMARY	knockdown of BRCA2 using small interference RNA suppression increased the sensitivity to 5-FU of human oral cancer cell lines
Literature	GENERIF_SUMMARY	depletion of genes within the ataxia telangiectasia and Rad3 related (ATR) and BRCA1/BRCA2 pathways, chromatin modification, and transcription-coupled repair reduced the synergy between mitomycin C and MLN4924
Literature	GENERIF_SUMMARY	this study argues that tumor suppression by BRCA1 and BRCA2 originates from their fundamental role in controlling the assembly and activity of macromolecular complexes that monitor chromosome duplication, maintenance, and segregation across the cell cycle.
Literature	GENERIF_SUMMARY	By sequencing, one BRCA2 LGR, the Portuguese founder mutation c.156_157insAluYa5.
Literature	GENERIF_SUMMARY	May be associated with cancer risk in BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Mutations were most frequent in BRCA1 and BRCA2, with 84 EOC cases (3.8%) carrying a BRCA1 mutation and 94 epithelial ovarian cancer cases (4.2%) carrying a BRCA2 mutation.
Literature	GENERIF_SUMMARY	The BRCA2 p.Asn3124Ile (BRCA2 c.9371A > T) variant is a rare mutation with a damaging effect on the BRCA2 protein that is strongly associated with familial breast and ovarian cancer risk, indicating its likely pathogenic nature and clinical relevance.
Literature	GENERIF_SUMMARY	A model designed to yield optimal intervention strategies including prophylactic surgeries and breast screening for BRCA1/2 mutations carriers is presented.
Literature	GENERIF_SUMMARY	The negative interplay between Aur A/B and BRCA1/2

Main Category	Category	Terms
		inversely controls the cell proliferation, cell cycle progression, cell multinuclearity, and tetraploidization to modulate tumorigenesis.
Literature	GENERIF_SUMMARY	this work gives the first evidence of a BRCA2 germline pathogenic mutation associated with CRC risk.
Literature	GENERIF_SUMMARY	BRCA2 facilitates Rad51 phosphorylation by Plk1, which promotes genome stability.
Literature	GENERIF_SUMMARY	Data indicate that BRCA mutation status did not influence the generational difference in age at diagnosis .
Literature	GENERIF_SUMMARY	BRCA2 p.Lys3326X is associated with lung cancer.
Literature	GENERIF_SUMMARY	Findings contribute significantly to the types of germline BRCA1/BRCA2 mutations and their biological effects in Turkish women.
Literature	GENERIF_SUMMARY	results of this study suggest that homozygosity for the BRCA2 T(-1144) allele is associated with a longer disease-free interval in Chinese women with breast cancer
Literature	GENERIF_SUMMARY	TNBC is a strong predictor for the presence of a BRCA1 mutation in this population, but additional risk factors should also be evaluated to ascertain a 10 % or higher prior probability of BRCA1/2 mutation testing.
Literature	GENERIF_SUMMARY	TREX-2 complex prevents genome instability, as determined by the accumulation of gamma-H2AX and 53BP1 foci and single-cell electrophoresis in cells depleted of the TREX-2 subunits PCID2, GANP and DSS1
Literature	GENERIF_SUMMARY	BRCA2 protein is modulator of anoikis sensitivity through an evolutionarily-conserved molecular mechanism involving regulation of reactive oxygen spessies production and/or detoxification by BRCA2 during PCD processes.
Literature	GENERIF_SUMMARY	Stratification of patients and controls based on gender revealed no association between rs1799943 of the BRCA2and the risk of CVD in either gender.
Literature	GENERIF_SUMMARY	conclude that contralateral risk-reducing mastectomy is associated with improved overall survival in BRCA1/2 mutation carriers with a history of primary breast cancer
Literature	GENERIF_SUMMARY	Intentions to undergo PND or PGD do not appear to change after disclosure of BRCA results. Additional counseling for patients who have undergone BRCA testing may be warranted to educate patients about available fertility preservation options.
Literature	GENERIF_SUMMARY	Data suggest that the Fanconi anemia group F protein/ BRCA1/2 proteins pathway may be a new target to reverse adriamycin (ADR) resistance in leukemia treatment.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Exon 11 of BRCA2, an important genetic determinant for hereditary breast and ovarian cancers, was sequenced from diverse primate species including humans; found that this gene, like BRCA1, bears the surprising signature of positive selection.
Literature	GENERIF_SUMMARY	absence of BRCA/FMR1 correlations in women with ovarian cancers
Literature	GENERIF_SUMMARY	BRCA2 directly represses the expression of IFN-related genes
Literature	GENERIF_SUMMARY	BRCA2 mutation is not associated with nodal metastasis in breast cancer.
Literature	GENERIF_SUMMARY	Mutations in the BRCA2 gene is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Prospective breast cancer risks in women in the UK are high especially for BRCA2 families ascertained on the basis of high risk.
Literature	GENERIF_SUMMARY	Genetic analysis revealed 32 different pathogenic germline BRCA1/2 mutations
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 contribute to a genetic predisposition for breast cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	Loss-of-function mutations in PALB2 are an important cause of hereditary breast cancer. Our data suggest the breast-cancer risk for PALB2 mutation carriers may overlap with that for BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	The survival of men with screen-detected prostate cancer and a BRCA2 mutation is much poorer than expected.
Literature	GENERIF_SUMMARY	Five BRCA2 mutations and ten variants of unknown significance (VUS) were identified in familial and aggressive prostate cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers are at risk for multiple cancers, including contralateral breast cancer
Literature	GENERIF_SUMMARY	This review is aimed at exploring the current knowledge on epidemiological and molecular aspects of BRCA-associated (BRCA1 and BRCA2) Ovarian cancer predisposition
Literature	GENERIF_SUMMARY	BRCA2 mutated intraductal carcinoma of the prostate is common in patients with familial prostate cancer and is associated with poor outcomes.
Literature	GENERIF_SUMMARY	conclude that the woman has two potential disease-causing mutations and that predictive testing of family members should include both the RAD51C and BRCA2 mutation

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Study showed that large genomic rearrangements were found in 3.7% of the patients who had mutations in BRCA1 or BRCA2, and 7.5% of patients with mutations in BRCA1 only.
Literature	GENERIF_SUMMARY	the analysis of studies to date that have evaluated the risk of BRCA1 and BRCA2 mutations for colorectal cancer (Review)
Literature	GENERIF_SUMMARY	Data indicate that tumour suppressor RASSF1A triggers large tumor suppressor kinase 1 (LATS)-CDK2 interaction and restricts CDK2 kinase activity towards BRCA2.
Literature	GENERIF_SUMMARY	BRCA2 rs10492396 (AG vs. GG
Literature	GENERIF_SUMMARY	The observed age difference may allow a greater understanding of mechanisms associated with the differences in cancer penetrance in BRCA1/2 mutation carriers, some of which may depend on paternal origin.
Literature	GENERIF_SUMMARY	The observed rate of 3 of 5754 cases (0.052%) of head and neck cancers in BRCA-positive probands warrants further investigation into the nature of a possible linkage between germline BRCA mutations and salivary gland cancer.
Literature	GENERIF_SUMMARY	We hereby report two cases of male breast cancer with neurofibromatosis.
Literature	GENERIF_SUMMARY	BRCA2 exists as a dimer and that two oppositely oriented sets of RAD51 molecules bind the dimer RAD51 to facilitate rad51-mediated homologous-recombinational repair.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with triple-negative breast cancer.
Literature	GENERIF_SUMMARY	Our analyses suggest that BRCA mutations(BRCA1 and BRCA2) are robust predictors of outcomes in both ovarian and breast cancers and these mutations should be taken into account when devising appropriate therapeutic strategies
Literature	GENERIF_SUMMARY	Well-designed and large-scale studies are needed to substantiate the association between BRCA2 N372H polymorphism and cancer risk.
Literature	GENERIF_SUMMARY	Germ-line DNA samples from 727 unrelated probands with positive family history (521 met criteria for familial pancreatic cancer) were tested for mutations in BRCA1 and BRCA2, PALB2, and CDKN2A.
Literature	GENERIF_SUMMARY	BRCA2 polymorphisms show no detectable endometriosis risk in Indian population.
Literature	GENERIF_SUMMARY	Identification of pathogenic BRCA2 variants resulting in alternative splicing.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Our data provide initial evidence for a novel putative ovarian cancer cluster region in BRCA2 exons 13-14.
Literature	GENERIF_SUMMARY	analysis of effect of germline mutations in BRCA1 and BRCA2 on mortality in ovarian cancer patients; BRCA1/2 mutations are associated with better short-term survival, but this advantage decreases over time and, in BRCA1 carriers is eventually reversed
Literature	GENERIF_SUMMARY	targeting relevant exons in BRCA1 and BRCA2 genes allows detection of a substantial percentage of mutations in the Tunisian population.
Literature	GENERIF_SUMMARY	Mutations in BRCA1 and BRCA2 or another breast cancer gene occur in one in four African American breast cancer patients with early onset disease, family history of breast or ovarian cancer, or TNBC
Literature	GENERIF_SUMMARY	genomic instability score in discriminating nonequivalent outcomes of BRCA1/2 mutations and in predicting outcomes of ovarian cancer treated with platinum-based chemotherapy
Literature	GENERIF_SUMMARY	The study shows evidence of the modulation effect of BRCA mutation on the expression of microRNA-200c in Triple negative breast cancer progression.
Literature	GENERIF_SUMMARY	Prostate cancer patients with germline BRCA mutations had worse outcomes than noncarriers when conventionally treated with surgery or radiation therapy.
Literature	GENERIF_SUMMARY	Non-founder mutations make up between 7.2 and 13.0 % of all BRCA1 and BRCA2 mutations in Ashkenazi Jews. A wide range of mutations are present, most of which are also seen in non-AJ individuals
Literature	GENERIF_SUMMARY	Fresh insight into the mechanism by which the HMG20b-BRCA2 complex controls mitotic cell division, and implicate heterozygous HMG20b mutations affecting cytokinesis regulation in the genesis of human cancers
Literature	GENERIF_SUMMARY	BRCA1/BRCA2 mutation carriers experienced a sense of disorientation about the most appropriate way of dealing with genetic risk
Literature	GENERIF_SUMMARY	Tumor-associated inflammation density was the strongest predictor for recurrence-free survival in this series of BRCA1/2 breast cancer patients
Literature	GENERIF_SUMMARY	Germline mutations in DICER1 are rare or absent in familial breast cancer patients with BRCA1/BRCA2-negative familial breast cancers.
Literature	GENERIF_SUMMARY	Data indicate that three microRNAs hsa-miR-146a, hsa-miR-148a and hsa-miR-545 targeting BRCA1/2 and were associated with overall survival and progression-free survival in patients with wild-type BRCA1/2.
Literature	GENERIF_SUMMARY	Report roles BRCA1 status and tubual intraepithelial carcinoma on prognosis in women with high grade serous carcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The deletion c.1792delA, in exon 10, and the duplication c.5869dupA, in exon 11 of BRCA2 are novel mutations in Spanish breast cancer patients.
Literature	GENERIF_SUMMARY	Taking into account modifier factors, location of mutation might be important for the clinical management of BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	Studies indicate that each of the three neoplasm protein variants in BRCA1, BRCA2 and MSH2 would normally be reported as pathogenic based on widely accepted guidelines.
Literature	GENERIF_SUMMARY	FANCJ and BRCA2 share FANCD2's role in replication fork restart.
Literature	GENERIF_SUMMARY	BRCA2 gene mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	study identified a BRCA1 or BRCA2 germline mutation in 74 of 774 (9.6 %) triple-negative breast cancer patients; mutation prevalence was 9.3 % in Australia and was 9.9 % in Poland
Literature	GENERIF_SUMMARY	Localized correlated spectroscopy recorded significant changes in women with BRCA1 and BRCA2 gene mutations when compared with control subjects. If these changes are ultimately proven to be a premalignant stage, this method may prove useful in screening
Literature	GENERIF_SUMMARY	BRCA2 gene mutations are associated with breast cancer.
Literature	GENERIF_SUMMARY	FANCD1 plays a role in gemcitabine drug resistance in biliary tract cancer.
Literature	GENERIF_SUMMARY	BRCA2 tumors were more likely to be ER and HER2 positive than other non-BRCA familial breast cancers.
Literature	GENERIF_SUMMARY	acinar cell carcinomas show a distinct mutation pattern and often harbor somatic or germline mutations of BRCA2 and FAT genes.
Literature	GENERIF_SUMMARY	FBN1 acts downstream of Aur A and BRCA2, promoting ovarian cancer metastasis through p53 and SLUG-associated signaling.
Literature	GENERIF_SUMMARY	BRCA1 gene mutation is associated with gynecologic cancer.
Literature	GENERIF_SUMMARY	A prevalence rate of 5.6% for having deleterious BRCA1 or BRCA2 mutations among Lebanese Arab women with breast cancer
Literature	GENERIF_SUMMARY	Study shows that Jewish BRCA1/2 mutation carriers are at an increased risk for breast and pancreatic, but not prostate cancer.
Literature	GENERIF_SUMMARY	Fertility treatments are not associated with an increased risk of invasive epithelial ovarian cancer in Jewish Israeli BRCA1/2 mutation carriers.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	By binding and destabilizing a histone acetyltransferase, p300, beta-HPV 5 and 8 E6 reduce the enrichment of the transcription factor at the promoter of two genes critical to the homology dependent repair of DSBs (BRCA1 and BRCA2)
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers are not associated with worse breast cancer prognosis.
Literature	GENERIF_SUMMARY	BRCA2 gene mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	Mutations in BRCA2 gene is associated with breast cancer.
Literature	GENERIF_SUMMARY	a causative link between altered function of AURKA-HMMR-TPX2-TUBG1 and breast carcinogenesis in BRCA1/2 mutation carriers
Literature	GENERIF_SUMMARY	Two novel heterozygous mutations predicted to alter the function of PALB2 were identified (c.2014G>C, p.E672Q and c.2993G>A, p.G998E). Notably, both of these mutations co-existed in BRCA1 and BRCA2 families
Literature	GENERIF_SUMMARY	The study found a robust association between a truncating BRCA2 variant and upper aerodigestive tract cancer risk.
Literature	GENERIF_SUMMARY	Breast and ovarian cancer risks varied by type and location of BRCA1/2 mutations. With appropriate validation, these data may have implications for risk assessment and cancer prevention decision making for carriers of BRCA1 and BRCA2 mutations.
Literature	GENERIF_SUMMARY	The Russian carriers of BRCA1, BRCA2 and CHEK2 germline mutations have genetic predisposition to breast, ovarian, and colorectal cancer.
Literature	GENERIF_SUMMARY	From an analysis of the mutation spectrum, 63 BRCA1 and 90 BRCA2 different mutations, including 44 novel mutations, were identified.BRCA mutations in familial breast cancer patients is similar to that among Western cohorts
Literature	GENERIF_SUMMARY	Report autoimmune response to BRCA2 in tumors.
Literature	GENERIF_SUMMARY	our findings demonstrate that among young Black breast cancer survivors, socioeconomic factors and physician referral patterns contribute to disparities in access to BRCA testing(BRCA1 and BRCA2 mutation detection)
Literature	GENERIF_SUMMARY	A rare BRCA2 genetic variant, known as rs11571833 (K3326X), is strongly associated with squamous-cell cancers of the upper aerodigestive tract (UADT)
Literature	GENERIF_SUMMARY	Our results demonstrate in a subset of cases that decline in BRCA1 expression that may be associated with potentially compensatory increase in BRCA2 protein, which may depend on tumor grade as well as

Main Category	Category	Terms
		menopausal status.
Literature	GENERIF_SUMMARY	Women with mutations in BRCA1 and BRCA2 have different risks of developing ovarian and breast cancer, depending on the nature of the mutations
Literature	GENERIF_SUMMARY	differences in associations of common breast cancer susceptibility alleles between BRCA1 and BRCA2 carriers and the general population are explained to a large extent by differences in the prevalence of ER-positive and ER-negative tumors
Literature	GENERIF_SUMMARY	Prevalence of founder mutations in the BRCA1 and BRCA2 genes among unaffected women from the Bahamas was studied.
Literature	GENERIF_SUMMARY	In response to replication stress, BRCA2 recruits RAD51 onto nascent DNA at stalled forks, protecting nascent DNA from nucleolitic cleavage.
Literature	GENERIF_SUMMARY	Pathogenic BRCA1/2 mutations were identified in 4.6% of a large cohort of clinic-based pancreatic adenocarcinoma patients.
Literature	GENERIF_SUMMARY	analysis of novel A1/2 mutations in Polish patients with familial breast/ovarian cancer
Literature	GENERIF_SUMMARY	Men carrying genetic mutations in the BRCA gene(BRCA1 and BRCA2) have a greater risk than the general population of developing certain types of cancer, including breast, pancreatic, and prostate cancer.
Literature	GENERIF_SUMMARY	the prevalence of BRCA1 and BRCA2 mutations is high in triple-negative breast cancer patients and BRCA1/2 mutations are not restricted to young women or patients with a positive family history.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with pancreatico-biliary neoplasms.
Literature	GENERIF_SUMMARY	IGF1 level is significantly lower in cancer-free BRCA mutation carriers versus BRCA-negative controls, and there is a potential association between estradiol and IGF1 in cancer-free BRCA mutation carriers
Literature	GENERIF_SUMMARY	BRCA pathway germline mutation (in BRCA1, BRCA2 or BARD1 genes) was present in 10 of the 42 investigated triple-negative breast cancer patients rendering them susceptible to homologous recombination deficiency.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with familial breast and ovarian cancer syndrome.
Literature	GENERIF_SUMMARY	Breast and ovarian cancer predisposition due to de novo BRCA1 and BRCA2 mutations
Literature	GENERIF_SUMMARY	BRCA mutations that survive selection pressure seem to provide significant fertility advantages.

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Literature	GENERIF_SUMMARY	BRCA2 Arg372His polymorphism is not associated with ovarian cancer risk.
Literature	GENERIF_SUMMARY	by targeting RPA and mimicking DNA, DSS1 functions with BRCA2 in a two-component homologous recombination mediator complex in genome maintenance and tumor suppression
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian cancers.
Literature	GENERIF_SUMMARY	Did not observe a significant decline in mortality associated with oophorectomy in breast cancer patients with BRCA2 mutations. b
Literature	GENERIF_SUMMARY	Mutation profile of the BRCA1/2 genes in breast cancer Bulgarian patients show 13 disease causing mutations with 4 new. Mutations in BRCA1 gene were found in 14% and in BRCA2, in 5.5% of the cases. The inherited predisposition was identified in 19.5%.
Literature	GENERIF_SUMMARY	the current pathogenic mutation spectrum of the BRCA2 gene in patients with breast cancer in various Asian populations (review).
Literature	GENERIF_SUMMARY	Breast cancer risk for Korean women with germline mutations in BRCA2.
Literature	GENERIF_SUMMARY	BRCA2 Mutation is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	Mutation in the BRCA2 gene is associated with Breast and/or Ovarian Cancers.
Literature	GENERIF_SUMMARY	BRCA2 single nucleotide polymorphism is associated with ontralateral breast cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	BRCA1 or BRCA2 overexpression is a significant predictive factor for biochemical recurrence (BCR), in prostate cancer (PC) patients.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with risk of breast and ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers demonstrated normal ovarian response in IVF cycles.
Literature	GENERIF_SUMMARY	BRCA2-mutated prostate cancer patients have an adverse prognosis and poor outcome compared to non-mutated.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with invasive breast cancer.
Literature	GENERIF_SUMMARY	cyclin D1 and other cyclins such as cyclin A regulates DNA integrity through RAD51 interaction with the BRCA2 C-terminal domain
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with Korean ovarian cancer.

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Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian Neoplasms.
Literature	GENERIF_SUMMARY	In this unselected Japanese population, approximately 13% of the cases with ovarian cancer appeared to be associated with an inherited risk, BRCA1/2 mutation, regardless of a family history.
Literature	GENERIF_SUMMARY	BRCA2 mutations are associated with increased plasma levels of thrombo-coagulating-related proteins, which are independent to breast cancer development
Literature	GENERIF_SUMMARY	The BRCA2 rs11571833, an A > T transversion in the final exon is the risk allele for the development of a familial breast cancer.
Literature	GENERIF_SUMMARY	Risk assessment for healthy BRCA2 mutation carriers should consider the ages of breast cancers diagnosed in first-degree relatives
Literature	GENERIF_SUMMARY	In this cohort of BRCA1/2-associated breast cancer (BC) patients, no association between radiotherapy for primary BC and risk of (BC) contralateral breast cancer was observed in the total group, nor in the patients irradiated before the age of 40 years
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with Pancreatic Cancer.
Literature	GENERIF_SUMMARY	There was no significant difference in disease-free survival between BRCA2 mutation carriers and patients with sporadic disease. (Review, meta-analysis)
Literature	GENERIF_SUMMARY	the N372H polymorphism is associated with susceptibility of ovarian cancer (Meta-Analysis)
Literature	GENERIF_SUMMARY	BRCA1-associated DCIS did not express ER, PR or HER2, while most BRCA2-associated DCIS did express ER and PR. BRCA1- as well as BRCA2-associated DCIS had expression of HER3 and C-MET
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with perturbation in telomere biology resulting in breast cancer.
Literature	GENERIF_SUMMARY	Our results provide novel insights regarding the physician-patient interaction and the organizational aspects of the health-care system that may significantly impact the cancer screening practices of BRCA1/2 noncarriers.
Literature	GENERIF_SUMMARY	Data suggest that, similar to Caucasians, a significant proportion of Asian ovarian cancer was attributed to germline mutations in BRCA1 and to a lesser extent in BRCA2.
Literature	GENERIF_SUMMARY	DNA from 120 subjects was used to sequence the entirety of BRCA1 and BRCA2 coding regions and splice sites, and validate pathogenic mutations, with a total material cost of
Literature	GENERIF_SUMMARY	Recognition of different invasion patterns of metastatic extrauterine high grade serous carcinoma with BRCA germline mutations has prognostic implications.

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Literature	GENERIF_SUMMARY	The aim of this study was to determine the frequency and spectrum of germline mutations in BRCA1, BRCA2 and PALB2 and to evaluate the presence of the CHEK2 c.1100delC allele in these patients.
Literature	GENERIF_SUMMARY	Using human and murine cells defective in FANCD2 or FANCA and primary bone marrow cells derived from FANCD2 deficient mice, we show that the FA pathway removes R loops and that many DNA breaks accumulated in FA cells are R loop-dependent.
Literature	GENERIF_SUMMARY	The K3326X variant of BRCA2 is associated with risk of developing breast and ovarian cancers independent of other pathogenic variants in BRCA2.
Literature	GENERIF_SUMMARY	Studies suggest that the clinical management of cancer risk in breast cancer 1/2 proteins (BRCA1 and BRCA2) mutation carriers is complex and should consider patient preferences.
Literature	GENERIF_SUMMARY	study to estimate the carrier frequency of BRCA1 185delAG and BRCA2 6174delT mutations in eastern Indian breast cancer patients; findings suggest that these BRCA mutations may not have a strong recurrent effect on breast cancer among the eastern Indian populations
Literature	GENERIF_SUMMARY	Osteoprotegerin (OPG), The Endogenous Inhibitor of Receptor Activator of NF-kappaB Ligand (RANKL), is Dysregulated in BRCA Mutation Carriers
Literature	GENERIF_SUMMARY	The strongest predictors are the BRCA2 mutations, and the highly penetrant G84E mutation in HOXB13
Literature	GENERIF_SUMMARY	There was no difference in overall response rate, progression-free survival and overall survival according to BRCA 1-2 status nor any association between single-nucleotide polymorphisms of genes involved in DNA repair and nucleotide excision repair machinery and response to trabectedin was reported.
Literature	GENERIF_SUMMARY	Treatment for infertility does not significantly increase the risk of ovarian cancer among women with a BRCA2 mutation.
Literature	GENERIF_SUMMARY	BRCA2 Mutation is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	three patients had biallelic inactivation of BRCA2, a tumor suppressor gene critical for homologous DNA repair. two had germline BRCA2 mutations. The third patient had somatic BRCA2 homozygous copy loss. Biallelic BRCA2 inactivation in Metastatic Castration-resistant Prostate Cancer warrants further exploration as a predictive biomarker for sensitivity to platinum c
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with Familial Pancreatic Cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Hypermethylation and low Expression of BRCA2 is

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		associated with Epithelial Ovarian Cancers
Literature	GENERIF_SUMMARY	Data indicate that pathogenic mutations were found in 13/47 (28%) ovarian cancers
Literature	GENERIF_SUMMARY	G4-quadruplex stabilizing drugs target tumors with BRCA1/2 deficiencies.
Literature	GENERIF_SUMMARY	39 were BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	BRCA2 protein expression at the cell membrane and lack of C-terminal expression in the cytoplasm were associated with a reduced risk of rapidly fatal prostate cancer
Literature	GENERIF_SUMMARY	The RNA effect of 28 BRCA1 and BRCA2 VUS identified in our set of HBOCS families.
Literature	GENERIF_SUMMARY	One rare variant in BRCA2 3'UTR was identified in 716 breast cancer cases and 619 controls. Identified variant gives no convincing evidence of potential pathogenicity.
Literature	GENERIF_SUMMARY	Germline mutation in BRCA2 gene is associated with familial breast cancer.
Literature	GENERIF_SUMMARY	Founder mutations in BRCA2 contribute to an increased risk of ovarian and breast cancer in the western Danish population.
Literature	GENERIF_SUMMARY	Results show that XPG partners with BRCA1 and BRCA2 to maintain genomic stability through homologous recombination, and its loss causes DNA breaks, chromosome aberrations, and replication fork stalling.
Literature	GENERIF_SUMMARY	Lung adenocarcinomas in both male and female patients were associated with (a) genotypic polymorphisms of FANCC and FANCD1.
Literature	GENERIF_SUMMARY	BRCA2 mutations strongly increase the in vitro- and in vivo-sensitivity of cancer cells towards TRAIL-R-mediated apoptosis.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations accounted for a considerable proportion of hereditary breast/ovarian cancer patients from eastern China and the spectrum of the mutations of these two genes exhibited some unique features
Literature	GENERIF_SUMMARY	BRCA1/2 male breast cancers display distinct pathologic characteristics compared with BRCA1/2 female breast cancers , and we identified a specific BRCA2-associated MBC phenotype characterised by a variable suggesting greater biological aggressiveness
Literature	GENERIF_SUMMARY	BRCA2 germ-line mutation is associated with neoplasm recurrence in breast cancer patients.
Literature	GENERIF_SUMMARY	Rates of BRCA1 and BRCA2 mutation testing are increasing in young women with breast cancer. Given that knowledge and concern about genetic risk

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		influence surgical decisions and may affect systemic therapy trial eligibility, all young women with breast cancer should be counseled and offered genetic testing, consistent with the National Comprehensive Cancer Network guidelines.
Literature	GENERIF_SUMMARY	The number of patients with uninformative BRCA genetic test(BRCA1 and BRCA2) results in this study is typical of reported high-risk screening populations. Next-generation sequencing has ushered in a new era of broad-spectrum testing for the growing list of potential genetic etiologies for breast cancer beyond BRCA
Literature	GENERIF_SUMMARY	The study concluded that true HER2-amplified breast cancers are rare amongst BRCA1 mutation carriers and are less common in BRCA2 than background rates
Literature	GENERIF_SUMMARY	we generated a Brca2 knock-in mouse model lacking exons 4-7 and demonstrated that these exons are dispensable for viability as well as tumor-free survival. This study provides the first in vivo evidence of the functional significance of a minor transcript of BRCA2 that can play a major role in the survival of humans who are homozygous for a clearly pathogenic mutation.
Literature	GENERIF_SUMMARY	The overall BRCA1/2 mutation prevalence was 24.0% (95% CI 23.4% to 24.6%).
Literature	GENERIF_SUMMARY	High BRCA2 expression is associated with drug resistance in ovarian cancer.
Literature	GENERIF_SUMMARY	Durable responses to olaparib were observed in patients with relapsed germline BRCA1/2 mutation ovarian cancer who had received >/=3 lines of prior chemotherapy
Literature	GENERIF_SUMMARY	BRCA2 protein stimulates DMC1-mediated DNA strand exchange between RPA-ssDNA complexes and duplex DNA, thus identifying BRCA2 as a mediator of DMC1 recombination function.
Literature	GENERIF_SUMMARY	The rs144848 mutation of BRCA2 may have a protective effect against lung cancer among women
Literature	GENERIF_SUMMARY	BRCA1/2 mutation carriers who have been treated with PS have a substantially reduced breast cancer incidence and mortality.
Literature	GENERIF_SUMMARY	Genomic profiling of the tumors showed specific alterations associated to BRCA1 or 2 mutation status, and BRCA1 expression in the tumors, affecting relevant cellular processes. Similar cellular functions were found affected in BRCA1 not expressing and BRCA1 or 2 mutated tumors.
Literature	GENERIF_SUMMARY	Study suggests an increased risk of leukaemia in women with a BRCA2 mutation who receive chemotherapy for breast cancer.
Literature	GENERIF_SUMMARY	Comparison of male breast cancer tumors with female breast cancer tumors demonstrates that male breast

Main Category	Category	Terms
		cancer tumors arise more frequently from BRCA2 mutations and less frequently from BRCA1 mutations than female breast cancer tumors.
Literature	GENERIF_SUMMARY	Structure-activity relationship of the peptide binding-motif mediating the BRCA2
Literature	GENERIF_SUMMARY	Pre-test genetic counselling prior to BRCA1/2 testing.
Literature	GENERIF_SUMMARY	Findings show eight distinct pathogenic mutations in eight unrelated families and suggest that BRCA1/2 mutations are responsible for a significant proportion of hereditary and sporadic breast cancer among Algerian young women.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast and ovarian cancer.
Literature	GENERIF_SUMMARY	that BRCA1/2 multicenter mutation testing is well feasible in formalin-fixed and paraffin-embedded tumor tissue
Literature	GENERIF_SUMMARY	Reduction in BRCC2 expression induced by miR-603 was responsible for the enhanced colony formation and proliferative ability noted in the MG-63 and U2OS cells.
Literature	GENERIF_SUMMARY	In this review, we will discuss the emerging importance of somatic and epigenetic alterations of BRCA- and BRCA-related genes(BRCA1 and BRCA2) in the management of ovarian cancer.
Literature	GENERIF_SUMMARY	CDNA representing BRCA2 alternate splice sites was amplified.
Literature	GENERIF_SUMMARY	Association of BRCA2 single nucleotide polymorphisms with the breast cancer risk in the Bangladeshi population
Literature	GENERIF_SUMMARY	Germline mutation in BRCA2 gene is associated with melanoma.
Literature	GENERIF_SUMMARY	Study provides evidence that the prevalence of BRCA1 and BRCA2 mutations is high in triple-negative breast cancer patients.
Literature	GENERIF_SUMMARY	BRCA2 Mutations are associated with Aggressive Prostate Tumors.
Literature	GENERIF_SUMMARY	Germline mutations in BRCA2 are not associated with reduced ovarian reserve as measured by Anti-Mullerian hormone.
Literature	GENERIF_SUMMARY	Sixteen VUS from 51 BRCA1 variants of uncertain significance (VUS) carriers and 28 VUS from 62 BRCA2 VUS carriers were analyzed. There was a slight agreement between the two analyses, with a kappa value of 0.14 (95% confidence interval (CI) = -0.34 to 0.62) for the BRCA1 VUS and a kappa value of 0.17 (95% CI = -0.10 to 0.49) for the BRCA2 VUS
Literature	GENERIF_SUMMARY	Data show that 11 breast cancer patients were mutated in BRCA1/2 genes, and 29 were not associated to

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		BRCA1/2 mutations.
Literature	GENERIF_SUMMARY	By monitoring DNA double-strand breaks mis-repair using a sensitive bioassay, it was found that depletion of homologous recombination proteins, including BRCA2, BRCA1 or RPA1, resulted in a distinct mutational signature associated with significant increases in break-induced mutation frequencies.
Literature	GENERIF_SUMMARY	ER-expression appears to be an effect modifier in patients with BRCA1 mutations, but not among those with BRCA2 mutations
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	The heterogeneity of the detected mutations confirms the necessity of simultaneous analysis of BRCA1/2 genes in all patients diagnosed with serous ovarian carcinoma. Moreover, the use of tumor tissue for mutational analysis allowed the detection of both somatic and germline BRCA1/2 mutations.
Literature	GENERIF_SUMMARY	the primary goal of this study was to use multiplex PCR with morphological and immunohistochemical methods for the detection of BRCA mutation(BRCA1 or BRCA2), mtDNA 4977 deletion, and HPV genotyping in familial and nonfamilial breast cancers
Literature	GENERIF_SUMMARY	In a population without strong germline founder mutations, the likelihood of a BRCA1/2 mutation found in a breast carcinoma being somatic was approximately 1/3 and germline 2/3. This may have implications for treatment and genetic counseling
Literature	GENERIF_SUMMARY	These findings suggest that R2787H variant of BRCA2 could have potential functional impact.
Literature	GENERIF_SUMMARY	BRCA2 Mutation is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	In this review, BRCA2 mutation carriers are more likely to have poor risk of prostate cancer and exhibit worse outcomes.
Literature	GENERIF_SUMMARY	Our aim was to set up a cost-effective and low time-consuming BRCA1-BRCA2 mutation detection workflow using the Ion Torrent PGM technology. A retrospective cohort of 40 patients with familial breast/ovarian cancer previously tested by Sanger sequencing and a prospective cohort of 72 patients (validation set) were analyzed.
Literature	GENERIF_SUMMARY	Using Kaplan-Meier analysis, we assessed rates of BRCA testing over time. We found substantial differences in testing rates according to year of diagnosis; for example, only 35% of women who were age 40 years or younger when they received a breast cancer diagnosis in 2005 had undergone BRCA testing by 5 years after their surgical date
Literature	GENERIF_SUMMARY	radiosensitization was at least partly due to the induction of apoptosis, which was only observed in the RKO cells and which may have been induced by BRCA2

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		degradation or different types of chromosomal aberrations.
Literature	GENERIF_SUMMARY	For BRCA1 mutation carriers, cumulative estimates of developing cancer by age 70 vary by population, ranging from 40% to 85% for breast cancer and 10% to 59% for ovarian cancer. For BRCA2 mutation carriers, cumulative estimates of developing cancer are 45% to 57% for breast cancer and 11% to 18% for ovarian cancer.
Literature	GENERIF_SUMMARY	These results reveal a synthetic lethal relationship between FANCD2 and BRCA1/2.
Literature	GENERIF_SUMMARY	results suggests that BRCA1 mutation carriers with breast cancer are more likely to have P53 overexpression compared with BRCA2 mutation carriers and non-carriers (Meta-analysis).
Literature	GENERIF_SUMMARY	No association between the thyroid hormones and breast cancer in BRCA1/2-mutation carriers was found
Literature	GENERIF_SUMMARY	have confirmed previous reports of an association between breast, ovarian, and pancreatic cancers with BRCA1 and 2 mutations
Literature	GENERIF_SUMMARY	Of note is the recent U.S. Food and Drug Administration breakthrough therapy designation of olaparib for the treatment of BRCA1/2- or ATM-mutated metastatic castration-resistant prostate cancer. The implications of this new knowledge for clinical practice now and in the future are discussed.
Literature	GENERIF_SUMMARY	miR-151-5p, targeting chromatin remodeler SMARCA5, may act as a useful biomarker for BRCA-related and sporadic breast cancers.
Literature	GENERIF_SUMMARY	Results show the overall prevalence of BRCA1/2 mutations at 3.9 % in the current large cohort of Chinese women with breast cancer, which is comparable with previous findings. Also, 41 % of the BRCA1/2 mutations detected have not been previously reported, suggesting that these mutations could be specific for Chinese women.
Literature	GENERIF_SUMMARY	uptake of risk-reducing salpingo-oophorectomy among BRCA1/2 mutation carriers who were counseled at our Family Cancer Clinic is extremely high
Literature	GENERIF_SUMMARY	These results suggest that cytoplasmic dynein 1 binds to BRCA2 through the latter's centrosomal localization signal and BRCA2 mediates the cohesion between centrosomes during the S phase, potentially serving as a cell-cycle checkpoint.
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers are at increased risk developing insulin resistance and type-II diabetes mellitus; have an altered expression of circulating proteins associated with venous thrombo-embolism (Review)
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers one region, up to 45 kb

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		upstream of BNC2, and containing 100 correlated SNPs was identified as candidate causal
Literature	GENERIF_SUMMARY	Study have shown that mutations in BRCA1, BRCA2, and PALB2 account for more than 10 % of breast cancer in Trinidad and Tobago. 25 different mutations identified; of these, four mutations were seen in two patients each.
Literature	GENERIF_SUMMARY	Patientswith BRCA1/2 mutation had longer OS than thosewith BRCA1/2wild type. Patients with BRCA1/2 mutation and BRCA1/2 VUS displayed similar prognoses.
Literature	GENERIF_SUMMARY	we describe a genetic approach to examine the functional significance of the interaction between BRCA2 and PALB2 by generating a knock-in mouse model of Brca2 carrying a single amino acid change (Gly25Arg, Brca2G25R) that disrupts this interaction. In addition, we have combined Brca2G25R homozygosity as well as hemizygosity with Palb2 and Trp53 heterozygosity .
Literature	GENERIF_SUMMARY	BRCA2 c.68-7T>A pathogenic mutation from Norwegian breast or ovarian cancer cohort
Literature	GENERIF_SUMMARY	Increasing evidence on the molecular role of the BRCA2 protein in the homologous recombination of DNA damages suggest that BRCA2-related PDAC are sensitive to agents causing DNA cross-linking damage, such as platinum salts, and treatments targeting rescue DNA repair pathways, such as poly(ADP-ribose) polymerase inhibitors that are currently under investigation.[review]
Literature	GENERIF_SUMMARY	the purified RAD51 G151D protein directly and significantly enhances DNA strand exchange activity in the presence of RPA. In concordance with this result, co-incubation of G151D with BRCA2 resulted in a much higher level of strand-exchange activity compared to WT RAD51. Strikingly, the RAD51 G151D variant confers resistance to multiple DNA damaging agents, including ionizing radiation, mitomycin C, and doxorubicin.
Literature	GENERIF_SUMMARY	In the very high-risk Bladder Cancer patients , several genes had a higher frequency of mutations than reported in the The Cancer Genome Atlas database, including BRCA2 . Mutation associations with receipt of neoadjuvant chemotherapy, nodal involvement, metastatic disease development, and survival were analyzed.
Literature	GENERIF_SUMMARY	The current model places BRCA2 as a central regulator of genome stability by repairing DNA double strand breaks and limiting replication stress. [review]
Literature	GENERIF_SUMMARY	A patient with an ampulla of Vater carcinoma was incidentally found to carry the BRCA2 c.156_157insAlu mutation. Further testing of a consecutive series of additional 15 ampullary carcinomas for BRCA1/BRCA2 mutations using a combination of direct founder

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		mutation testing and full gene analysis with next generation sequencing. BRCA2 mutations were observed with a frequency of 14.3% in ampulla of Vater carcinomas.
Literature	GENERIF_SUMMARY	WTIP interacts with BRCA2 and might be responsible for BRCA2 centrosome localization in cervical cancer cell.
Literature	GENERIF_SUMMARY	BRCA2 germline mutation is associated with triple-negative breast cancer.
Literature	GENERIF_SUMMARY	We examined the intratumor variability of BRCA1/2 mutation, BRCA1 promoter methylation status, and the overall HRD score in small samples from different areas of the same cancer. We observed highly concordant HRD scores across different regions of the same cancer suggesting that HRD, when present, is a homogeneous feature of primary breast cancers
Literature	GENERIF_SUMMARY	This is the first reported case of adrenocortical carcinoma associated with a BRCA2 germline mutation.
Literature	GENERIF_SUMMARY	region in the N terminus exhibits DNA binding activity and promotes RAD51-mediated homologous recombination
Literature	GENERIF_SUMMARY	BRCA2 and CHEK2 play an important role in the genetic susceptibility to urinary tract cancers.
Literature	GENERIF_SUMMARY	study demonstrates that the individual and combined expression patterns of the DDR molecules PARP1, gammaH2AX, BRCA1, and BRCA2 could be predictive of the prognosis of STS patients and suggests that controlling the activity of these DDR molecules could be employed in new therapeutic stratagems for the treatment of STS
Literature	GENERIF_SUMMARY	These results suggested that LIMD1 is a novel BRCA2-interacting protein and is involved in the centrosome localization of BRCA2 and suppression of LIMD1, causing abnormal cell division in EC cells.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 mutations were associated with poor OS in patients with breast cancer.
Literature	GENERIF_SUMMARY	Embryos carrying paternally inherited BRCA1 or 2 pathogenic variants were shown to develop more slowly compared with the embryos with maternally inherited BRCA1 or 2 pathogenic mutations.
Literature	GENERIF_SUMMARY	In patients with ovarian cancer, BRCA mutation, irrespective of its subtypes (BRCA1/2, BRCA1 and BRCA2) carriers had better overall survival and progression-free survival than non-carriers.[meta-analysis]
Literature	GENERIF_SUMMARY	soluble alpha-fetoprotein is a marker for Fanconi anemia; FANC proteins form multi-protein complexes with BRCA proteins and are involved in homologous DNA repair [review]

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Literature	GENERIF_SUMMARY	Letter
Literature	GENERIF_SUMMARY	60% of BRCA1/2 carriers, but only in 8% of non-carriers. These findings show that BRCA1/2 mutations overwhelmingly lead to a contralateral disease when breast cancer is diagnosed at early age.
Literature	GENERIF_SUMMARY	No association was found between genetic ancestry and BRCA1/2 mutational status in a Brazilian population sample at-risk for hereditary breast cancer.
Literature	GENERIF_SUMMARY	IMPACT data were consistent with increased risks of onset among BRCA1 and BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	BRCA1/2-associated EOC patients with a previous BC have a worse outcome.
Literature	GENERIF_SUMMARY	Results show that amplification of targetable oncogenes is frequent in BRCA1/2 mutation negative MBCs and may identify MBC subsets characterized by aggressive phenotype that may benefit from potential targeted therapeutic approaches.
Literature	GENERIF_SUMMARY	we show that this combined approach allows the rapid and reliable detection of both germline and somatic aberrations affecting BRCA1 and BRCA2 in DNA derived from FFPE OCs, enabling improved hereditary cancer risk assessment and clinical treatment of ovarian cancer patients.
Literature	GENERIF_SUMMARY	Report reliable detection of BRCA1 and BRCA2 mutations in fresh frozen ovarian and breast tumor samples.
Literature	GENERIF_SUMMARY	polygenic risk scores (PRS) may be used to refine risk assessment for women at increased familial risk who test negative/have low likelihood of BRCA1/2 mutations. They may alter the recommended prevention strategy for many women attending family history clinics.
Literature	GENERIF_SUMMARY	Findings suggest that the miRNA-binding SNPs in BRCA1/BRCA2 and their interaction with reproductive factors might contribute to BC risk, and miR-627 might down-regulate BRCA2 expression in MCF-7 and MDA-MB-231 cells.
Literature	GENERIF_SUMMARY	Women at high familial risk and BRCA1/2 mutation carries develop tumors with different clinical-pathological characteristics and, consequently, are influenced by different protective and risk factors.
Literature	GENERIF_SUMMARY	This is the largest study in Cyprus evaluating the prevalence and spectrum of BRCA1/2 mutations. Thirteen percent of the patients tested were positive for deleterious mutations which are within the range reported in other countries
Literature	GENERIF_SUMMARY	Data indicate a significant inverse relationship between plasma osteoprotegerin (OPG) levels and breast cancer risk among women with an inherited BRCA1 or BRCA2 mutation.

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Literature	GENERIF_SUMMARY	Statistically significant correlations were identified between breast cancer and 3 not described previously single nucleotide polymorphisms of homologous recombination repair genes BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	BRCA1/2 mutations are common in Taiwanese patients with ovarian carcinoma and occur at mutation rates similar to those seen in other ethnic groups.
Literature	GENERIF_SUMMARY	Study showed that the prevalence of BRCA1/2 germline mutations was 19% in epithelial ovarian cancer, mostly detected in different gene locations. Two novel frameshift mutations were identified, one in BRCA1 and one in BRCA2, as well as two large deletions.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 pathogenic mutations and their prediction have been characterized in the Italian breast and ovarian cancer pedigrees using two genetic models.
Literature	GENERIF_SUMMARY	ubiquitination of RAD51 hinders RAD51-BRCA2 interaction, while deubiquitination of RAD51 facilitates RAD51-BRCA2 binding and RAD51 recruitment and thus is critical for proper homologous recombination
Literature	GENERIF_SUMMARY	Studies indicate almost 20% of women with high-grade serous ovarian cancer are now recognized to harbour a germ line BRCA mutation, and of these, >40% might not have a family history of cancer and would not have received BRCA1 and BRCA2 testing.
Literature	GENERIF_SUMMARY	The Single-molecule molecular inversion probes (smMIP)-based genetic testing enables automated and reliable analysis of the coding sequences of BRCA2.
Literature	GENERIF_SUMMARY	Anticancer activity was observed in BC and OC and in gBRCAm (BRCA1 or BRCA2)and gBRCA wild-type (gBRCAwt) patient. BKM120 and olaparib can be co-administered, but the combination requires attenuation of the BKM120 dose. Clinical benefit was observed in both gBRCAm and gBRCAwt pts.
Literature	GENERIF_SUMMARY	Prevalence of pathogenic and likely pathogenic variants in the hotspots regions of BRCA2 was 23 and 6.3 % respectively in this cohort
Literature	GENERIF_SUMMARY	data from our systematic review and meta-analysis suggests a causal relationship between germline BRCA1/2-pathogenic mutations and the development of Uterine serous carcinoma.
Literature	GENERIF_SUMMARY	BRCA2-truncated protein resulted from the frame shift mutation was found to lose two putative nuclear localization signals and a Rad51-binding motif in the extreme C-terminal region by bioinformatic prediction
Literature	GENERIF_SUMMARY	we saw no increased risk of cardiotoxicity among breast cancer patients with BRCA1 and/or BRCA2 gene mutations treated with standard doses of anthracycline compared to the general population.
Literature	GENERIF_SUMMARY	Evidence has been provided that the majority of the

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		Cas9-induced single nicks at the target DNA strand rely on RAD51 and BRCA2 for efficient and scar-less DNA repair.
Literature	GENERIF_SUMMARY	It has been shown that BRCA2-mutant prostate cancer cells harbor increased genomic instability and a mutational profile that more closely resembles metastatic than localized disease.
Literature	GENERIF_SUMMARY	These data demonstrate the added benefit of combining Topo I inhibition mediated by IMMU-132 with synthetic lethality provided by PARPi in triple-negative breast cancer, regardless of BRCA1/2 status, thus supporting the rationale for such a combination clinically
Literature	GENERIF_SUMMARY	the present systematic review collects and analyses the role of BRCA2 alterations to be used in early diagnosis of an inherited syndrome associated with familiar cancer and as a prognostic and predictive biomarker for the management of pancreatic cancer patients.
Literature	GENERIF_SUMMARY	Data suggest that RAD51, BRCA2, and BRCA1 promote stability of nascent DNA at replication forks; RAD51 prevents MRE11 nuclease-mediated degradation of nascent ssDNA; BRCA2 displaces RPA (replication protein A) complex by recruiting RAD51 to protect ssDNA; BRCA1 promotes repair foci following DNA damage and is essential for cell survival. (BRCA = breast cancer recombination protein; RAD51 = Rad51 recombinase) [REVIEW]
Literature	GENERIF_SUMMARY	In this study, we evaluated this novel mechanism of drug resistance in newly diagnosed, early-stage BRCA1/2-mutant breast cancer patients who had a poor response to platinum-based neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	We found an approximately approximately 3 fold increased rate of high RS among BRCA1 and 2 carriers with ER positive breast cancer (BC)compared with the general BC population.
Literature	GENERIF_SUMMARY	Architectural plasticity of human BRCA2-RPA-RAD51 complexes in DNA break repair has been described.
Literature	GENERIF_SUMMARY	We recommend BRCA1/2 genetic testing to all Chinese women diagnosed with EOC to identify HBOC families, to provide genetic counseling and clinical management for at-risk relatives. Mutation carriers may also benefit from PARP-targeted therapies.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with pancreatic cancer.
Literature	GENERIF_SUMMARY	BRCA2 mutation is associated with breast cancer.
Literature	GENERIF_SUMMARY	We present two cases of black South African patients with FA diagnosed with biallelic BRCA2 mutations and discuss the phenotypic consequences and implications for them and their families
Literature	GENERIF_SUMMARY	Even if a BRCA2 mutation is already identified within a

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		family, the presence of early onset breast cancer affected non-carriers hampers accurate risk estimates for both mutation carriers and other negative family members.
Literature	GENERIF_SUMMARY	In this study, Multiplex ligation-dependent probe amplification (MLPA) assay was used for detection of large rearrangements of BRCA1 and BRCA2 genes in 36 unrelated high-risk breast/ovarian cancer patients negative for BRCA1/2 point mutations. Overall BRCA1/2 LGRs prevalence among high-risk Tunisian patients was 5.5%.
Literature	GENERIF_SUMMARY	In the olaparib LT group, 244 genetic alterations were detected, with TP53, BRCA1, and BRCA2 mutations being most common (90%, 25%, and 35%, respectively). BRCA2 mutations were enriched among the LT responders. BRCA methylation was not associated with response duration. High myriad HRD score (>42) and/or BRCA1/2 mutation was associated with LT response to olaparib
Literature	GENERIF_SUMMARY	We studied effect of food on ABT-767 bioavailability in subjects with advanced BRCA1 or BRCA2 mutated solid tumors or high grade serous ovarian, primary peritoneal or fallopian tube cancer
Literature	GENERIF_SUMMARY	Rucaparib was tolerable and had activity in patients with platinum-sensitive germline BRCA1/2-mutated high-grade ovarian carcinoma
Literature	GENERIF_SUMMARY	the study included all BRCA1/2 germline missense mutations identified in breast and ovarian cancer patients; the Asian population was the most affected by BRCA2 mutant patterns (Systematic Review)
Literature	GENERIF_SUMMARY	Patients with Fanconi anemia have higher incidences of ectopic neurohypophysis, adenohypophysis hypoplasia, platybasia and other midline central nervous system skull base posterior fossa abnormalities than age- and sex-matched controls. Patients with posterior fossa abnormalities, including pontocerebellar hypoplasia, are more likely to have biallelic BRCA2 mutations.
Literature	GENERIF_SUMMARY	BRCA2 reversion mutation is associated with resistance to PARP inhibitor-based therapy in a pancreatic ductal adenocarcinoma.
Literature	GENERIF_SUMMARY	This study shows that BRCA mutation carriers could be frequently identified among breast cancer patients with multiple risk factors. Importantly, we established an NGS-based pipeline for BRCA1/2 testing in clinical practice and strongly suggest that breast cancer patients of premier- and moderate-grade risks receive BRCA1/2 mutations testing in China
Literature	GENERIF_SUMMARY	Risk-reducing bilateral salpingo-oophorectomy is currently the most effective method for reducing the risk of ovarian cancer in BRCA mutation (BRCA1 and BRCA2) carriers.
Literature	GENERIF_SUMMARY	Results indicate that aberrant splicing is especially

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		prevalent in breast cancer 2 protein (BRCA2) exons 17 and 18.
Literature	GENERIF_SUMMARY	Breast cancers in BRCA1 mutation carriers are associated with more aggressive tumor characteristics compared to BRCA2 and are less well seen on mammography. Mammography rarely identified cancers not visible on MRI. Thus, the omission of mammography in BRCA1 mutation carriers screened with MRI can be considered.
Literature	GENERIF_SUMMARY	22 different BRCA2 uncertain significance were detected and p.M784V in BRCA2 was the most common uncertain significance.
Literature	GENERIF_SUMMARY	BRCA1- and BRCA2- (BRCA)-associated tumors respond to both platinum compounds and PARP inhibitors (PARPi). Potential synergism for carboplatin and the PARPi veliparib was demonstrated in preclinical studies. In a companion phase II trial, veliparib was demonstrated to be active as a single agent in both BRCA1- and BRCA2-associated MBC patients
Literature	GENERIF_SUMMARY	Patients with BRCA2 mutation were almost 25 times more likely to have chronic pancreatitis-like changes compared with sex-matched controls.
Literature	GENERIF_SUMMARY	Data show that there are large differences in the absolute cancer risks between breast cancer 1, early onset protein (BRCA1) and breast cancer 2, early onset protein (BRCA2) mutation carriers with higher vs lower values of the polygenic risk scores (PRS).
Literature	GENERIF_SUMMARY	The results of Ion PGM with OTG-snpcaller, a pipeline based on Torrent mapping alignment program and Genome Analysis Toolkit, from 75 clinical samples and 14 reference DNA samples were compared with Sanger sequencing for BRCA1/BRCA2.
Literature	GENERIF_SUMMARY	Implementing germline BRCA testing in all patients with ovarian cancer would be cost-effective in the United Kingdom.
Literature	GENERIF_SUMMARY	This meta-analysis showed that a BRCA2 mutation predicted poor survival outcomes in patients with prostate cancer, especially in those undergoing treatments with radiotherapy. Therefore, the use of BRCA2 mutation as a clinical prognostic factor could help stratify the high-risk patients and provide clinical strategies for more effective targeted treatments for patients with prostate cancer.
Literature	GENERIF_SUMMARY	Five patients were identified with intragenic mutations predicted to restore BRCA1/2 open reading frames, including two patients with multiple independent reversion alleles.
Literature	GENERIF_SUMMARY	BRCA2 rs144848 polymorphism is associated with cancer.
Literature	GENERIF_SUMMARY	Data suggest that modulation of histone deacetylase (HAT) activity by an SNP in BRCA2 (breast cancer type

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		2 susceptibility protein; rs144848, 1342A>C, N372H) is a plausible mechanism of paclitaxel resistance in breast cancer; after HAT inhibitor treatment, HAT activity, and paclitaxel sensitivity is restored in heterozygous BRCA2 variant breast cancer cells.
Literature	GENERIF_SUMMARY	Germline BRCA1/2 gene mutations could result in genomic instability and an elevated gene mutation rate (such as the p53 gene) in breast luminal cells compared with the general population, predisposing BRCA carriers to develop p53-positive/triple-negative breast carcinomas.
Literature	GENERIF_SUMMARY	Knowing a patient's BRCA mutation status is important for prevention and treatment decision-making. Improving the characterization of mutations will lead to better management, treatment, and BCa prevention efforts in African Americans who are disproportionately affected with aggressive BCa and may inform future precision medicine genomic-based clinical studies.
Literature	GENERIF_SUMMARY	polygenic risk scores may provide informative cancer risk stratification for male carriers of BRCA1/2 mutations that might enable these men and their physicians to make informed decisions on the type and timing of breast and prostate cancer risk management.
Literature	GENERIF_SUMMARY	BRCA2 mutation carriers showed earlier onset age of breast tumor and higher risk of developing contra lateral breast cancer in women from Murcia (south-eastern Spain).
Literature	GENERIF_SUMMARY	Clinical outcomes between placebo- and olaparib-treated patients with somatic BRCA1/2 mutations were similar to those with germline BRCA1/2 mutations, indicating that patients with somatic BRCA1/2 mutations benefit from treatment with olaparib.
Literature	GENERIF_SUMMARY	Women with BRCA1/2 mutations show at diagnosis higher peritoneal tumor load and increased frequency of bulky lymph nodes compared to patients without germline BRCA mutations. Primary debulking surgery seems to ensure a longer progression-free survival in women with BRCA wild type genotype compared to neoadjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Being a male BRCA1 or a BRCA2 mutation carrier or even being clinically defined as having high cancer risk with no known BRCA mutation in men is not associated with any clinically significant risk for developing this rare entity.
Literature	GENERIF_SUMMARY	The diagnostic value of BRCA promoter methylation analysis in distinguishing BRCA1/2-related and sporadic breast carcinomas is considerably dependent on the targeted CpG sites. These findings are important for adequate use of BRCA methylation analysis as a prescreening tool for germline genetic testing
Literature	GENERIF_SUMMARY	Study reports that exposure to naturally occurring concentrations of formaldehyde or acetaldehyde

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		selectively unmasks genomic instability in cells heterozygous for multiple, clinically relevant, truncating BRCA2 mutations.
Literature	GENERIF_SUMMARY	Data indicate that c.1310_1313delAAGA mutation of BRCA2 gene is recurrent with high frequency in patients from the North-East region of Morocco.
Literature	GENERIF_SUMMARY	Male carrying BRCA mutations have significantly lower QMAX than healthy men. BRCA1 patients have on average larger prostate glands and higher PSA than BRCA2 patients.
Literature	GENERIF_SUMMARY	This study implicates novel risk loci as well as highlights the clinical utility for retesting BRCA1/2 negative ovarian cancer patients by genomic sequencing and analysis of genes in relevant pathways.
Literature	GENERIF_SUMMARY	Results show frequent BRCA2, EGFR, and NTRK1/2/3 mutations in mismatch repair-deficient colorectal cancers , sugggesting personalized medicine strategies to treat the patients with advanced disease who may have no remaining treatment options.
Literature	GENERIF_SUMMARY	The clustered regularly interspaced short palindromic repeats (CRISPR)/Cas9 system was employed to target and correct a FANCD1 gene deletion. The study shows the ability to correct a patient mutation in primary FANCD1 cells in a precise manner.
Literature	GENERIF_SUMMARY	role of BRCA2 splicing for acquired chemo-resistance in BRCA2 mutation associated malignancy
Literature	GENERIF_SUMMARY	Results, including a comprehensive co-segregation analysis, indicate that the novel duplication identified has a pathogenic role and would be considered a causing-disease variant in genetic and oncologic counseling.
Literature	GENERIF_SUMMARY	IGH/MYC-positive Burkitt lymphoma/leukemia cells have decreased BRCA2 and are sensitive to PARP1 inhibition alone or in combination with other chemotherapies. This study postulates that IGH/MYC-induced BRCA2 deficiency may predispose Burkitt lymphoma cells to synthetic lethality triggered by PARP1 inhibitors.
Literature	GENERIF_SUMMARY	This report summarizes the results of BRCA testing in South-Eastern Norway according to these revised Norwegian guidelines from 1st of January 2014 to 31st of August 2015 -it was to calculate the prevalence of BRCA mutations in this cohort of breast cancer (BC)patients that as a whole had an a priori low risk of being mutation carriers anddescribe the spectrum of mutations
Literature	GENERIF_SUMMARY	Results identified a novel large BRCA2 deletion in 0.9% of the screened Colombian families in addition to 6310delGA and the recurrent 1991del4 mutations. Among unselected breast cancer cases, 1.1% tested positive for BRCA2/3034del4, and 0.4% for BRCA2/1991del4.

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Literature	GENERIF_SUMMARY	Inactivation of homologous recombination factors BRCA1, BRCA2, or RAD51 hypersensitizes cells to acetaldehyde treatment, in spite of the Fanconi anemia pathway being functional.
Literature	GENERIF_SUMMARY	radiosensitization was evaluated using the glioblastoma cell line, U87MG-E6, which harbors inactivated p53, in comparison with the cell line, HCT116 p53 (-/-). We conclude that radiosensitization by arsenite is related to ROS and BRCA2 function.
Literature	GENERIF_SUMMARY	Putative BRCA1/2 reversion mutations can be detected by cfDNA sequencing analysis in patients with ovarian and breast cancer. Our findings warrant further investigation of cfDNA sequencing to identify putative BRCA1/2 reversion mutations and to aid the selection of patients for PARP inhibition therapy
Literature	GENERIF_SUMMARY	BRCA2 Mutation is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	BRCA2 Mutation is associated with Breast Cancer.
Literature	GENERIF_SUMMARY	Carboplatin and talazoparib showed efficacy in DNA damage mutation carriers, but hematologic toxicity was more pronounced in gBRCA(gBRCA1/2) carriers. Carboplatin is best combined with intermittent talazoparib dosing differentiated by germline and somatic DNA damage mutation carriers
Literature	GENERIF_SUMMARY	Studies suggest that people with high risk of breast cancer and negative BRCA1/BRCA2 should be tested for PALB2 mutations.
Literature	GENERIF_SUMMARY	Evidence does not support routine risk-reducing hysterectomy for BRCA1 and BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	this study identifies the importance of TDP1 as a novel determinant of response to CNDAC across various cancer types (especially non-small cell lung cancers), and demonstrates the differential involvement of BRCA2, PARP1, and TDP1 in the cellular responses to CNDAC, AraC, and CPT
Literature	GENERIF_SUMMARY	This study detected monoallelic L1053X mutation causing the same stop codon in BRCA2 protein sequence at the same position in four Sudanese female breast cancer patients out of nine from different families.
Literature	GENERIF_SUMMARY	The work-flow represents a robust and easy-to-use method for full BRCA1/2 screening, which can be easily implemented in routine diagnostic testing and adapted to genes other than BRCA1/2.
Literature	GENERIF_SUMMARY	Data suggest that locus-specific LOH may be a clinically useful biomarker to predict primary resistance to DNA damaging agents in patients with germline BRCA1 and BRCA2 mutations.
Literature	GENERIF_SUMMARY	Ovarian response to stimulation, expressed as the number of mature oocytes, was reduced in BRCA1 but not in BRCA2 mutation carriers. Although oocyte yield

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		was in correspondence to a normal response in all subgroups, this finding points to a possible negative influence of the BRCA1 gene on ovarian reserve
Literature	GENERIF_SUMMARY	We performed the genotyping of the polymorphisms BRCA1/P871L and BRCA1/Q356R by Polymerase Chain Reaction-Restriction Fragment Length Polymorphism, and of variant allele BRCA2/N372H through direct sequencing.
Literature	GENERIF_SUMMARY	Germline mutations involving the Fanconi anemia pathway, such as BRCA2 are often implicated in Invasive Pancreatic Ductal Adenocarcinoma.
Literature	GENERIF_SUMMARY	Tumours arising in BRCA2 mutation carriers showed significantly higher methylation of candidate genes, than those arising in non-BRCA2 familial MBCs (average AMI 23.6 vs 16.6, p = 0.01, 45% of genes hypermethylated vs 34%, p
Literature	GENERIF_SUMMARY	Data show that mammary epithelial cells are inviable upon BRCA2 loss, which leads to replication stress associated with under replication, causing mitotic abnormalities and G1 arrest.
Literature	GENERIF_SUMMARY	Among BRCA mutation(BRCA1 or BRCA2) carriers, the mortality benefit of preventive mastectomy at age 25 is substantial, but the expected benefit declines rapidly with increasing age at surgery.
Literature	GENERIF_SUMMARY	Although no founder mutations were identified in this study, two recurrent mutations, BRCA1 (c.3607C>T) and BRCA2 (c.5164_5165 delAG), were found
Literature	GENERIF_SUMMARY	The present data in this small cohort of 68 mutation carriers suggest that smoking and low physical activity during adolescence are risk factors for developing breast cancer in women with BRCA1 or BRCA2 mutation.
Literature	GENERIF_SUMMARY	We demonstrated an association between six previously published single nucleotide polymorphisms (rs15869 [BRCA2], rs1805389 [LIG4], rs8079544 [TP53], rs25489 [XRCC1], rs1673041 [POLD1], and rs11615 [ERCC1]) and subsequent CNS tumors in survivors of childhood cancer treated by radiation therapy.
Literature	GENERIF_SUMMARY	The present study shows that the G allele carriers of BRCA2 rs9534275 were associated with increased serum total cholesterol and ApoB levels in the coronary artery disease patients and increased risk of coronary artery disease and ischemic stroke.
Literature	GENERIF_SUMMARY	Data show that BRCA2 was required for HDAC2/3 association with acetylated BubR1 in nocodazole (Noc)-arrested cells.
Literature	GENERIF_SUMMARY	Five per cent of unselected Asian patients with breast cancer carry deleterious variants in BRCA2 in Malaysia.
Literature	GENERIF_SUMMARY	We describe twenty-nine novel BRCA1 and BRCA2

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		variants detected in Italian individuals suffering from hereditary breast and ovarian cancer syndrome (HBOC).
Literature	GENERIF_SUMMARY	the physiological significance of GIPC3 as a genetic interactor of BRCA2 is supported by the observation that Brca2-null embryos with Gipc3 overexpression are developmentally more advanced than their control littermates. Taken together, we have uncovered a novel role for GIPC3 as a BRCA2 genetic interactor.
Literature	GENERIF_SUMMARY	14 had pathogenic variants in the BRCA1 gene and 5 had pathogenic variants in the BRCA2 gene
Literature	GENERIF_SUMMARY	observations indicate that nascent DNA degradation in BRCA1/2-deficient cells occurs as a consequence of MRE11-dependent nucleolytic processing of reversed forks generated by fork remodelers
Literature	GENERIF_SUMMARY	Among ovarian cancer patients, BRCA2 carriers with mutations located in the RAD51-BD (exon 11) have prolonged PFS, PFI, and OS.
Literature	GENERIF_SUMMARY	Overall, 5152 oncogenetic tests were reviewed in the present study, of which 4452 had no a priori known familial mutation. The majority of participants (68.6%) were genotyped because of personal history of cancer; 20.6% were tested because of family history of cancer, and details for the remaining 10.7% were missing. Overall, 256/4452 (5.8%) carriers were detected, 141 BRCA1 and 115 BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	patient-derived xenografts capture the molecular and phenotypic heterogeneity of triple-negative breast cancer . Here we show that PARP inhibition can have activity beyond germline BRCA1/2 altered tumors, causing regression in a variety of molecular subtypes. These models represent an opportunity for the discovery of rational combinations with targeted therapies and predictive biomarkers
Literature	GENERIF_SUMMARY	The present study demonstrates a clear protective effect of early first pregnancy on breast cancer risk in both BRCA1 and BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	These results will form the basis for a larger randomized trial (LIBRE-2) with an estimated sample size of 600 BRCA1 and BRCA2 mutation carriers aiming at improvement of BMI, aerobic and maximal exercise capacity and adherence to a Mediterranean diet
Literature	GENERIF_SUMMARY	We conclude that BRCA1 and BRCA2 could be used as clinicopathological biomarkers to evaluate the prognosis of digestive system cancers.
Literature	GENERIF_SUMMARY	The authors propose that BRCA2 antagonizes 53BP1, RIF1, and Artemis-dependent c-nonhomologous end-joining and alt-nonhomologous end-joining to prevent gross genomic instability in a RAD51-independent manner.
Literature	GENERIF_SUMMARY	This is the largest study of comprehensive BRCA testing(BRCA1 and BRCA2) among Brazilians to date.

Main Category	Category	Terms
		Several criteria that are not included in the NCCN achieved a higher predictive value. Identification of the most informative criteria for each population will assist in the development of a rational approach to genetic testing
Literature	GENERIF_SUMMARY	We show a strong association between Triple Negative Breast Cancer and mutations in BRCA1/2 genes and the poor outcome of these patients. The survival curve analysis showed that the presence of AKT1, TP53, KDR, KIT, BRCA1 and BRCA2 mutations is correlated with a poor prognosis.
Literature	GENERIF_SUMMARY	The estimated penetrance of breast cancer to age 80 years was 60.8% for BRCA1 and 63.1% for BRCA2. For all BRCA carriers, the penetrance of breast cancer to age 80 for those with no first-degree relative with breast cancer was 60.4% and 63.3% for those with at least 1 first-degree relative with breast cancer
Literature	GENERIF_SUMMARY	The 2 identified likely pathogenic Unclassified variants in BRCA1 and BRCA2 require further verification with clinical evidence.
Literature	GENERIF_SUMMARY	Both BRCA1 and BRCA2 mutations are associated with an increased risk for Prostate cancer(PC). BRCA2 in particular confers a more aggressive PC phenotype with a higher probability of locally advanced and metastatic disease, and should be considered a prognostic marker associated with poorer survival
Literature	GENERIF_SUMMARY	We demonstrate that homologous recombination deficiency (HRD)mutation signatures may offer clinically relevant information independently of BRCA1/2 mutation status and hope this work will guide the development of clinical trials
Literature	GENERIF_SUMMARY	All patients in our centre were tested using next-generation sequencing (NGS) panels that included full gene sequencing as well as coverage for large deletions/duplications in BRCA1/2. We calculated MSS1-3 scores for index patients between 2014 and 2017 who had undergone BRCA1/2 genetic testing and recorded their genetic test results
Literature	GENERIF_SUMMARY	Among BRCA1/2 mutation wild-type patients, homologous recombination (HR) deficient(HRD) patients were more likely to achieve a pathologic complete response (OR 16, 95% CI 1.65-160.41, p = 0.0041) compared with HR non-deficient patients. Further, HRD scores were highly concordant pre- and post-therapy
Literature	GENERIF_SUMMARY	The overall frequencies of the BRCA germline mutations (BRCA1 and BRCA2)were 10.2% in breast and 30.7% in ovarian cancer patients. These data shed new light into the prevalence of BRCA mutations in the Arab women population.
Literature	GENERIF_SUMMARY	BRCA2 SNP is associated with Enhanced response rate to pegylated liposomal doxorubicin in high grade serous

Main Category	Category	Terms
		ovarian carcinomas.
Literature	GENERIF_SUMMARY	It is possible that mutant BRCA1/2 genes are associated with a reduction in ovarian reserve (review)
Literature	GENERIF_SUMMARY	Carriers of BRCA1/BRCA2 mutation with sporadic Pancreatic Ductal Adenocarcinoma had a worse survival after pancreatectomy than their BRCA wild-type counterparts. However, platinum-based chemotherapy regimens were associated with markedly improved survival in patients with BRCA1/BRCA2 mutations, with survival differences no longer appreciated with wild-type patients.
Literature	GENERIF_SUMMARY	Eligible patients for this study were healthy women with >/= 17% lifetime risk of breast cancer or with a mutation in BRCA1 or BRCA2. A total of 632 women were screened between 2002 and 2012 (each for 6 years). During the study, 30 women were diagnosed with breast cancer, with 10 of these diagnoses occurring between screening visits, and six of the 10 diagnosed women were gene carriers
Literature	GENERIF_SUMMARY	This study provides a comprehensive view of known and novel BRCA1/2 mutations in breast cancer (BC)patients from the Republic of Macedonia and contributes to the global spectrum of BRCA1/2 mutations in breast cancer
Literature	GENERIF_SUMMARY	Our findings, using the largest gene panel for Male breast cancer (MBC) patients so far, indicate that BRCA testing (BRCA1 and BRCA2)should be the primary concern for MBC patients.
Literature	GENERIF_SUMMARY	In this study, we used comprehensive multigene panels that included 35 known or suspected cancer susceptibility genes to examine BRCA1/2 mutation-negative Korean patients who had clinical features indicative of hereditary breast cancer
Literature	GENERIF_SUMMARY	No evidence for a an association between the genotype at BRCA2 c.6937 + 594T>G and breast cancer risk.
Literature	GENERIF_SUMMARY	Assessment of the clinical relevance of BRCA2 missense variants by functional and computational approaches has been presented.
Literature	GENERIF_SUMMARY	Neither the patients tested nor the control subjects showed germline hypermethylation of the BRCA1 and BRCA2 promoter regions analyzed
Literature	GENERIF_SUMMARY	BRCA1/2 mutations are not uncommon among selected Jordanian females with breast cancer.
Literature	GENERIF_SUMMARY	male BRCA1/2 mutation carriers with breast and prostate cancer demonstrated a favorable 5-year survival.
Literature	GENERIF_SUMMARY	This study comprehensively describes the characteristics of the 1,650 unique BRCA1 and 1,731 unique BRCA2 deleterious (disease-associated) mutations identified in the CIMBA database.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	These findings might be exploited to assess the relevance for cancer risk of other BRCA2 spliceogenic variants.
Literature	GENERIF_SUMMARY	Neoadjuvant versus adjuvant with standard anthracycline- and taxane-containing regimens results in similar disease-free survival and overall survival among patients with stage I and II triple-negative breast cancer regardless of BRCA status
Literature	GENERIF_SUMMARY	Our findings indicate that in the Indian population, there is a high prevalence of mutations in the high-risk breast cancer genes
Literature	GENERIF_SUMMARY	Results showed that BRCA2 is down-regulated in epithelial ovarian cancer by lncRNA RP11-552M11.4 which promotes cell proliferation, migration and invasion.
Literature	GENERIF_SUMMARY	BRCA1/2 mutations were associated with improved overall survival (OS) and progression-free survival (PFS), but the improved OS was only seen in patients with primary and recurrent ovarian cancer, not in those with advanced state disease. When examined separately, BRCA1 mutations remained significantly associated with improved OS, but not PFS, and BRCA2 mutations alone were not associated with either improved OS or PFS.
Literature	GENERIF_SUMMARY	association between family history (FH) of Breast cancer(BC) and ovarian cancer(OC) risks was slightly smaller in magnitude (HR
Literature	GENERIF_SUMMARY	We identified a speculative association of germ-line BRCA1/2 alterations with brain metastases in pancreas ductal adenocarcinoma
Literature	GENERIF_SUMMARY	BRCA2 germ line mutation is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 germ line mutation is associated with unilateral triple-negative breast cancer.
Literature	GENERIF_SUMMARY	Our results did not provide evidence for elevated risks of invasivebreast cancer (BC) or epithelial ovarian cancer in BRCA1/BRCA2 predictive test negatives
Literature	GENERIF_SUMMARY	we believe we are the first to describe this germline mutation of BRCA2 (c.4211C>G, p.Ser1404Ter) in a patient with PCa, which was effectively treated with ADT and radiotherapy.
Literature	GENERIF_SUMMARY	Different mechanisms inactivating the wild type allele are present within the same tumor sample at various extents. Results indicate that BRCA1/2-linked breast and ovarian cancer cells are predominantly characterized by LOH, but harbor a complex combination of second hits at various frequencies.
Literature	GENERIF_SUMMARY	This study reports familial communication in a BRCA1/BRCA2 screening setting... This suggests that universal BRCA screening does not differentially affect

Main Category	Category	Terms
		communication or the use of familial genetic information
Literature	GENERIF_SUMMARY	chimerism in breast/ovarian cancers in BRCA non-carriers from BRCA-positive families
Literature	GENERIF_SUMMARY	Among Korean women with the BRCA1/2 mutation, the location of the mutations may influence the risk of breast and ovarian cancers according to the putative functional domain regions
Literature	GENERIF_SUMMARY	Authors found that BRCA1/2 germline mutations in China exhibit distinct characteristics compared to those in Western populations.
Literature	GENERIF_SUMMARY	homologous recombination deficiency (HRD), tumor BRCA1/2 mutations, and absence of CCNE1 amplification are associated with improved survival of ovarian cancer patients treated with platinum monotherapy and HRD-positive patients may benefit from platinum dose intensification.
Literature	GENERIF_SUMMARY	BRCA1/2 mutation status leads to better responses to neoadjuvant chemotherapy in breast cancer. Pathological complete response is the main predictor of disease-free survival and overall survival, independently of BRCA1/2 mutation status.
Literature	GENERIF_SUMMARY	In 7657 consecutive unselected breast cancer patients without BRCA1/2 mutations.
Literature	GENERIF_SUMMARY	Numbers for BRCA2 were insufficient to draw conclusions. The rate of local recurrences after breast-conserving therapy did not differ between BRCA1 carriers (10-year risk = 7.3%) and noncarriers (10-year risk = 7.9%).
Literature	GENERIF_SUMMARY	Pre-menopausal BRCA1/2 mutation carriers aged 30 to 47years chose screening, RRSO, or BS/DO. For those undergoing BS/DO, the delayed oophorectomy was recommended at age 40years for BRCA1 and age 45years for BRCA2 patients.
Literature	GENERIF_SUMMARY	Despite calls for BRCA1/2 population screening, there remains a critical need to identify those most at risk who should receive cancer genetics services. B-RSTtrade mark version 3.0 demonstrates high sensitivity for BRCA1/2 mutations, yet remains a simple and quick screening tool for at-risk individuals.
Literature	GENERIF_SUMMARY	BRCA2 expression is highly sensitive to promoter variations as most of them induced relevant changes. Moreover, we mapped critical regions of the BRCA2 promoter that may constitute potential targets for regulatory variants.
Literature	GENERIF_SUMMARY	our study shows that the K3326* variant in BRCA2 associates primarily with cancer types that have strong environmental genotoxic risk factors. We report for the first time an association between the variant and risk of SCLC and SQCSC.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The prevalence of germline BRCA1/2 mutations in patients with peritoneal carcinoma or fallopian tube carcinoma is comparable to that of BRCA1/2 mutations in epithelial ovarian cancer patients.
Literature	GENERIF_SUMMARY	FoxM1 activation occurs secondary to FoxO3a suppression in Idiopathic pulmonary fibrosis (IPF) fibroblasts while restoration of FoxO3a function sensitizes IPF fibroblasts to radiation-induced cell death and downregulates FoxM1, RAD51, and BRCA2.
Literature	GENERIF_SUMMARY	Based on a cumulative risk of 0.55% to age 35 for BRCA1 mutation carriers and of 0.56% to age 45 for BRCA2 mutation carriers, we recommend bilateral salpingo-oophorectomy before age 40, but by age 35, for women with a BRCA1 mutation and by age 45 for those with a BRCA2 mutation to maximize prevention and to minimize adverse effects.
Literature	GENERIF_SUMMARY	Mutations in BRCA1, BRCA2, and PALB2, and a panel of 50 cancer-associated genes in pancreatic ductal adenocarcinoma
Literature	GENERIF_SUMMARY	BRCA1/2 mutation testing thus has important and expanding roles in treatment planning for subsets of patients with breast cancer..Determining BRCA1/2 mutation status in this breast cancer subgroup could potentially expand treatment options beyond the current standard of taxane and anthracycline-based chemotherapy.
Literature	GENERIF_SUMMARY	In adjusted analysis, age >/=80 years (odds ratio [OR] 0.10; P = 0.002), psychiatric disorders (OR 0.46; P = 0.006), occupation requiring at least 3 years of university or college education (OR 2.03; P = 0.003), and breast cancer or ovarian cancer in first-degree or second-degree relatives (OR 1.66; P = 0.02) were independently associated with uptake of BRCA1/2 testing.
Literature	GENERIF_SUMMARY	The recalibrated algorithms and new metapredictors significantly improved upon current models for predicting the impact of variants in cancer risk-associated domains of BRCA1 and BRCA2. Prediction of the functional impact of all possible variants in BRCA1 and BRCA2 provides important information about the clinical relevance of variants in these genes.
Literature	GENERIF_SUMMARY	BRCA methylation is rare in breast and ovarian carcinomas of BRCA germline mutation carriers, although the frequency of BRCA promoter methylation may be underestimated. This could have major implications for clinical practice, including referral for genetic testing and BRCAness analysis for treatment decision-making.
Literature	GENERIF_SUMMARY	The germline mutational landscape of BRCA1 and BRCA2 in Brazil.
Literature	GENERIF_SUMMARY	Protein truncating variants (PTVs) in specific DNA repair genes were significantly overrepresented among

Main Category	Category	Terms
		patients with the aggressive phenotype, with BRCA2, ATM and NBN the most frequently mutated genes.
Literature	GENERIF_SUMMARY	These results suggest a modest association between BRCA1 and BRCA2 and Nonsyndromic cleft lip with or without cleft palate.
Literature	GENERIF_SUMMARY	BRCA1/2 genes are the most commonly mutated pancreatic cancer susceptibility genes that should be considered in all pancreatic cancer cases with young age at onset or a family history of cancer.
Literature	GENERIF_SUMMARY	oursplicing analysis performed in three independent carriers show thatBRCA2c.7976+5G>T alleles produce a major in-frame transcript Delta17 predicted to encode a non-functional protein, and that even though variable proportions of additional transcripts (Delta16-18, Delta17,18,Delta18 and 17q224) have been detected, none of them arepredicted to rescue BRCA2 functionality.
Literature	GENERIF_SUMMARY	findings provided the evidence that gBRCA1/2 mutation was not associated with survival in Chinese EOC patients, which possibly attributed to more than 37% of the patients without gross residual disease. Survival benefit of gBRCA1/2 mutation was prominent in ovarian cancer patients with gross residual disease.
Literature	GENERIF_SUMMARY	The present study aimed to clarify the clinicopathological features, including the level of p53 protein expression and BRCA mutations, of primary fallopian tube cancer (PFTC) in Japanese women.
Literature	GENERIF_SUMMARY	The overall yield of comprehensive BRCA1/2 testing in ethnically diverse high-risk Israeli individuals is 3.3%. This is lower than expected by probability models. A slightly higher rate of BRCA1/2 carriers was seen among ovarian cancer cases. These data should guide BRCA1/2 optimal testing strategy in Israel.
Literature	GENERIF_SUMMARY	Overall, 65/648 (10%) study participants were BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	our study provides novel data on ovarian BRCA1/2 mutation prevalence worldwide, enhances adequate patient selection for family counseling and prevention, and sheds light on the benefits of iPARP treatment.
Literature	GENERIF_SUMMARY	Data found higher BRCA1/2 mRNA-expression in ovarian cancer (OC). BRCA1 mutated OCs exhibited lower BRCA1 but higher BRCA2 mRNA-expression. Low BRCA1-expression was associated with favorable overall survival and low BRCA2-expression with better progression-free and overall survival.
Literature	GENERIF_SUMMARY	BRCA1/2 mutation prevalence in unselected breast cancer patients was 1.8%.
Literature	GENERIF_SUMMARY	Study identified rare single nucleotide variants and small indels mapping to the 5' non-coding region of BRCA2 through targeted sequencing of over 6000 early onset/familial breast cancer patients. Four variants in

Main Category	Category	Terms
		minimal promoter regions seem to alter gene activity; two others disrupt the binding with PAX5.
Literature	GENERIF_SUMMARY	these results provide a significant insight into certain mutations and proteins involved in the interaction network of BRCA1 and BRCA2, which may have common roles in breast cancer and ovarian cancer.
Literature	GENERIF_SUMMARY	Identified two BRCA2 variants of unknown significance.
Literature	GENERIF_SUMMARY	Study confirmed the clinical relevance of BRCA1/2 mutations in Korean ovarian cancer (KoOC) patients but not for the early age-of-onset. Possible inconsistency in the ratio of BRCA1-to-BRCA2 mutations and the most common mutation between KoOC and Korean breast cancer (KoBC) may probably suggest presence of mutation sequence-associated penetrance tendency in hereditary Korean breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Study shows that BRCA1, PALB2 and BRCA2 can all play a significant role in both checkpoint activation and checkpoint maintenance, depending on cell type and context, and that PALB2 links BRCA1 and BRCA2 in the checkpoint response. The BRCA1-PALB2 interaction can be important for checkpoint activation, whereas PALB2-BRCA2 complex appears to be more critical for checkpoint maintenance.
Literature	GENERIF_SUMMARY	The aims of our study were to assess the clinical validity of BRCA1/2 promoter methylation and BRCA1-like genomic profile to identify BRCAness and to evaluate its correlations with clinicopathological features in triple-negative breast cancers
Literature	GENERIF_SUMMARY	The total number of Ovarian Cancer patients captured was 802; of these, 53 and five families carry 22 mutations in BRCA1/2 genes. Of these, eight mutations were unique to the Arab populations, and five mutations were commonly circulated among Arabs (BRCA1
Literature	GENERIF_SUMMARY	The study of BRCA1/2 gene in Southern Mediterranean countries revealed low penetrance recurrent mutations in sporadic and familial breast cancer.
Literature	GENERIF_SUMMARY	The meta-analysis results provide clinicians and health-care regulatory agencies with evidence of the increased risk of colorectal cancer in BRCA1 mutation carriers, but not in BRCA2.
Literature	GENERIF_SUMMARY	BRCA2 variant (rs80359182, 319T/C) is significantly associated with high breast cancer risk in the Pakistani population. BRCA2-tryptophan > arginine substitutions result in altered interaction of BRCA1/PALB2/BRCA2/ protein complex and impaired homology-directed DNA repair pathway.
Literature	GENERIF_SUMMARY	Among BRCA carrier(BRCA1/2) patients treated for early breast cancer, the addition of contralateral breast irradiation was associated with a significant reduction of subsequent contralateral breast cancers and a delay in their onset.

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Literature	GENERIF_SUMMARY	Results found that Israeli BRCA1 or BRCA2 mutation carriers are at an increased risk for developing uterine cancer, especially serous papillary and sarcoma.
Literature	GENERIF_SUMMARY	BRCA carriers with and without malignancy exhibit comparable ovarian reserve and responses to ovarian stimulation compared with women with BRCA-negative cancers and cancer-free controls.
Literature	GENERIF_SUMMARY	Germline variants of unknown significance in BRCA1/2 are more frequent in African-American than Caucasian-American prostate cancer patients; however, the prevalence of pathogenic mutations were similar across the races. Patients carrying BRCA2 pathogenic mutations are more likely to progress to metastasis.
Literature	GENERIF_SUMMARY	Study show that damage-induced long non-coding RNAs (dilncRNAs) play a role in DNA double-strand breaks (DSBs) repair by homologous recombination (HR) by contributing to the recruitment of the HR proteins BRCA1, BRCA2, and RAD51, without affecting DNA-end resection. BRCA2 directly interacts with RNase H2, mediates its localization to DSBs in the S/G2 cell-cycle phase, and controls DNA
Literature	GENERIF_SUMMARY	Although BRCA mutations are known to alter DNA repair mechanism, BRCA 1/2 gene mutations do not affect the capacity of oocytes from breast cancer candidates for fertility preservation to mature in vitro.
Literature	GENERIF_SUMMARY	RAD52 protein, just as BRCA2, interacts with pCHK1 checkpoint protein and helps maintain the checkpoint control in BRCA2 deficient cells during DNA damage response
Literature	GENERIF_SUMMARY	Prevalence of BRCA1/2 germline pathogenic (P) and likely pathogenic (LP) variants is slightly higher than previously described by the largest occidental studies, with a high prevalence of variant c.5266dupC (p.Gln1756Profs*74) in BRCA1 observed. Moreover, we identified a new LP variant.
Literature	GENERIF_SUMMARY	Frequency of BRCA2 breast cancer pathogenic mutations in the Mexican population.
Literature	GENERIF_SUMMARY	Detection of disease-associated variants in the BRCA1 and BRCA2 (BRCA1/2) genes allows for cancer prevention and early diagnosis in high-risk individuals.
Literature	GENERIF_SUMMARY	Applying established referral criteria for genetic testing in Serbia will help identify BRCA1/2 mutation carriers but will not help identify mutations in other cancer susceptibility genes. Until better predictors emerge we should be performing wider genetic testing of EOC in order to identify all mutation carriers.
Literature	GENERIF_SUMMARY	Germline BRCA2 K3326X and CHEK2 I157T mutations increase risk for sporadic pancreatic ductal adenocarcinoma.
Literature	GENERIF_SUMMARY	Identified APEX2 and FEN1 as synthetic lethal genes with both BRCA1 and BRCA2 loss of function in tumor

Main Category	Category	Terms
		cell lines.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 play essential roles in maintaining the genome stability.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 Germline Mutation Analysis in Hereditary Breast/Ovarian Cancer Families from the Aures Region (Eastern Algeria)
Literature	GENERIF_SUMMARY	A common SNP in the UNG gene decreases ovarian cancer risk in BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	we conducted a prospective study of the relationship between oophorectomy and the risk of contralateral breast cancer in 1781 BRCA1 and 503 BRCA2 mutation carriers with breast cancer.
Literature	GENERIF_SUMMARY	Analyses reveal that BRCA(wt) and BRCA(mut) TNBC are similar except for a significant increase of TIL(pos) tumors in the BRCA(mut) group. While BRCA gene mutations may not directly drive immune infiltration, the greater number of TIL(pos) tumors could signal greater immunogenicity in this group.
Literature	GENERIF_SUMMARY	two novel heterozygous pathogenic variants in exons 18 and 11 of the BRCA2 gene in two Lebanese families, are reported.
Literature	GENERIF_SUMMARY	Patients who harbour germline BRCA2 mutations have worse clinical outcomes than noncarriers when treated with surgery or radiotherapy
Literature	GENERIF_SUMMARY	In pancreatic ductal adenocarcinoma, analysis of persevered tissue, analysis of BRCA expression offered the benefit of yielding results in a short period of time and at a low cost.
Literature	GENERIF_SUMMARY	12 pathogenic variants (PVs) were identified in 26 cases (9%) , 22 cases harboring BRCA1, and four cases in BRCA2. Eight out of these twelve PVs were reported. novel mutation (c.5760_5770delTTTTGCTGACA) in BRCA2 gene.
Literature	GENERIF_SUMMARY	RAD51 can dissociate the very D-loops they create if in stoichiometric excess. BRCA2 regulates HR at a post-synaptic stage by modulating RAD51-mediated D-loop dissociation. Provides a biochemical mechanistic underpinning of homeostasis between RAD51 and BRCA2 previously reported in cellular experiments.
Literature	GENERIF_SUMMARY	A large fraction of BRCA2 variants (approximately 64%) provoked splicing anomalies lending further support to the high prevalence of this disease-mechanism in hereditary breast cancer.
Literature	GENERIF_SUMMARY	The BRCA2 is the most prevalent type mutation among hereditary breast and/or ovarian cancer patients in the Arab region, with an estimated prevalence was 17% with a higher pooled prevalence of BRCA mutations in the Levant Region 28%. When compared with studies from different parts of the world the Arab world appear to have different profiles with varying BRCA

Main Category	Category	Terms
		frequencies.[meta-analysis]
Literature	GENERIF_SUMMARY	Meta-analysis of cohort, 7 case-control, 4 case-series, 28 frequency, and 11 survival studies shows being a BRCA mutation carrier (BRCA1 and/or BRCA2) was associated with a significant increase in prostate cancer (PCa) risk (OR = 1.90, 95% CI = 1.58-2.29), with BRCA2 mutation being associated with a greater risk of PCa than BRCA1 (OR = 1.35, 95% CI = 1.03-1.76).
Literature	GENERIF_SUMMARY	Heterozygous c.5722_5723del mutation in the BRCA2 exon 11 of a large Han-Chinese family with breast cancer.This mutation may cause DNA double-strand breaks repair dysfunction by disturbing homologous recombination, further resulting in breast cancer.
Literature	GENERIF_SUMMARY	This review summarizes the current evidence for screening/imaging protocols, as well as screening for breast and ovarian cancer susceptibility protein 1 (BRCA1) and breast and ovarian cancer susceptibility protein 2(BRCA2) gene mutations, in breast and ovarian cancer diagnosis in the clinical setting.
Literature	GENERIF_SUMMARY	These findings suggested that the BRCA2 gene mutation is a good prognostic factor and can be used as a gene to predict the prognosis in the bladder cancer patients.
Literature	GENERIF_SUMMARY	BRCA 1/2 mutations are associated with a higher hematologic toxicity in patients with ovarian cancer who underwent platinum-based chemotherapy.
Literature	GENERIF_SUMMARY	There was no statistically significant association between sperm DNA fragmentation and the methylation status of the promoter of either BRCA1 or BRCA2 genes in infertile men with oligoasthenoteratozoospermia.
Literature	GENERIF_SUMMARY	This is the largest study to date comparing outcomes in patients with EOC and BM by mutation status. mBRCA1 and mBCRA2 patients were more likely to have isolated BM, which may be a factor in their long survival.
Literature	GENERIF_SUMMARY	The BRCA2 c.67+2T>C splice-site variant may contribute to the pathogenesis of familial breast cancer.
Literature	GENERIF_SUMMARY	Screening, Active Surveillance, and Treatment of Localized Prostate Cancer Among Carriers of Germline BRCA Mutations.
Literature	GENERIF_SUMMARY	Previving
Literature	GENERIF_SUMMARY	systematic genomic analysis of breast tumors with BRCA1 and BRCA2 mutations
Literature	GENERIF_SUMMARY	These data assign a definite deficiency to BRCA2 as a primary ovarian insufficiency driver.
Literature	GENERIF_SUMMARY	observation of a positive association between body mass index and ovarian cancer risk in premenopausal BRCA1/2 mutation carriers is consistent with findings in the general population

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	This report is the first to describe the highly pathogenic variant in the BRCA2 gene (rs483353122) and the likely damaging germline variant in the MUTYH gene (rs35352891) in Russian Mongoloid BC patients with young-onset and/or bilateral and/or familial BC.
Literature	GENERIF_SUMMARY	mutant BRCA is an indispensable founding event for some tumours, but in a considerable proportion of other cancers, it appears to be biologically neutral--a difference predominantly conditioned by tumour lineage--with implications for disease pathogenesis, screening, design of clinical trials and therapeutic decision-making
Literature	GENERIF_SUMMARY	BRRM was associated with lower mortality than surveillance for BRCA1 mutation carriers, but for BRCA2 mutation carriers, BRRM may lead to similar BC-specific survival as surveillance. Our findings support a more individualized counseling based on BRCA mutation type.
Literature	GENERIF_SUMMARY	BRCA1/2 pathogenic variants have become associated with familial breast and ovarian cancers, and hereditary cancer-predisposition syndrome.
Literature	GENERIF_SUMMARY	The role of BRCA1 and BRCA2 mutations in cancer susceptibility is REVIEWED.
Literature	GENERIF_SUMMARY	Outcome of Pancreatic Cancer Surveillance Among High-Risk Individuals Tested for Germline Mutations in BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	carrying a BRCA mutation have an increased risk of developing breast and ovarian cancer. The most effective strategy to reduce this risk is the bilateral salpingo-oophorectomy, with or without additional risk-reducing mastectomy. Risk-reducing bilateral salpingo-oophorectomy (RRBSO) is recommended between age 35 and 40 and between age 40 and 45 years for women carriers of BRCA1 and BRCA2 mutations, respectively. Consequen
Literature	GENERIF_SUMMARY	The study highlights these recent discoveries, which include mechanisms of RFP regulation, DNA damage checkpoint proteins, as well as kinases that regulate BRCA1/2 function. [review]
Literature	GENERIF_SUMMARY	Findings are the first to provide compelling evidence of the role of BRCA2 in ESCC genetic susceptibility in Chinese, suggesting defective homologous recombination is an underlying cause in ESCC pathogenesis.
Literature	GENERIF_SUMMARY	BRCA1/2 mutation prevalence in Chinese ovarian cancer patients is higher than the international rate.
Literature	GENERIF_SUMMARY	Chromatin Remodeling in Response to BRCA2-Crisis.
Literature	GENERIF_SUMMARY	BRCA2, but not BRCA1 mutations may have a role in uveal melanoma susceptibility that represents a rare source of increased risk

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Literature	GENERIF_SUMMARY	Two mutations in BRCA1 c.224_227delAAAG and c.5431C > T and one mutation in BRCA2 c.5576_5579delTTAA were identified.
Literature	GENERIF_SUMMARY	Germline BRCA-Associated Endometrial Carcinoma Is a Distinct Clinicopathologic Entity.
Literature	GENERIF_SUMMARY	Germline BRCA1 and BRCA2 testing for breast cancer survivors.
Literature	GENERIF_SUMMARY	In the randomized, double-blind, phase III POLO trial progression-free survival was significantly longer with maintenance olaparib, a poly(ADP-ribose) polymerase inhibitor, than placebo in patients with a germline BRCA1 and/or BRCA2 mutation (gBRCAm) and metastatic pancreatic cancer whose disease had not progressed during first-line platinum-based chemotherapy. The prespecified HRQoL evaluation is reported here.
Literature	GENERIF_SUMMARY	Novel BRCA2 pathogenic genotype and breast cancer phenotype discordance in monozygotic triplets.
Literature	GENERIF_SUMMARY	Amplification of the Mutation-Carrying BRCA2 Allele Promotes RAD51 Loading and PARP Inhibitor Resistance in the Absence of Reversion Mutations.
Literature	GENERIF_SUMMARY	in the present study we set out to characterize the clinical features of platinum sensitivity that are associated with BRCA 1/2 mutations in ovarian cancer in both primary debulking and interval debulking surgery.
Literature	GENERIF_SUMMARY	The gBRCA mutation detection rate in serous tubal intra-epithelial carcinoma-positive patients was nearly double that of serous tubal intra-epithelial carcinoma-negative patients. Factors such as a positive family history of BRCA-related cancers were seen at a higher proportion in the mutation positive women.
Literature	GENERIF_SUMMARY	Clinicopathologic features and genetic characteristics of the BRCA1/2 mutation in Turkish breast cancer patients.
Literature	GENERIF_SUMMARY	We identified point mutations in 9 (23%) of 40 T-ALL cases analyzed, with variant allele fractions consistent with heterozygous mutations early in tumor evolution. BRCA2 was the most commonly mutated gene.
Literature	GENERIF_SUMMARY	Association of Genomic Domains in BRCA1 and BRCA2 with Prostate Cancer Risk and Aggressiveness.
Literature	GENERIF_SUMMARY	Women with ovarian cancer at an older age, with few or no affected family members, or with a lower Manchester Score are more likely to carry a BRCA2 PV than a BRCA1 PV, irrespective of the overall higher risk of ovarian cancer associated with BRCA1 PVs.
Literature	GENERIF_SUMMARY	Do BRCA1 and BRCA2 gene mutation carriers have a reduced ovarian reserve? Protocol for a prospective observational study.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Anti-mullerian hormone curves according to BRCA status and adjusted by age showed that BRCA2 mutation-positive patients have lower levels of anti-mullerian hormone as compared with BRCA-negative and BRCA1 mutation-positive.
Literature	GENERIF_SUMMARY	Identification of a novel germline BRCA2 variant in a Chinese breast cancer family.
Literature	GENERIF_SUMMARY	consensus guide represents a collection of technical recommendations to address the detection of BRCA1/2 mutations in the molecular diagnostic testing strategy for ovarian carcinomas[review]
Literature	GENERIF_SUMMARY	Our results demonstrated that germline BRCA2 and CHEK2 mutations are independent unfavorable predictors in patients with mPCa which are associated with decreased time to castration resistance (HR 3.04, 95% CI 1.63-5.66, p
Literature	GENERIF_SUMMARY	Study revealed a potential novel gene and multiple disruptive variants of BRCA2 for breast cancer risk, which may identify high-risk women in Chinese populations.
Literature	GENERIF_SUMMARY	Mutations of genes BRCA1 and BRCA2 in women with ovarian cancer exposed to factors of Chornobyl nuclear accident have been reported.
Literature	GENERIF_SUMMARY	Analysis of clinicopathologic characteristics of gastric cancer in patients /=40 years of age.
Literature	GENERIF_SUMMARY	Impact of BRCA1 and BRCA2 mutations on ovarian reserve and fertility preservation outcomes in young women with breast cancer.
Literature	GENERIF_SUMMARY	Is BRCA2 involved in early onset colorectal cancer risk?
Literature	GENERIF_SUMMARY	Impact of proactive high-throughput functional assay data on BRCA1 variant interpretation in 3684 patients with breast or ovarian cancer.
Literature	GENERIF_SUMMARY	BRCA2 gene mutation and prostate cancer risk. Comprehensive review and update.
Literature	GENERIF_SUMMARY	Germline and somatic mutations in prostate cancer
Literature	GENERIF_SUMMARY	BRCA1-BRCA2 mutation analysis results in 910 individuals
Literature	GENERIF_SUMMARY	The role of the redox/miR-6855-3p/PRDX5A axis in reversing SLUG-mediated BRCA2 silencing in breast cancer cells.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 germline testing in Cretan isolates reveals novel and strong founder effects.
Literature	GENERIF_SUMMARY	Multifocal breast cancers are more prevalent in BRCA2 versus BRCA1 mutation carriers.
Literature	GENERIF_SUMMARY	Longitudinal outcomes with cancer multigene panel testing in previously tested BRCA1/2 negative patients.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Association of Germline Genetic Testing Results With Locoregional and Systemic Therapy in Patients With Breast Cancer.
Literature	GENERIF_SUMMARY	Germline mutations of multiple breast cancer-related genes are differentially associated with triple-negative breast cancers and prognostic factors.
Literature	GENERIF_SUMMARY	Contralateral breast cancer risk in BRCA1 and BRCA2 mutation carriers in a large cohort of unselected Chinese breast cancer patients.
Literature	GENERIF_SUMMARY	The implications of BRCA loss of heterozygosity (LOH) and deficient mismatch repair gene (dMMR) expression in the breast cancer of a patient with both inherited breast and ovarian cancer syndrome (BRCA2) and Lynch syndrome (MLH1).
Literature	GENERIF_SUMMARY	Skipping Nonsense to Maintain Function
Literature	GENERIF_SUMMARY	BRCA and PALB2 mutations in a cohort of male breast cancer with one bilateral case.
Literature	GENERIF_SUMMARY	Aneuploidy and a deregulated DNA damage response suggest haploinsufficiency in breast tissues of BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	Homologous recombination deficiency status-based classification of high-grade serous ovarian carcinoma.
Literature	GENERIF_SUMMARY	Salvage lymphadenectomy in recurrent ovarian cancer patients
Literature	GENERIF_SUMMARY	miRNA expression profiling of hereditary breast tumors from BRCA1- and BRCA2-germline mutation carriers in Brazil.
Literature	GENERIF_SUMMARY	next-generation sequencing BRCA1/2 assay could detect most large genomic rearrangements including BRCA1 promoter-region deletion
Literature	GENERIF_SUMMARY	Germline BRCA, chemotherapy response scores, and survival in the neoadjuvant treatment of ovarian cancer.
Literature	GENERIF_SUMMARY	Mutation landscape of germline and somatic BRCA1/2 in patients with high-grade serous ovarian cancer.
Literature	GENERIF_SUMMARY	The RAD52 S346X variant reduces risk of developing breast cancer in carriers of pathogenic germline BRCA2 mutations.
Literature	GENERIF_SUMMARY	Executive function after risk-reducing salpingo-oophorectomy in BRCA1 and BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Knowledge and attitude of hereditary breast cancer among Japanese university female students.
Literature	GENERIF_SUMMARY	Germline BRCA Mutation Rates in Latina Women Presenting for Gynecologic Oncology Care.
Literature	GENERIF_SUMMARY	Low BRCA2 expression predicts poor prognoses in patients with rectal cancer receiving

Main Category	Category	Terms
		chemoradiotherapy.
Literature	GENERIF_SUMMARY	BRCA2 bound to PLK1 forms a complex with the phosphatase PP2A and phosphorylated-BUBR1.
Literature	GENERIF_SUMMARY	Risk of colorectal cancer associated with BRCA1 and/or BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Founder BRCA1/BRCA2/PALB2 pathogenic variants in French-Canadian breast cancer cases and controls.
Literature	GENERIF_SUMMARY	Clinical outcome of breast cancer in carriers of BRCA1 and BRCA2 mutations according to molecular subtypes.
Literature	GENERIF_SUMMARY	Distinct roles of BRCA2 in replication fork protection in response to hydroxyurea and DNA interstrand cross-links.
Literature	GENERIF_SUMMARY	Case report
Literature	GENERIF_SUMMARY	The contribution of large genomic rearrangements in BRCA1 and BRCA2 to South African familial breast cancer.
Literature	GENERIF_SUMMARY	E3 ligase RFWD3 is a novel modulator of stalled fork stability in BRCA2-deficient cells.
Literature	GENERIF_SUMMARY	Functional evaluation of five BRCA2 unclassified variants identified in a Sri Lankan cohort with inherited cancer syndromes using a mouse embryonic stem cell-based assay.
Literature	GENERIF_SUMMARY	Predictors of long-term cancer-related distress among female BRCA1 and BRCA2 mutation carriers without a cancer diagnosis
Literature	GENERIF_SUMMARY	Alternative mRNA splicing can attenuate the pathogenicity of presumed loss-of-function variants in BRCA2.
Literature	GENERIF_SUMMARY	Our results suggest that single-nucleotide variant rs2910164 does not influence the colorectal cancer risk in Brazilian patients; however, the GG genotype could act as a factor of worse prognosis in patients with advanced disease due to suppression of BRCA1/2 modulated by miR-146a.
Literature	GENERIF_SUMMARY	High-throughput functional evaluation of BRCA2 variants of unknown significance.
Literature	GENERIF_SUMMARY	Increased prevalence of BRCA1/2 mutations in women with macrotextured breast implants and anaplastic large cell lymphoma of the breast.
Literature	GENERIF_SUMMARY	The FANC/BRCA Pathway Releases Replication Blockades by Eliminating DNA Interstrand Cross-Links.
Literature	GENERIF_SUMMARY	Prevalence of BRCA1/BRCA2 pathogenic variation in Chinese Han population.
Literature	GENERIF_SUMMARY	Similar response rates and survival with PARP inhibitors for patients with solid tumors harboring somatic versus

Main Category	Category	Terms
		Germline BRCA mutations
Literature	GENERIF_SUMMARY	Association of gBRCA1/2 mutation locations with ovarian cancer risk in Japanese patients from the CHARLOTTE study.
Literature	GENERIF_SUMMARY	Effects of BRCA gene mutation on female reproductive potential
Literature	GENERIF_SUMMARY	Validation of a next generation sequencing assay for BRCA1, BRCA2, CHEK2 and PALB2 genetic testing.
Literature	GENERIF_SUMMARY	Clinical Characteristics and Exploratory Genomic Analyses of Germline BRCA1 or BRCA2 Mutations in Breast Cancer.
Literature	GENERIF_SUMMARY	Retrospective reinterpretation and reclassification of BRCA1/2 variants from Chinese population.
Literature	GENERIF_SUMMARY	Identification of Somatically Acquired BRCA1/2 Mutations by cfDNA Analysis in Patients with Metastatic Breast Cancer.
Literature	GENERIF_SUMMARY	Germline BRCA2 Truncating Mutation in Familial Esophageal Squamous Cell Carcinoma
Literature	GENERIF_SUMMARY	Characterization and in silico analyses of the BRCA1/2 variants identified in individuals with personal and/or family history of BRCA-related cancers.
Literature	GENERIF_SUMMARY	DSS1 and ssDNA regulate oligomerization of BRCA2.
Literature	GENERIF_SUMMARY	Characterization of the Cancer Spectrum in Men With Germline BRCA1 and BRCA2 Pathogenic Variants
Literature	GENERIF_SUMMARY	Complex Characterization of Germline Large Genomic Rearrangements of the BRCA1 and BRCA2 Genes in High-Risk Breast Cancer Patients-Novel Variants from a Large National Center.
Literature	GENERIF_SUMMARY	BReast CAncer susceptibility gene 2 deficiency exacerbates oxidized LDL-induced DNA damage and endothelial apoptosis.
Literature	GENERIF_SUMMARY	Calibration of Pathogenicity Due to Variant-Induced Leaky Splicing Defects by Using BRCA2 Exon 3 as a Model System.
Literature	GENERIF_SUMMARY	Presymptomatic Awareness of Germline Pathogenic BRCA Variants and Associated Outcomes in Women With Breast Cancer.
Literature	GENERIF_SUMMARY	A Phase I/II Study of Veliparib (ABT-888) in Combination with 5-Fluorouracil and Oxaliplatin in Patients with Metastatic Pancreatic Cancer.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 pathogenic variant carriers and endometrial cancer risk
Literature	GENERIF_SUMMARY	Germline and Somatic BRCA1/2 Gene Mutational Status and Clinical Outcomes in Epithelial Peritoneal, Ovarian, and Fallopian Tube Cancer

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Microcephaly family protein MCPH1 stabilizes RAD51 filaments.
Literature	GENERIF_SUMMARY	Olaparib and durvalumab in patients with germline BRCA-mutated metastatic breast cancer (MEDIOLA)
Literature	GENERIF_SUMMARY	Genetic testing in Poland and Ukraine
Literature	GENERIF_SUMMARY	Conformational flexibility and oligomerization of BRCA2 regions induced by RAD51 interaction.
Literature	GENERIF_SUMMARY	Rucaparib in Men With Metastatic Castration-Resistant Prostate Cancer Harboring a BRCA1 or BRCA2 Gene Alteration.
Literature	GENERIF_SUMMARY	An overview of cancer genetics with focus on involvement of BRCA1/2 genes in breast carcinomas.
Literature	GENERIF_SUMMARY	Segregation analysis of the BRCA2 c.9227G>T variant in multiple families suggests a pathogenic role in breast and ovarian cancer predisposition.
Literature	GENERIF_SUMMARY	A Preclinical Trial and Molecularly Annotated Patient Cohort Identify Predictive Biomarkers in Homologous Recombination-deficient Pancreatic Cancer.
Literature	GENERIF_SUMMARY	DNA Strand Exchange to Monitor Human RAD51-Mediated Strand Invasion and Pairing.
Literature	GENERIF_SUMMARY	Next generation sequencing analysis of BRCA1 and BRCA2 identifies novel variations in breast cancer.
Literature	GENERIF_SUMMARY	Several novel variations in BRCA1 and BRCA2 genes have been identified.
Literature	GENERIF_SUMMARY	Recurrent BRCA1 Mutation, but no BRCA2 Mutation, in Vietnamese Patients with Ovarian Carcinoma Detected with Next Generation Sequencing.
Literature	GENERIF_SUMMARY	Prevalence of Tissue BRCA Gene Mutation in Ovarian, Fallopian Tube, and Primary Peritoneal Cancers
Literature	GENERIF_SUMMARY	Genetic screening results of individuals with high risk BRCA-related breast/ovarian cancer in Trakya region of Turkey.
Literature	GENERIF_SUMMARY	Mutations in ATM, NBN and BRCA2 predispose to aggressive prostate cancer in Poland.
Literature	GENERIF_SUMMARY	Identification of Recurrent Variants in BRCA1 and BRCA2 across Multiple Cancers in the Chinese Population.
Literature	GENERIF_SUMMARY	Contribution of BRCA1 and BRCA2 germline mutations to early onset breast cancer
Literature	GENERIF_SUMMARY	Prevalence of pancreaticobiliary cancers in Irish families with pathogenic BRCA1 and BRCA2 variants.
Literature	GENERIF_SUMMARY	Breast cancer survival in Nordic BRCA2 mutation carriers-unconventional association with oestrogen receptor status.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	RAD52 aptamer regulates DNA damage repair and STAT3 in BRCA1/BRCA2deficient human acute myeloid leukemia.
Literature	GENERIF_SUMMARY	Ethnic-specific BRCA1/2 variation within Asia population
Literature	GENERIF_SUMMARY	Pathogenicity reclassification of two BRCA1/BRCA2 exonic duplications after identification of genomic breakpoints and tandem orientation.
Literature	GENERIF_SUMMARY	DSS1 allosterically regulates the conformation of the tower domain of BRCA2 that has dsDNA binding specificity for homologous recombination.
Literature	GENERIF_SUMMARY	Germline BRCA 1-2 status prediction through ovarian ultrasound images radiogenomics
Literature	GENERIF_SUMMARY	Incidence of contralateral and ipsilateral breast cancers and prognosis in BRCA1/2 pathogenic variant carriers based on the Japanese HBOC Consortium registration.
Literature	GENERIF_SUMMARY	Retrospective evaluation of risk-reducing salpingo-oophorectomy for BRCA1/2 pathogenic variant carriers among a cohort study in a single institution.
Literature	GENERIF_SUMMARY	Genetic and clinical landscape of breast cancers with germline BRCA1/2 variants.
Literature	GENERIF_SUMMARY	Prevalence of germline BRCA mutations in HER2-negative metastatic breast cancer
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 associated breast cancer and the roles of current modelling systems in drug discovery.
Literature	GENERIF_SUMMARY	Elevated CXorf67 Expression in PFA Ependymomas Suppresses DNA Repair and Sensitizes to PARP Inhibitors.
Literature	GENERIF_SUMMARY	Impact of BRCA1/2 Mutations on the Efficacy of Secondary Cytoreductive Surgery.
Literature	GENERIF_SUMMARY	Novel poly (ADP-ribose) polymerases inhibitor DHC-1 exhibits in vitro and in vivo anticancer activity on BRCA-deficient pancreatic cancer cells.
Literature	GENERIF_SUMMARY	BRCAness as a prognostic indicator in patients with early breast cancer.
Literature	GENERIF_SUMMARY	Five Italian Families with Two Mutations in BRCA Genes.
Literature	GENERIF_SUMMARY	Epidemiological and ES cell-based functional evaluation of BRCA2 variants identified in families with breast cancer.
Literature	GENERIF_SUMMARY	Detection of novel infiltrating ductal carcinoma-associated BReast CAncer gene 2 mutations which alter the deoxyribonucleic acid-binding ability of BReast CAncer gene 2 protein.
Literature	GENERIF_SUMMARY	Novel BRCA2 c.8434_8435insTT (p. Gly2812Valfs*10)

Main Category	Category	Terms
		mutation in a family with multiple hematologic malignancies and solid tumors.
Literature	GENERIF_SUMMARY	The BRCA2 p.N372 H i.a.1342A>C Could Regulate the Sensitivity of Ovarian Cancer Cells to Platinum-Based Drugs.
Literature	GENERIF_SUMMARY	Germinal BRCA1-2 pathogenic variants (gBRCA1-2pv) and pancreatic cancer
Literature	GENERIF_SUMMARY	Reduced translation efficiency due to novel splicing variants in 5' untranslated region and identification of novel cis-regulatory elements in canine and human BRCA2.
Literature	GENERIF_SUMMARY	BRCA2, ATM, and CDK12 Defects Differentially Shape Prostate Tumor Driver Genomics and Clinical Aggression.
Literature	GENERIF_SUMMARY	RADX controls RAD51 filament dynamics to regulate replication fork stability.
Literature	GENERIF_SUMMARY	Prevalence and reclassification of BRCA1 and BRCA2 variants in a large, unselected Chinese Han breast cancer cohort.
Literature	GENERIF_SUMMARY	Interpretation of BRCA2 Splicing Variants
Literature	GENERIF_SUMMARY	Four novel BRCA variants found in Chinese hereditary breast cancer patients by next-generation sequencing.
Literature	GENERIF_SUMMARY	Homologous recombination deficiency
Literature	GENERIF_SUMMARY	Identification of Variants (rs11571707, rs144848, and rs11571769) in the BRCA2 Gene Associated with Hereditary Breast Cancer in Indigenous Populations of the Brazilian Amazon.
Literature	GENERIF_SUMMARY	Germline copy number variations in BRCA1/2 negative families
Literature	GENERIF_SUMMARY	Transcriptome Patterns of BRCA1- and BRCA2- Mutated Breast and Ovarian Cancers.
Literature	GENERIF_SUMMARY	The disease sites of female genital cancers of BRCA1/2-associated hereditary breast and ovarian cancer
Literature	GENERIF_SUMMARY	Satisfaction with prophylactic risk-reducing salpingo-oophorectomy in BRCA mutation carriers is very high and little dependent on the participants' characteristics at surgery
Literature	GENERIF_SUMMARY	Genomic Analysis of Circulating Tumor DNA in 3,334 Patients with Advanced Prostate Cancer Identifies Targetable BRCA Alterations and AR Resistance Mechanisms.
Literature	GENERIF_SUMMARY	Surgical decision making in premenopausal BRCA carriers considering risk-reducing early salpingectomy or salpingo-oophorectomy

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Uptake and efficacy of bilateral risk reducing surgery in unaffected female BRCA1 and BRCA2 carriers.
Literature	GENERIF_SUMMARY	Nipple Sparing Mastectomy as a Risk-Reducing Procedure for BRCA-Mutated Patients.
Literature	GENERIF_SUMMARY	A case-only study to identify genetic modifiers of breast cancer risk for BRCA1/BRCA2 mutation carriers.
Literature	GENERIF_SUMMARY	Treatment-emergent neuroendocrine prostate cancer with a germline BRCA2 mutation
Literature	GENERIF_SUMMARY	Strong functional data for pathogenicity or neutrality classify BRCA2 DNA-binding-domain variants of uncertain significance.
Literature	GENERIF_SUMMARY	Association between BRCA2 alterations and intraductal and cribriform histologies in prostate cancer.
Literature	GENERIF_SUMMARY	BRCA2 promotes DNA-RNA hybrid resolution by DDX5 helicase at DNA breaks to facilitate their repairdouble dagger.
Literature	GENERIF_SUMMARY	Endometrial Cancer Risk in Women With Germline BRCA1 or BRCA2 Mutations
Literature	GENERIF_SUMMARY	Novel and recurrent BRCA1/BRCA2 germline mutations in patients with breast/ovarian cancer
Literature	GENERIF_SUMMARY	Structural Variants at the BRCA1/2 Loci are a Common Source of Homologous Repair Deficiency in High-grade Serous Ovarian Carcinoma.
Literature	GENERIF_SUMMARY	A comprehensive reference for BRCA1/2 genes pathogenic variants in Iran
Literature	GENERIF_SUMMARY	Frequency of Contralateral Prophylactic Mastectomy in Breast Cancer Patients with a Negative BRCA1 and BRCA2 Rapid Genetic Test Result.
Literature	GENERIF_SUMMARY	Mutational spectrum in clinically aggressive low-grade serous carcinoma/serous borderline tumors of the ovary-Clinical significance of BRCA2 gene variants in genomically stable tumors.
Literature	GENERIF_SUMMARY	Mutations in BRCA1 and BRCA2 differentially affect the tumor microenvironment and response to checkpoint blockade immunotherapy.
Literature	GENERIF_SUMMARY	Pathways and signatures of mutagenesis at targeted DNA nicks.
Literature	GENERIF_SUMMARY	Association of Germline BRCA Pathogenic Variants With Diminished Ovarian Reserve
Literature	GENERIF_SUMMARY	BRCA1/2 Mutation Detection in the Tumor Tissue from Selected Polish Patients with Breast Cancer Using Next Generation Sequencing.
Literature	GENERIF_SUMMARY	Detection of Molecular Signatures of Homologous Recombination Deficiency in Bladder Cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	High BRCA2 Gene Expression is Associated with Aggressive and Highly Proliferative Breast Cancer.
Literature	GENERIF_SUMMARY	A recurrent pathogenic BRCA2 exon 5-11 duplication in the Christian Arab population in Israel.
Literature	GENERIF_SUMMARY	Endometrial Cancer, BRCA1, and BRCA2 in the UK Biobank Cohort.
Literature	GENERIF_SUMMARY	Optimization of prediction methods for risk assessment of pathogenic germline variants in the Japanese population.
Literature	GENERIF_SUMMARY	BRCA1/2 signaling and homologous recombination deficiency in breast and ovarian cancer.
Literature	GENERIF_SUMMARY	Making Sense of a Health Threat
Literature	GENERIF_SUMMARY	Double heterozygotes of BRCA1/BRCA2 and mismatch repair gene pathogenic variants
Literature	GENERIF_SUMMARY	BRCA2
Literature	GENERIF_SUMMARY	Clinicopathological Features of BRCA1/2 Mutation-Positive Breast Cancer.
Literature	GENERIF_SUMMARY	Prevalence of BRCA1 and BRCA2 Mutations Among Patients With Ovarian, Primary Peritoneal, and Fallopian Tube Cancer in India
Literature	GENERIF_SUMMARY	Prevalence and clinical impact of tumor BRCA1 and BRCA2 mutations in patients presenting with localized or metastatic hormone-sensitive prostate cancer.
Literature	GENERIF_SUMMARY	The predictive ability of the 313 variant-based polygenic risk score for contralateral breast cancer risk prediction in women of European ancestry with a heterozygous BRCA1 or BRCA2 pathogenic variant.
Literature	GENERIF_SUMMARY	A homozygous nonsense mutation early in exon 5 of BRCA2 is associated with very severe Fanconi anemia.
Literature	GENERIF_SUMMARY	BRCA 1, 2 mutation and earlier menopause
Literature	GENERIF_SUMMARY	A novel BRCA1 duplication and new insights on the spectrum and frequency of germline large genomic rearrangements in BRCA1/BRCA2.
Literature	GENERIF_SUMMARY	Towards a New, Endophenotype-Based Strategy for Pathogenicity Prediction in BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	Mutations in BRCA-related breast and ovarian cancer in the South African Indian population
Literature	GENERIF_SUMMARY	Next-generation sequencing of BRCA1 and BRCA2 genes in Moroccan prostate cancer patients with positive family history.
Literature	GENERIF_SUMMARY	Germline and tumor BRCA1/2 pathogenic variants in Chinese triple-negative breast carcinomas.
Literature	GENERIF_SUMMARY	Role of BRCA2 DNA-binding and C-terminal domain in

Main Category	Category	Terms
		its mobility and conformation in DNA repair.
Literature	GENERIF_SUMMARY	CRIP1 cooperates with BRCA2 to drive the nuclear enrichment of RAD51 and to facilitate homologous repair upon DNA damage induced by chemotherapy.
Literature	GENERIF_SUMMARY	BRCA-mutant pancreatic ductal adenocarcinoma.
Literature	GENERIF_SUMMARY	Prevalence and predictors of germline BRCA1 and BRCA2 mutations among young patients with breast cancer in Jordan.
Literature	GENERIF_SUMMARY	Preliminary results of targeted sequencing of BRCA1 and BRCA2 in a cohort of breast cancer families
Literature	GENERIF_SUMMARY	Association of somatic mutations in BRCA2 BRC domain with chemotherapy sensitivity and survival in high grade serous ovarian cancer.
Literature	GENERIF_SUMMARY	Anti-mullerian hormone levels and antral follicle count in women with a BRCA1 or BRCA2 germline pathogenic variant
Literature	GENERIF_SUMMARY	Breast and Prostate Cancer Risks for Male BRCA1 and BRCA2 Pathogenic Variant Carriers Using Polygenic Risk Scores.
Literature	GENERIF_SUMMARY	BRCA2 binding through a cryptic repeated motif to HSF2BP oligomers does not impact meiotic recombination.
Literature	GENERIF_SUMMARY	Understanding BRCA2 Function as a Tumor Suppressor Based on Domain-Specific Activities in DNA Damage Responses.
Literature	GENERIF_SUMMARY	Intrinsic Disorder and Phosphorylation in BRCA2 Facilitate Tight Regulation of Multiple Conserved Binding Events.
Literature	GENERIF_SUMMARY	Tumour-infiltrating CD4-, CD8- and FOXP3-positive immune cells as predictive markers of mortality in BRCA1- and BRCA2-associated breast cancer.
Literature	GENERIF_SUMMARY	A transcription-based mechanism for oncogenic beta-catenin-induced lethality in BRCA1/2-deficient cells.
Literature	GENERIF_SUMMARY	Guardians of the Genome
Literature	GENERIF_SUMMARY	PARP Inhibitors Display Differential Efficacy in Models of BRCA Mutant High-Grade Serous Ovarian Cancer.
Literature	GENERIF_SUMMARY	Multitumor Case Series of Germline BRCA1, BRCA2 and CHEK2-Mutated Patients Responding Favorably on Immune Checkpoint Inhibitors.
Literature	GENERIF_SUMMARY	Vaginal NOTES approach for risk-reducing salpingo-oophorectomy in BRCA mutation carriers
Literature	GENERIF_SUMMARY	Analysis of mutation status and homologous recombination deficiency in tumors of patients with germline BRCA1 or BRCA2 mutations and metastatic breast cancer

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Literature	GENERIF_SUMMARY	Is olaparib cost effective in metastatic castration-resistant prostate cancer patients with at least one favorable gene mutation in BRCA1, BRCA2 or ATM?
Literature	GENERIF_SUMMARY	Association between pathogenic germline mutations in BRCA2 and ATM and tumor-infiltrating lymphocytes in primary prostate cancer.
Literature	GENERIF_SUMMARY	Physical activity and Mediterranean diet as potential modulators of osteoprotegerin and soluble RANKL in gBRCA1/2 mutation carriers
Literature	GENERIF_SUMMARY	Detection of hypermethylation BRCA1/2 gene promoter in breast tumours among Moroccan women.
Literature	GENERIF_SUMMARY	A complex of BRCA2 and PP2A-B56 is required for DNA repair by homologous recombination.
Literature	GENERIF_SUMMARY	The Prognostication Potential of BRCA Genes Expression in Gliomas
Literature	GENERIF_SUMMARY	Setting a diagnostic benchmark for tumor BRCA testing
Literature	GENERIF_SUMMARY	Tumor BRCA testing in ovarian cancer and EQA scheme
Literature	GENERIF_SUMMARY	BRCA2 associates with MCM10 to suppress PRIMPOL-mediated repriming and single-stranded gap formation after DNA damage.
Literature	GENERIF_SUMMARY	Prevalence and clinical characterization of BRCA1 and BRCA2 mutations in Korean patients with epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	Association of germline rare pathogenic mutations in guideline-recommended genes with prostate cancer progression
Literature	GENERIF_SUMMARY	The Importance of Extended Analysis Using Current Molecular Genetic Methods Based on the Example of a Cohort of 228 Patients with Hereditary Breast and Ovarian Cancer Syndrome.
Literature	GENERIF_SUMMARY	Genotype-cancer association in patients with Fanconi anemia due to pathogenic variants in FANCD1 (BRCA2) or FANCN (PALB2).
Literature	GENERIF_SUMMARY	Clinicopathological features and BRCA1 and BRCA2 mutation status in a prospective cohort of young women with breast cancer.
Literature	GENERIF_SUMMARY	Breast cancer in West Africa
Literature	GENERIF_SUMMARY	Basal expression of RAD51 foci predicts olaparib response in patient-derived ovarian cancer xenografts.
Literature	GENERIF_SUMMARY	Sirtuin inhibition is synthetic lethal with BRCA1 or BRCA2 deficiency.
Literature	GENERIF_SUMMARY	Some subgroups might get less benefit from adjuvant olaparib trial in high-risk, HER2-negative and germline BRCA2 BRCA1- or BRCA2-mutated early breast cancer patients.

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Literature	GENERIF_SUMMARY	WRN helicase safeguards deprotected replication forks in BRCA2-mutated cancer cells.
Literature	GENERIF_SUMMARY	Double heterozygous pathogenic variants in the BRCA1 and BRCA2 genes in a patient with bilateral metachronous breast cancer.
Literature	GENERIF_SUMMARY	Genomic mutation features identify distinct BRCA-associated mutation characteristics in endometrioid carcinoma and endometrioid ovarian carcinoma.
Literature	GENERIF_SUMMARY	BRCA Mutations in Prostate Cancer
Literature	GENERIF_SUMMARY	The risks of breast and ovarian cancer associated with the Ashkenazi Jewish founder allele BRCA2 6174delT.
Literature	GENERIF_SUMMARY	Comparing breast cancer imaging characteristics of CHEK2 with BRCA1 and BRCA2 gene mutation carriers.
Literature	GENERIF_SUMMARY	Risks of breast and ovarian cancer for women harboring pathogenic missense variants in BRCA1 and BRCA2 compared with those harboring protein truncating variants.
Literature	GENERIF_SUMMARY	Prevalence of BRCA1 and BRCA2 pathogenic sequence variants in ovarian cancer patients in the Gulf region
Literature	GENERIF_SUMMARY	BRCA1/2 NGS Somatic Testing in Clinical Practice
Literature	GENERIF_SUMMARY	BRCA2 represses the transcriptional activity of pS2 by E2-ERalpha.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 pathogenic variants and prostate cancer risk
Literature	GENERIF_SUMMARY	The BRCA2 missense mutation K2497R suppressed self-degradation and increased ATP production and cell proliferation.
Literature	GENERIF_SUMMARY	Prognostic Value of the Immunohistochemical Expression of RAD51 and BRCA2 in Gastric Adenocarcinoma.
Literature	GENERIF_SUMMARY	Analysis of pathogenic variants in BRCA1 and BRCA2 genes using next-generation sequencing in women with triple negative breast cancer from South India.
Literature	GENERIF_SUMMARY	Mutations of BRCA1/2 Genes in the West of Turkey and Genotype-Phenotype Correlations.
Literature	GENERIF_SUMMARY	Cancer Risks Associated With BRCA1 and BRCA2 Pathogenic Variants.
Literature	GENERIF_SUMMARY	Single-cell tumor-immune microenvironment of BRCA1/2 mutated high-grade serous ovarian cancer.
Literature	GENERIF_SUMMARY	Recurring pathogenic variants in the BRCA2 gene in the Ethiopian Jewish population. Founder mutations?
Literature	GENERIF_SUMMARY	RB1 loss overrides PARP inhibitor sensitivity driven by RNASEH2B loss in prostate cancer.

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Literature	GENERIF_SUMMARY	Epidemiology and geographic distribution of BRCA1-2 and DNA Damage response genes pathogenic variants in pancreatic ductal adenocarcinoma patients.
Literature	GENERIF_SUMMARY	Predictors of cardiopulmonary fitness in cancer-affected and -unaffected women with a pathogenic germline variant in the genes BRCA1/2 (LIBRE-1).
Literature	GENERIF_SUMMARY	Association of BRCA Mutations and Pancreatic Cancer
Literature	GENERIF_SUMMARY	BRCA1/2 Mutations in Vietnamese Patients with Hereditary Breast and Ovarian Cancer Syndrome.
Literature	GENERIF_SUMMARY	Pancreatic Cancer with Mutation in BRCA1/2, MLH1, and APC Genes
Literature	GENERIF_SUMMARY	Prevalence of specific and recurrent/founder pathogenic variants in BRCA genes in breast and ovarian cancer in North Africa.
Literature	GENERIF_SUMMARY	Prevalent versus incident breast cancers
Literature	GENERIF_SUMMARY	Incidence and Prevalence of Intraductal Papillary Mucinous Neoplasms in Individuals With BRCA1 and BRCA2 Pathogenic Variant.
Literature	GENERIF_SUMMARY	BRCA2 promoter hypermethylation as a biomarker for the leukemic transformation of myeloproliferative neoplasms.
Literature	GENERIF_SUMMARY	Immunohistochemical Evaluation of BAP1 Expression in Breast Cancer with Known BRCA1 and BRCA2 Mutations and Comparison with Histopathological Features.
Literature	GENERIF_SUMMARY	Common founder BRCA2 pathogenic variants and breast cancer characteristics in Ethiopian Jews.
Literature	GENERIF_SUMMARY	Genetic anticipation of breast cancer among BRCA1/ BRCA2 mutation carriers
Literature	GENERIF_SUMMARY	Germline Pathogenic Variants in BRCA1 and BRCA2
Literature	GENERIF_SUMMARY	Population-based screening of Uruguayan Ashkenazi Jews for recurrent BRCA1 and BRCA2 pathogenic sequence variants.
Literature	GENERIF_SUMMARY	Anxiety and depression among racial/ethnic minorities and impoverished women testing positive for BRCA1/2 mutations in the United States.
Literature	GENERIF_SUMMARY	Identification and Management of Pathogenic Variants in BRCA1, BRCA2, and PALB2 in a Tumor-Only Genomic Testing Program.
Literature	GENERIF_SUMMARY	Templated Insertions Are Associated Specifically with BRCA2 Deficiency and Overall Survival in Advanced Ovarian Cancer.
Literature	GENERIF_SUMMARY	Germline BRCA variants, lifestyle and ovarian cancer survival.

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Literature	GENERIF_SUMMARY	BRCA2 C-Terminal RAD51-Binding Domain Confers Resistance to DNA-Damaging Agents.
Literature	GENERIF_SUMMARY	Evaluation of conformational transitions of h-BRCA2 functional domain and unclassified variant Arg2502Cys using multimodal approach.
Literature	GENERIF_SUMMARY	Expansion of Cancer Risk Profile for BRCA1 and BRCA2 Pathogenic Variants.
Literature	GENERIF_SUMMARY	Common Variant in ALDH2 Modifies the Risk of Breast Cancer Among Carriers of the p.K3326* Variant in BRCA2.
Literature	GENERIF_SUMMARY	Divergent binding mode for a protozoan BRC repeat to RAD51.
Literature	GENERIF_SUMMARY	Do Female BRCA2 Pathogenic Variant Carriers Have an Increased Risk of Lung Cancer?
Literature	GENERIF_SUMMARY	Concurrent BRAFV600E and BRCA Mutations in MSS Metastatic Colorectal Cancer
Literature	GENERIF_SUMMARY	Mutational spectrum of BRCA1/2 genes in Moroccan patients with hereditary breast and/or ovarian cancer, and review of BRCA mutations in the MENA region.
Literature	GENERIF_SUMMARY	Prevalence of BRCA1 and BRCA2 Germline Mutations in Patients of African Descent with Early-Onset and Familial Colombian Breast Cancer.
Literature	GENERIF_SUMMARY	Discovery of BRCA1/BRCA2 founder variants by haplotype analysis.
Literature	GENERIF_SUMMARY	Understanding and predicting the functional consequences of missense mutations in BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	WRN rescues replication forks compromised by a BRCA2 deficiency
Literature	GENERIF_SUMMARY	Rare germline variants in PALB2 and BRCA2 in familial and sporadic chordoma.
Literature	GENERIF_SUMMARY	Risk reducing surgery with peritoneal staging in BRCA1-2 mutation carriers. A prospective study.
Literature	GENERIF_SUMMARY	Effect of BRCA1/2 Mutational Status on Survival Outcomes According to Secondary Cytoreductive Surgery and Maintenance Therapy in Platinum-Sensitive Relapsed Ovarian Cancer
Literature	GENERIF_SUMMARY	Physical activity and sleep behaviour in women carrying BRCA1/2 mutations.
Literature	GENERIF_SUMMARY	Prevalence of BRCA1 and BRCA2 pathogenic variants in 8627 unselected patients with breast cancer
Literature	GENERIF_SUMMARY	Heterozygous BRCA1 and BRCA2 and Mismatch Repair Gene Pathogenic Variants in Children and Adolescents With Cancer.

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Literature	GENERIF_SUMMARY	A comprehensive literature review and meta-analysis of the prevalence of pan-cancer BRCA mutations, homologous recombination repair gene mutations, and homologous recombination deficiencies.
Literature	GENERIF_SUMMARY	Analysis of BRCA2 Copy Number Loss and Genomic Instability in Circulating Tumor Cells from Patients with Metastatic Castration-resistant Prostate Cancer.
Literature	GENERIF_SUMMARY	Clinical characteristics of patients with Breast and / or Ovarian Cancer with mutations in the BRCA1 and BRCA2 genes in Cordoba, Argentina"", trans ""[Caracteristicas clinicas de pacientes con Cancer de Mama y / o Cancer de Ovario con mutaciones en los genes BRCA1 y BRCA2 en Cordoba, Argentina.]"
Literature	GENERIF_SUMMARY	Characterization of lung cancers in patients with BRCA germline variants
Literature	GENERIF_SUMMARY	BRCA1/2 Variants Identified Through Tumor Genomic Profiling
Literature	GENERIF_SUMMARY	Copy number variants as modifiers of breast cancer risk for BRCA1/BRCA2 pathogenic variant carriers.
Literature	GENERIF_SUMMARY	Emergence of BRCA Reversion Mutations in Patients with Metastatic Castration-resistant Prostate Cancer After Treatment with Rucaparib.
Literature	GENERIF_SUMMARY	The avoiding late diagnosis of ovarian cancer (ALDO) project; a pilot national surveillance programme for women with pathogenic germline variants in BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	Familial history and prevalence of BRCA1, BRCA2 and TP53 pathogenic variants in HBOC Brazilian patients from a public healthcare service.
Literature	GENERIF_SUMMARY	Prognostic impact of germline BRCA1/2 pathogenic variants in breast cancer.
Literature	GENERIF_SUMMARY	Characteristics of BRCA2 Mutated Prostate Cancer at Presentation.
Literature	GENERIF_SUMMARY	A spectrum of BRCA1 and BRCA2 germline deleterious variants in ovarian cancer in Russia.
Literature	GENERIF_SUMMARY	Racial and ethnic variation in BRCA1 and BRCA2 genetic test results among individuals referred for genetic counseling at a large urban comprehensive cancer center.
Literature	GENERIF_SUMMARY	Immune cells are increased in normal breast tissues of BRCA1/2 mutation carriers.
Literature	GENERIF_SUMMARY	Ethnic-specificity, evolution origin and deleteriousness of Asian BRCA variation revealed by over 7500 BRCA variants derived from Asian population.
Literature	GENERIF_SUMMARY	Association Between Family History and Risk of Pancreatic Cancer in Patients With BRCA1 and BRCA2 Pathogenic Variants.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	BRCA2 gene mutation in cancer.
Literature	GENERIF_SUMMARY	Elbracht M, Meyer R, Kricheldorf K, et al. Germline variants in DNA repair genes, including BRCA1/2, may cause familial myeloproliferative neoplasms. Blood Adv. 2021;5(17)
Literature	GENERIF_SUMMARY	BRCA1/2 Reversion Mutations in Patients Treated with Poly ADP-Ribose Polymerase (PARP) Inhibitors or Platinum Agents.
Literature	GENERIF_SUMMARY	Ovarian cancer onset across different BRCA mutation types
Literature	GENERIF_SUMMARY	Contralateral Breast Cancer Risk Among Carriers of Germline Pathogenic Variants in ATM, BRCA1, BRCA2, CHEK2, and PALB2.
Literature	GENERIF_SUMMARY	The expression and mutation of BRCA1/2 genes in ovarian cancer
Literature	GENERIF_SUMMARY	Distinct breast cancer phenotypes in BRCA 1/2 carriers based on ER status.
Literature	GENERIF_SUMMARY	PARylation of BRCA1 limits DNA break resection through BRCA2 and EXO1.
Literature	GENERIF_SUMMARY	BRCA1/2 Pathogenic Variants Are Not Common in Merkel Cell Carcinoma
Literature	GENERIF_SUMMARY	Cancer risk in relatives of BRCA1/2 pathogenic variant carriers in a large series of unselected patients with breast cancer.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 germline mutation analysis from a cohort of 1267 patients at high risk for breast cancer in Brazil.
Literature	GENERIF_SUMMARY	Increased prevalence of the founder BRCA1 c.5309G>T and recurrent BRCA2 c.1310_1313delAAGA mutations in breast cancer families from Northerstern region of Morocco
Literature	GENERIF_SUMMARY	The pathogenic role of the BRCA2 c.7847C>T (p.Ser2616Phe) variant in breast and ovarian cancer predisposition.
Literature	GENERIF_SUMMARY	Ovarian cancer pathology characteristics as predictors of variant pathogenicity in BRCA1 and BRCA2.
Literature	GENERIF_SUMMARY	Metastatic Prostate Cancers with BRCA2 versus ATM Mutations Exhibit Divergent Molecular Features and Clinical Outcomes.
Literature	GENERIF_SUMMARY	BRCA2 Germline Mutations Identify Gastric Cancers Responsive to PARP Inhibitors.
Literature	GENERIF_SUMMARY	An overview of the research progress of BRCA gene mutations in breast cancer.
Literature	GENERIF_SUMMARY	Overcoming therapeutic resistance in pancreatic cancer

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Literature	GENERIF_SUMMARY	Association of BRCA2 Gene Functional Polymorphisms with Nonsyndromic Cleft Lip With or Without Cleft Palate in a Chinese Population.
Literature	GENERIF_SUMMARY	Prevalence of BRCA1 and BRCA2 Mutations Among High-risk Bahraini Patients with Breast Cancer.
Literature	GENERIF_SUMMARY	Changes in Bone Density in Carriers of BRCA1 and BRCA2 Pathogenic Variants After Salpingo-Oophorectomy.
Literature	GENERIF_SUMMARY	Significant association of BRCA1 (rs1799950), BRCA2 (rs144848) and TP53 (rs1042522) polymorphism with breast cancer risk in Pashtun population of Khyber Pakhtunkhwa, Pakistan.
Literature	GENERIF_SUMMARY	Associations of height, body mass index, and weight gain with breast cancer risk in carriers of a pathogenic variant in BRCA1 or BRCA2
Literature	GENERIF_SUMMARY	Contralateral breast cancer risk in patients with breast cancer and a germline-BRCA1/2 pathogenic variant undergoing radiation.
Literature	GENERIF_SUMMARY	Functional annotation of variants of the BRCA2 gene via locally haploid human pluripotent stem cells.
Literature	GENERIF_SUMMARY	BRCA-deficient metastatic prostate cancer has an adverse prognosis and distinct genomic phenotype.
Literature	GENERIF_SUMMARY	Long-molecule scars of backup DNA repair in BRCA1- and BRCA2-deficient cancers.
Literature	GENERIF_SUMMARY	Exome sequencing identifies breast cancer susceptibility genes and defines the contribution of coding variants to breast cancer risk.
Literature	GENERIF_SUMMARY	Finding significance
Literature	GENERIF_SUMMARY	Recurrent BRCA2 exon 3 deletion in Assyrian families.
Literature	GENERIF_SUMMARY	BRCA1/2 Haploinsufficiency
Literature	GENERIF_SUMMARY	The Tumor Suppressor BRCA1/2, Cancer Susceptibility and Genome Instability in Gynecological and Mammary Cancers.
Literature	GENERIF_SUMMARY	Immunohistochemical Detection of the Expressed BRCA1 and BRCA2 Proteins in Microenvironment of Malignant Breast Cancerous Tissues Infected with Human Mammary Tumor Virus.
Literature	GENERIF_SUMMARY	Deleterious and ethnic-related BRCA1/2 mutations in tissue and blood of Egyptian colorectal cancer patients and its correlation with human papillomavirus.
Literature	GENERIF_SUMMARY	Subset of retinoblastoma tumours is associated with BRCA1/2 mutations.
Literature	GENERIF_SUMMARY	Causality and functional relevance of BRCA1 and BRCA2 pathogenic variants in non-high-grade serous ovarian carcinomas.

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Literature	GENERIF_SUMMARY	CRISPR/Cas9-mediated knock-in of BRCA1/2 mutations restores response to olaparib in pancreatic cancer cell lines.
Literature	GENERIF_SUMMARY	Low rate of complications in nipple-sparing mastectomy for patients with BRCA1 and BRCA2 mutation.
Literature	GENERIF_SUMMARY	Structural basis for stabilisation of the RAD51 nucleoprotein filament by BRCA2.
Literature	GENERIF_SUMMARY	PP2A inhibition causes synthetic lethality in BRCA2-mutated prostate cancer models via spindle assembly checkpoint reactivation.
Literature	GENERIF_SUMMARY	A Single nucleotide polymorphism in the ALDH2 gene modifies the risk of esophageal squamous cell carcinoma in BRCA2 p.K3326* carriers.
Literature	GENERIF_SUMMARY	Frequencies of Diagnostically Significant Polymorphisms of Hereditary Breast Cancer Forms in BRCA1 and BRCA2 Genes in the Kazakh Population.
Literature	GENERIF_SUMMARY	Risk-reducing mastectomy and breast cancer mortality in women with a BRCA1 or BRCA2 pathogenic variant
Literature	GENERIF_SUMMARY	[BRCA2 Mutation Profile in a Proband with Hereditary Breast and Ovarian Cancer-Two Germline Pathogenic Variants Aligned in the Cis Position].
Literature	GENERIF_SUMMARY	Presence of crown-like structures in breast adipose tissue; differences between healthy controls, BRCA1/2 gene mutation carriers and breast cancer patients.
Literature	GENERIF_SUMMARY	The BRCA2 R2645G variant increases DNA binding and induces hyper-recombination.
Literature	GENERIF_SUMMARY	BRCA1 and BRCA2 germline mutations in Chinese Hakka breast cancer patients.
Literature	GENERIF_SUMMARY	Beyond BRCA
Literature	GENERIF_SUMMARY	BRCA2 promotes genomic integrity and therapy resistance primarily through its role in homology-directed repair.
Literature	GENERIF_SUMMARY	Longitudinal profiling identifies co-occurring BRCA1/2 reversions, TP53BP1, RIF1 and PAXIP1 mutations in PARP inhibitor-resistant advanced breast cancer.
Literature	GENERIF_SUMMARY	Reclassifying BRCA1 c.4358-2A > G and BRCA2 c.475 + 5G > C variants from ""Uncertain Significance"" to ""Pathogenic"" based on minigene assays and clinical evidence."
Literature	GENERIF_SUMMARY	BRCA1, BRCA2, and TP53 germline and somatic variants and clinicopathological characteristics of Brazilian patients with epithelial ovarian cancer.
Literature	GENERIF_SUMMARY	Central nervous system metastases in breast cancer patients with germline BRCA pathogenic variants compared to non-carriers

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Prevalence and spectrum of germline BRCA1 and BRCA2 in a cohort of ovarian cancer patients from the Salento peninsula (Southern Italy)
Literature	GENERIF_SUMMARY	Functional analysis and clinical classification of 462 germline BRCA2 missense variants affecting the DNA binding domain.
Literature	GENERIF_SUMMARY	Preliminary insights on the mutational spectrum of BRCA1 and BRCA2 genes in Pakhtun ethnicity breast cancer patients from Khyber Pakhtunkhwa (KP), Pakistan.
Literature	GENERIF_SUMMARY	Germline biallelic BRCA2 pathogenic variants and medulloblastoma
Literature	GENERIF_SUMMARY	Mutation spectra of the BRCA1/2 genes in human breast and ovarian cancer and germline.
Literature	GENERIF_SUMMARY	Featuring BRCA1 and BRCA2 germline mutational landscape from Asturias (North Spain).
Literature	GENERIF_SUMMARY	Distinct mobility patterns of BRCA2 molecules at DNA damage sites.
Literature	GENERIF_SUMMARY	[Prostate cancer with BRCA2 pathogenic mutation
Literature	GENERIF_SUMMARY	Hereditary breast cancer next-generation sequencing panel evaluation in the south region of Brazil
Literature	GENERIF_SUMMARY	DSS1 restrains BRCA2's engagement with dsDNA for homologous recombination, replication fork protection, and R-loop homeostasis.
Literature	GENERIF_SUMMARY	Incidence of endometrial cancer in BRCA mutation carriers.
Literature	GENERIF_SUMMARY	Hypermethylation of the BRCA2 gene promoter and its co-hypermethylation with the BRCA1 gene promoter in patients with breast cancer.
Pathways	KEGG_PATHWAY	Homologous recombination
Pathways	KEGG_PATHWAY	Fanconi anemia pathway
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	Pancreatic cancer
Pathways	KEGG_PATHWAY	Breast cancer
Pathways	REACTOME_PATHWAY	Reproduction
Pathways	REACTOME_PATHWAY	Meiosis
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	HDR through MMEJ (alt-NHEJ)
Pathways	REACTOME_PATHWAY	HDR through Homologous Recombination (HRR)

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	DNA Double-Strand Break Repair
Pathways	REACTOME_PATHWAY	Resolution of D-Loop Structures
Pathways	REACTOME_PATHWAY	Homology Directed Repair
Pathways	REACTOME_PATHWAY	Resolution of D-loop Structures through Synthesis-Dependent Strand Annealing (SDSA)
Pathways	REACTOME_PATHWAY	HDR through Homologous Recombination (HRR) or Single Strand Annealing (SSA)
Pathways	REACTOME_PATHWAY	Resolution of D-loop Structures through Holliday Junction Intermediates
Pathways	REACTOME_PATHWAY	Homologous DNA Pairing and Strand Exchange
Pathways	REACTOME_PATHWAY	Presynaptic phase of homologous DNA pairing and strand exchange
Pathways	REACTOME_PATHWAY	DNA Repair
Pathways	REACTOME_PATHWAY	Meiotic recombination
Pathways	REACTOME_PATHWAY	Diseases of DNA repair
Pathways	REACTOME_PATHWAY	Diseases of DNA Double-Strand Break Repair
Pathways	REACTOME_PATHWAY	Defective homologous recombination repair (HRR) due to BRCA2 loss of function
Pathways	REACTOME_PATHWAY	Defective homologous recombination repair (HRR) due to BRCA1 loss of function
Pathways	REACTOME_PATHWAY	Defective homologous recombination repair (HRR) due to PALB2 loss of function
Pathways	REACTOME_PATHWAY	Defective HDR through Homologous Recombination Repair (HRR) due to PALB2 loss of BRCA1 binding function
Pathways	REACTOME_PATHWAY	Defective HDR through Homologous Recombination Repair (HRR) due to PALB2 loss of BRCA2/RAD51/RAD51C binding function
Pathways	REACTOME_PATHWAY	Impaired BRCA2 translocation to the nucleus
Pathways	REACTOME_PATHWAY	Impaired BRCA2 binding to RAD51
Pathways	REACTOME_PATHWAY	Impaired BRCA2 binding to PALB2
Pathways	REACTOME_PATHWAY	Impaired BRCA2 binding to SEM1 (DSS1)
Pathways	WIKIPATHWAYS	Homologous recombination
Pathways	WIKIPATHWAYS	Integrated breast cancer pathway
Pathways	WIKIPATHWAYS	Glioblastoma signaling
Pathways	WIKIPATHWAYS	DNA IR double strand breaks and cellular response via ATM
Pathways	WIKIPATHWAYS	DNA IR damage and cellular response via ATR

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Breast cancer pathway
Pathways	WIKIPATHWAYS	Pancreatic adenocarcinoma pathway
Pathways	WIKIPATHWAYS	Male infertility
Pathways	WIKIPATHWAYS	DNA repair pathways full network
Pathways	WIKIPATHWAYS	Cancer pathways
Pathways	WIKIPATHWAYS	Primary ovarian insufficiency
Pathways	WIKIPATHWAYS	Fanconi anemia
Protein Domains	INTERPRO	BRCA2_repeat
Protein Domains	INTERPRO	NA-bd_OB-fold
Protein Domains	INTERPRO	BRCA2_OB_1
Protein Domains	INTERPRO	BRCA2_OB_3
Protein Domains	INTERPRO	Tower_dom
Protein Domains	INTERPRO	BRCA2_hlx
Protein Domains	INTERPRO	BRCA2
Protein Domains	INTERPRO	BRCA2_hlx_sf
Protein Domains	INTERPRO	BRCA2_OB_2_dom
Protein Domains	INTERPRO	BRCA2_TR2
Protein Domains	PIR_SUPERFAMILY	BRCA2
Protein Domains	SMART	Tower
Protein Domains	UP_KW_DOMAIN	Repeat
Protein Domains	UP_KW_DOMAIN	Transmembrane
Protein Domains	UP_KW_DOMAIN	Transmembrane helix